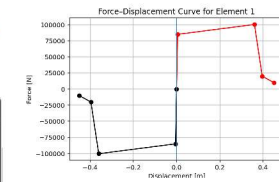
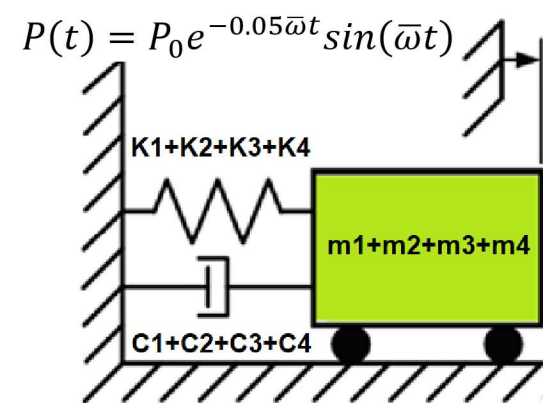


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

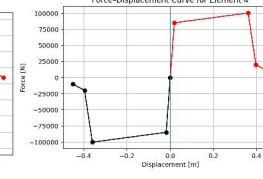
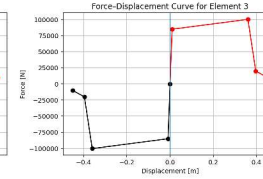
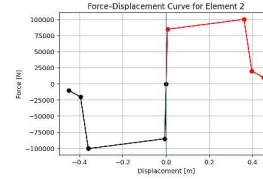
SENSITIVITY ANALYSIS OF SINGLE-DEGREE-FREEDOM (SDOF) STRUCTURES USING INCREMENTAL DYNAMIC ANALYSIS EFFECTS OF INITIAL DISPLACEMENT, STRUCTURAL ELASTIC STIFFNESS, STRUCTURAL DUCTILITY RATIO AND OVER-STRENGTH FACTOR ON OUTPUT KEY PARAMETERS SUCH AS EQUIVALENT VISCOUS DAMPING RATIO FROM NONLINEAR DYNAMIC ANALYSES USING PYTHON AND OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

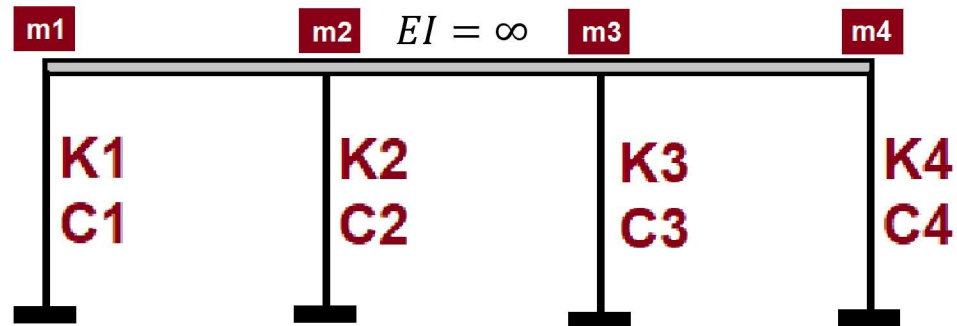
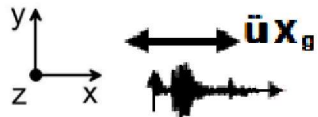
INCREMENTAL
DISPLACEMENT



Initial Displacement
Initial Velocity
Initial Acceleration



INCREMENTAL
DISPLACEMENT



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

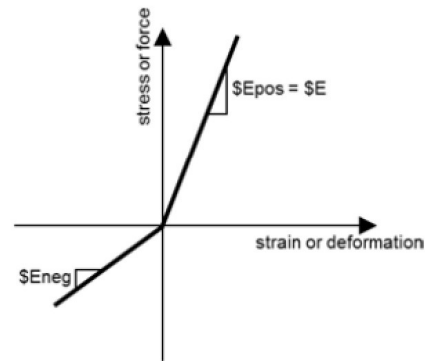
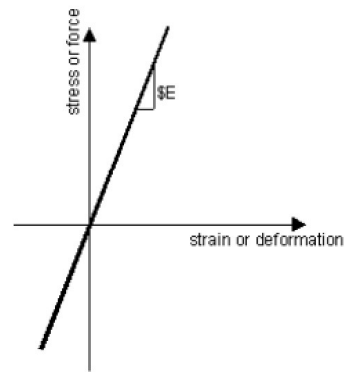
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

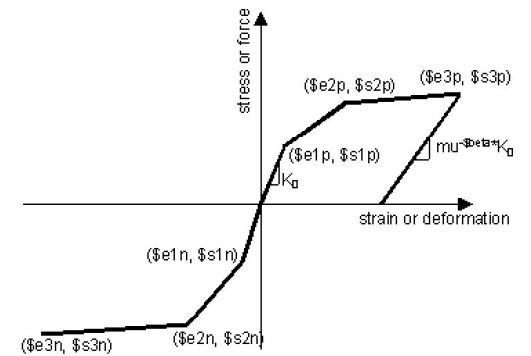
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL

Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

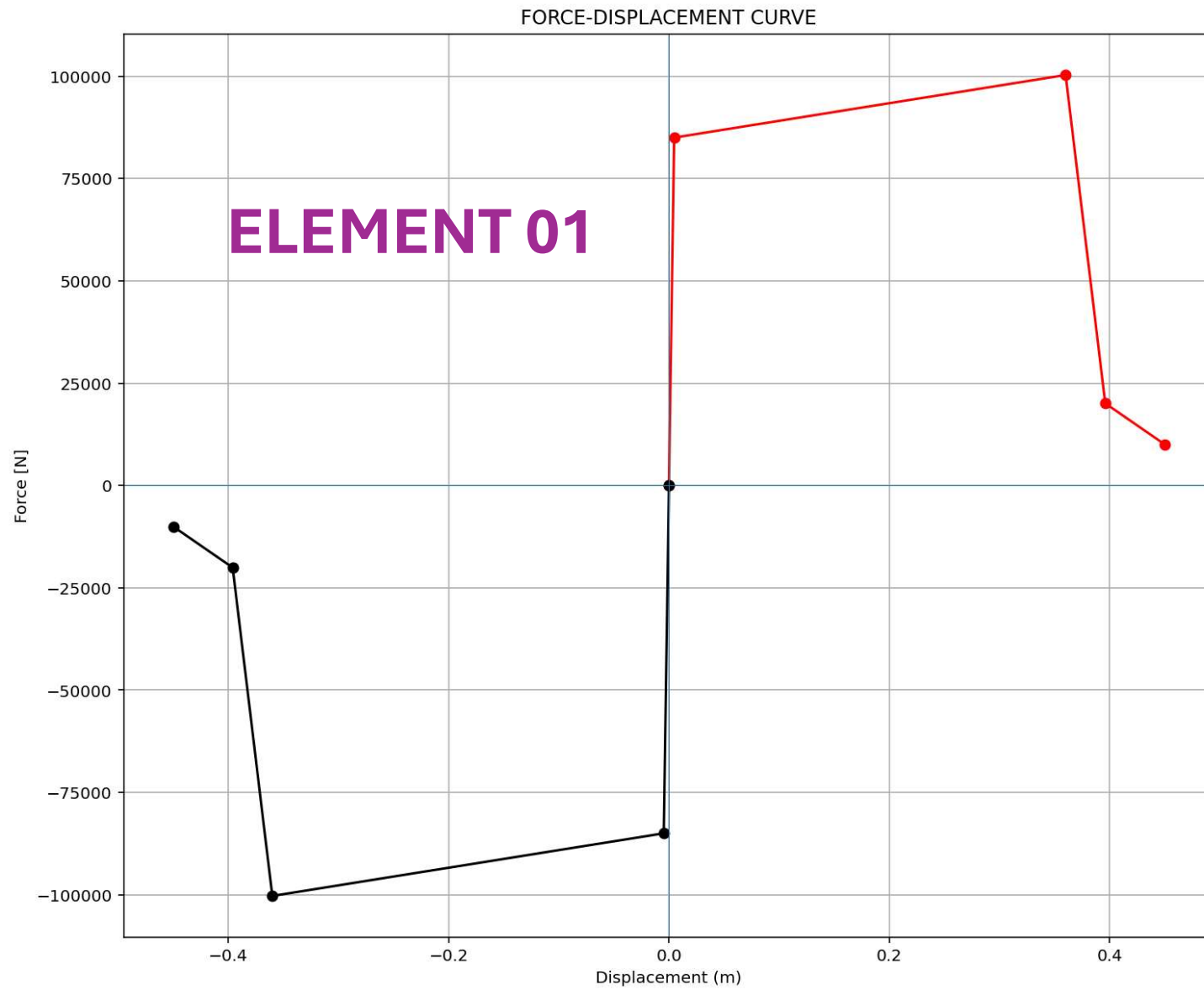
$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

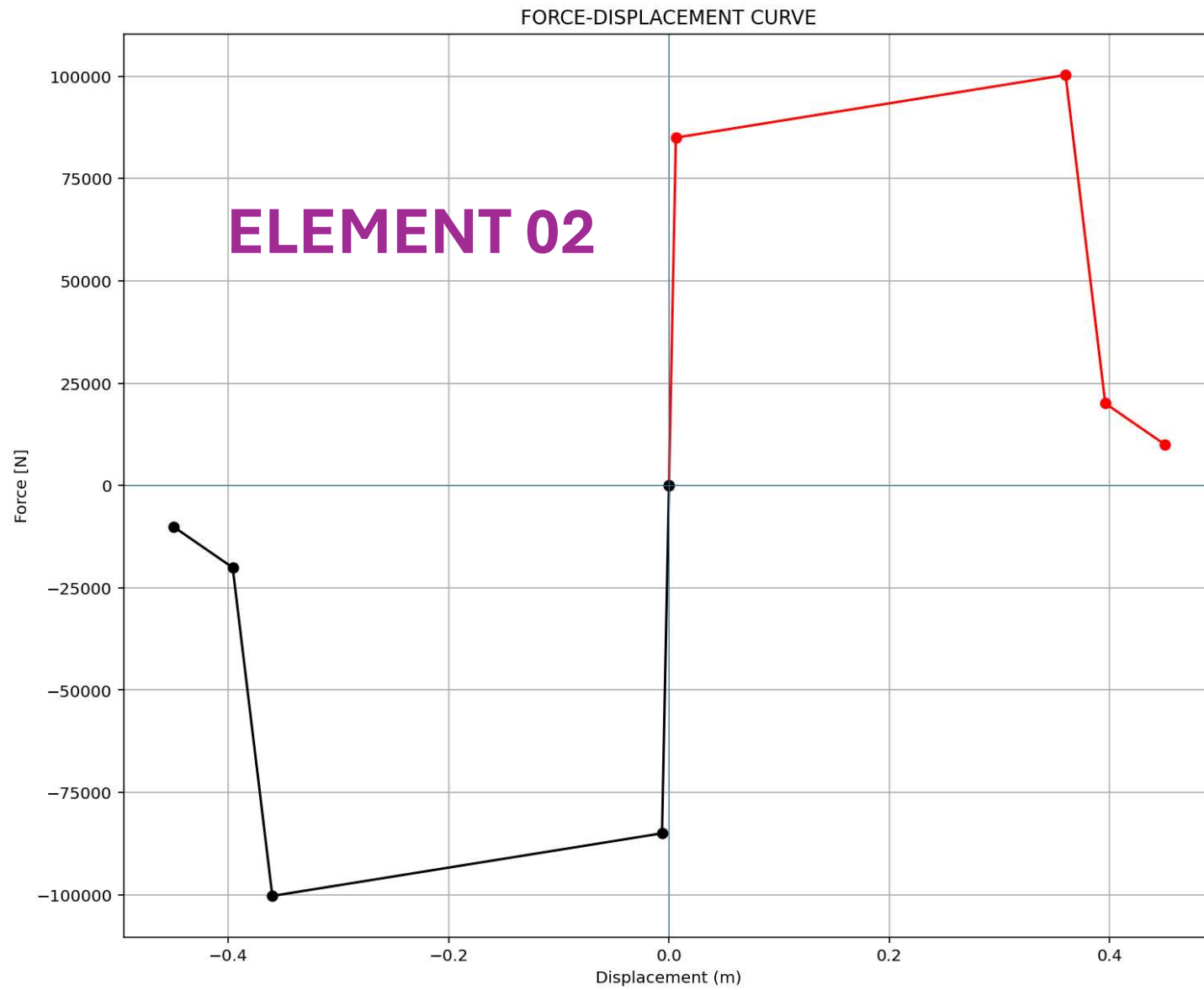
where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{\text{sec}} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

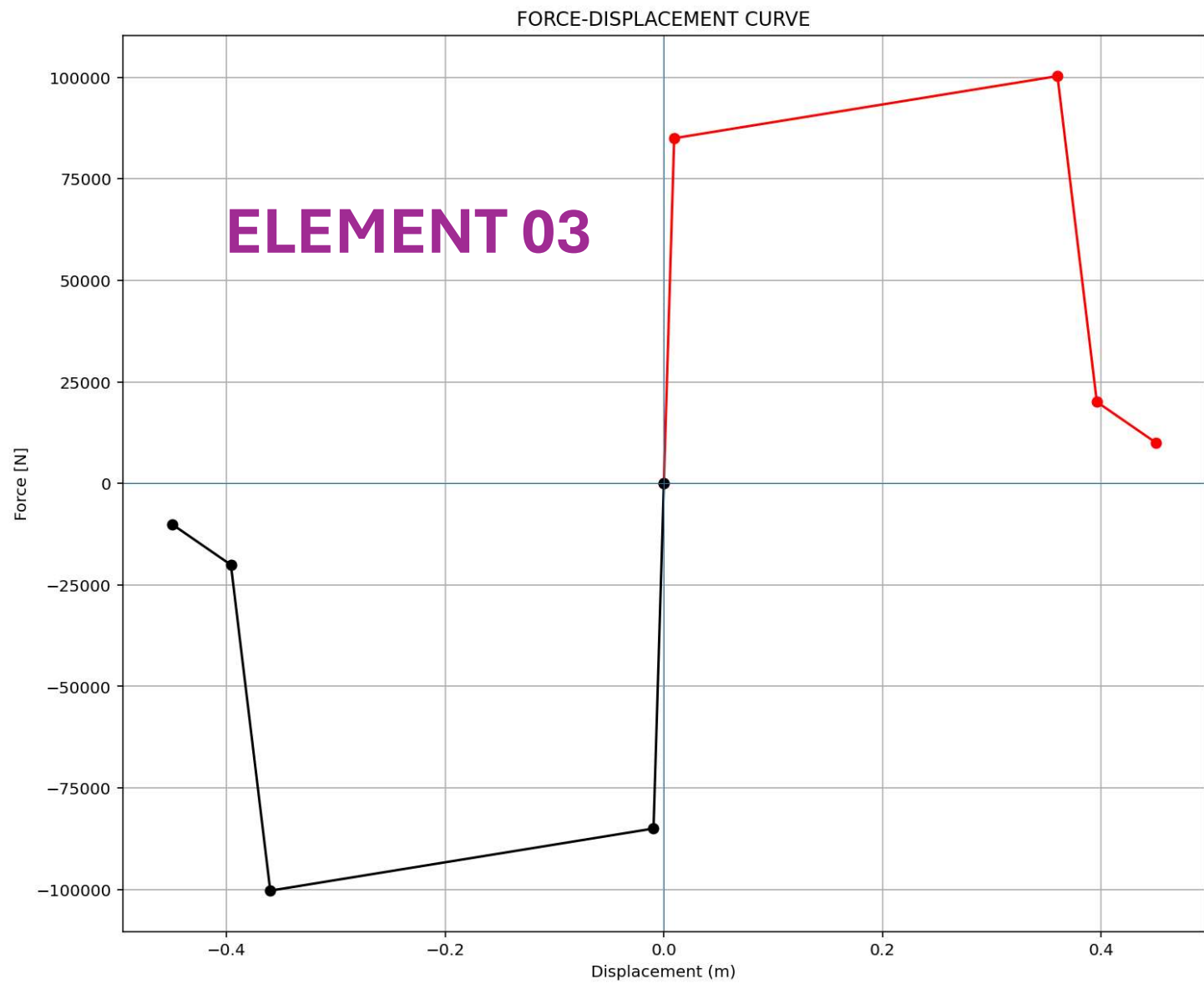
The equivalent viscous damping ratio is then:

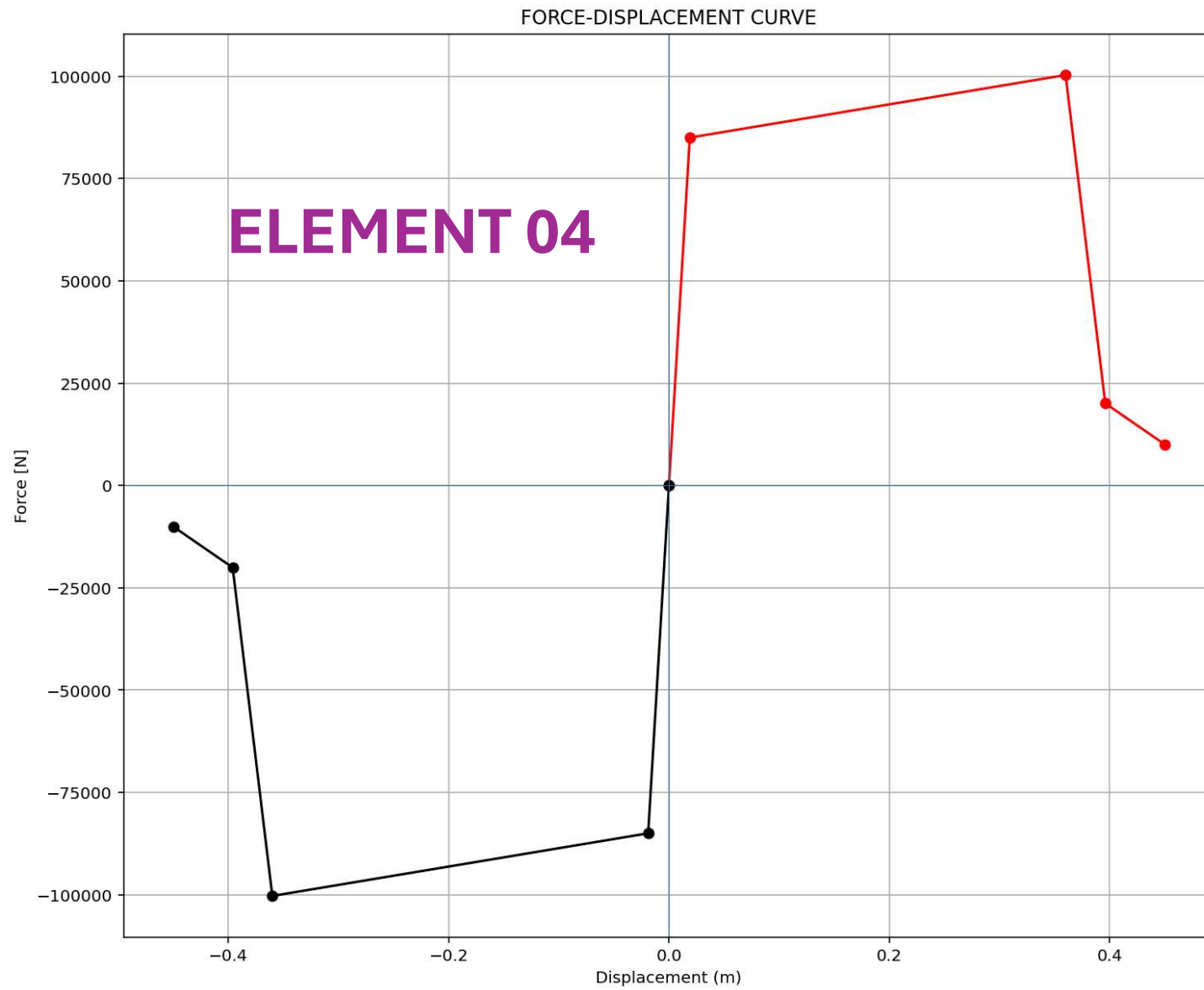
$$\xi_{eq} = \frac{100 E_d}{4\pi E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.









INCREMENTAL DYNAMIC ANALYSIS

Spyder (Python 3.12)

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P(t)_SENSITIVITY_A...UR_ELEMENTS_IDA.py x ANOVA_SENSITIVITY_FUN.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # SENSITIVITY ANALYSIS OF SINGLE-DEGREE-FREOM (SDOF) STRUCTURES USING INCREMENTAL DYNAMI
4 # EFFECTS OF STRUCTURAL MASS, ELASTIC STIFFNESS, STRUCTURAL DUCTILITY RATIO
5 # AND OVER-STRENGTH FACTOR ON OUTPUT KEY PARAMETERS SUCH AS EQUIVALENT VISCOUS DAMPING
6 # FROM NONLINEAR DYNAMIC ANALYSES USING PYTHON AND OPENSEES
7 #
8 #-----
9 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
10 #-----
11 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
12 # EMAIL: salar.d.ghashghaei@gmail.com
13 #####
14 1. This script performs a "sensitivity analysis of a single-degree-of-freedom (SDOF) structur
15 2. The objective is to study the effects of "initial displacement, sructural elastic stiffnes
16 3. Structural properties such as "yield force, ultimate force, elastic stiffness, hardening r
17 4. Elastic and plastic periods of the system are computed and reported.
18 5. Key seismic performance parameters are calculated, including "over-strength ( $\Delta_0$ )", "ductil
19 6. A nonlinear "hysteretic spring model" is used to represent structural behavior with streng
20 7. A "viscous damper" is added to simulate energy dissipation.
21 8. The analysis considers "Incremental Dynamic due to initial displacement" (no external grou
22 9. The function `ANALYSIS_SDOF` builds the OpenSees model, applies initial conditions, and ru
23 10. "Newmark time integration" and "Newton-Raphson iteration" are used for nonlinear solution
24 11. "Rayleigh damping coefficients" are computed from the first eigenvalue.
25 12. Time histories of "displacement, velocity, acceleration, base reaction, stiffness, and da
26 13. A "ductility-based damage index" is calculated at each time step.
27 14. The effective stiffness degradation is monitored during the response.
28 15. Multiple simulations are executed by looping over ranges of "damping ratio, initial displ
29 16. For each simulation, "maximum response values" are extracted.
30 17. Damage index values are limited between "0%" and "100%".
31 18. All results are stored for post-processing and statistical analysis.
32 19. The script reports completion of each simulation and total runtime.
33 20. Overall, the code provides a "parametric nonlinear dynamic assessment" of SDOF systems un
34 ""
```

2D CONTOUR: DUCTILITY RATIO [m/m] - Topographic Map

[Python Console] Plots Variable Explorer Help Debugger History Files

Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1098, Col 1 UTF-8 CRLF RW Mem 42%

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...ENTS_IDA+ \EQUIVALENT_VISCOUS_DAMPING_RATIO_FUN.py

```
1  """ EQUIVALENT VISCOUS DAMPING RATIO
2  """
3  [1] The function EQUIVALENT_VISCOUS_DAMPING_RATIO_FUN computes the dissipated energy (E_d) usi
4  [2] Both methods produce the same styled plot: the hysteresis loop in black, the enclosed area
5  [3] The bottom section generates a synthetic elliptical hysteresis loop (sinusoidal displaceme
6  THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
7  """
10 def EQUIVALENT_VISCOUS_DAMPING_RATIO_FUN(displacement, base_shear, method, XLABEL, YLABEL, TIT
11     import numpy as np
12     from scipy.spatial import ConvexHull
13     import matplotlib.pyplot as plt
14     if method == 1:
15
16         """
17         Compute the equivalent viscous damping ratio from a hysteresis loop.
18
19         The dissipated energy per cycle (E_d) is the area enclosed by the loop,
20         computed using the convex hull (Shoelace formula on the hull). The maximum
21         strain energy (E_s) is estimated as 0.5 * F_max * u_max, where u_max is the
22         maximum absolute displacement and F_max is the absolute base shear at that
23         same displacement.
24
25         Parameters
26         -----
27         displacement : array-like
28             Displacement values (assumed to form a closed hysteresis loop).
29         base_shear : array-like
30             Corresponding base shear values.
31         XLABEL : str
32             Label for the x-axis.
33         YLABEL : str
```

Equivalent viscous damping ratio - Seismic Hysteresis

Base Reaction [N]

Displacement [m] $1e-5$

Legend:

- Hysteresis Loop
- (u_{max}, F_{max})
- $E_d = 0.007 \text{ N-m}$

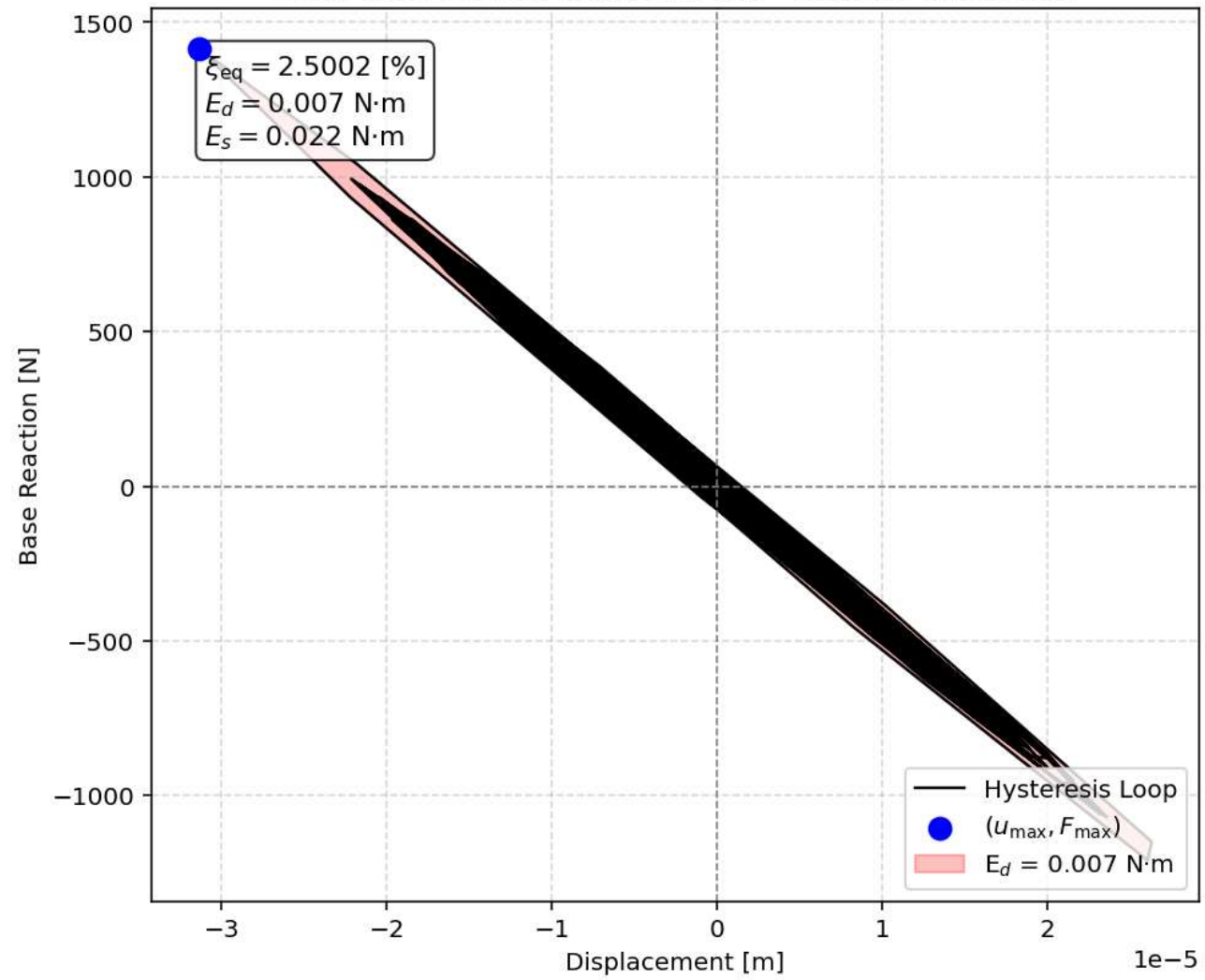
Annotations:

- $\xi_{eq} = 2.5002 \text{ [%]}$
- $E_d = 0.007 \text{ N-m}$
- $E_s = 0.022 \text{ N-m}$

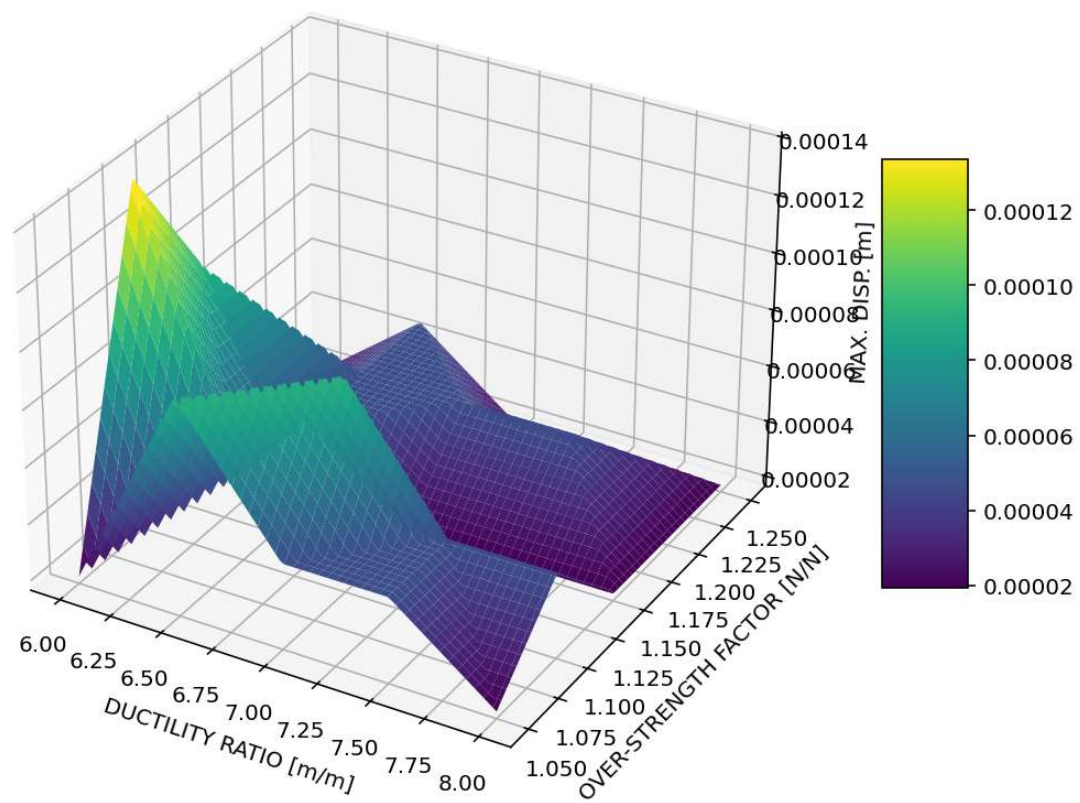
Python Console Plots Variable Explorer Help Debugger History Files

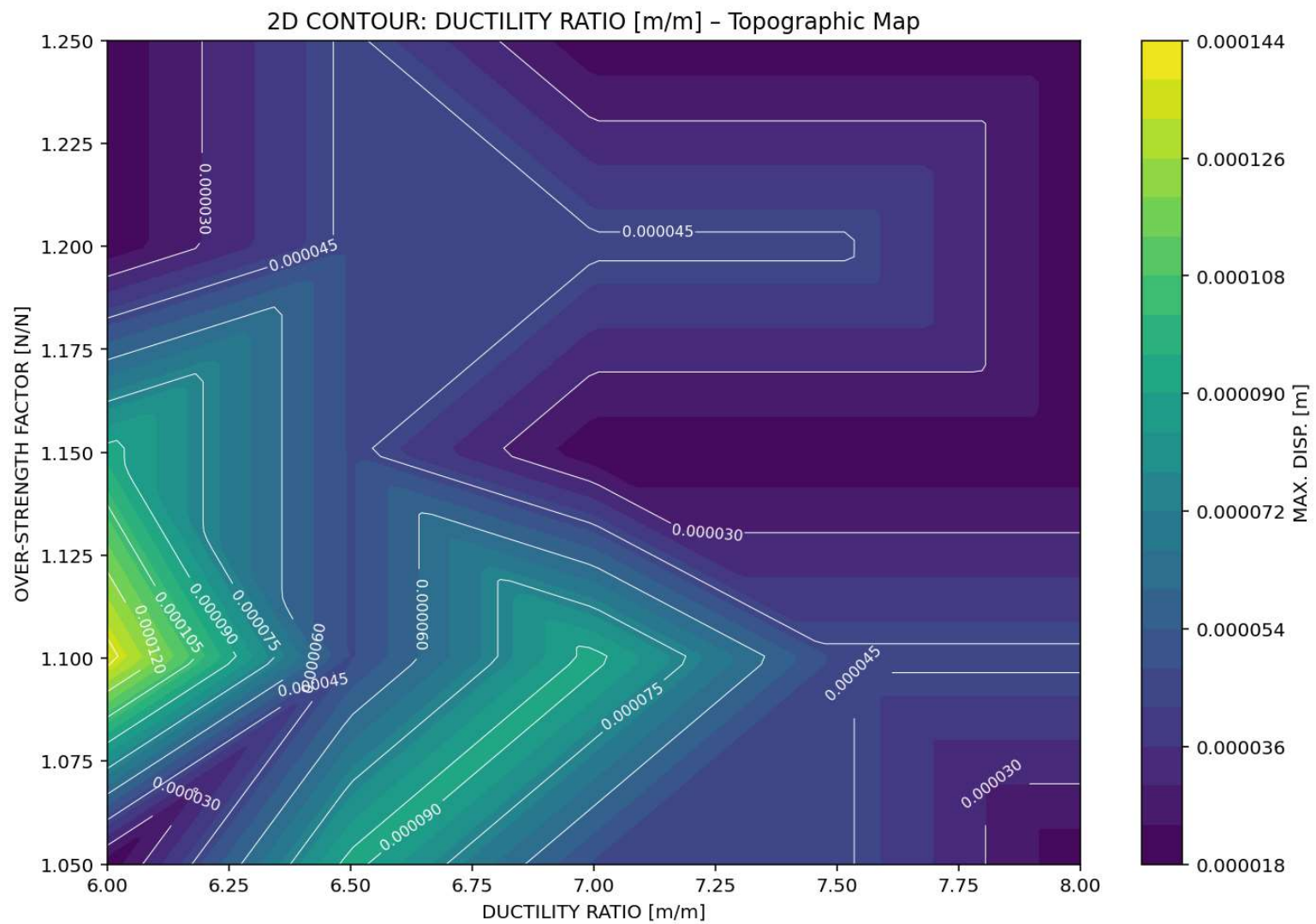
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Equivalent viscous damping ratio - Seismic Hysteresis

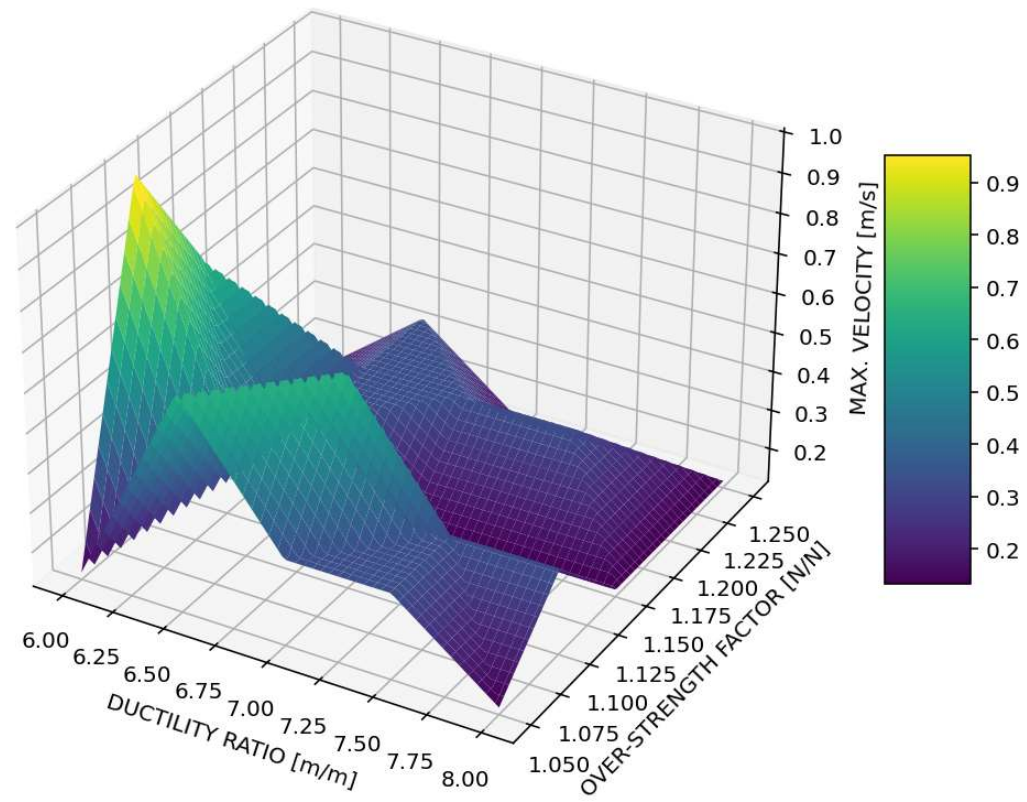


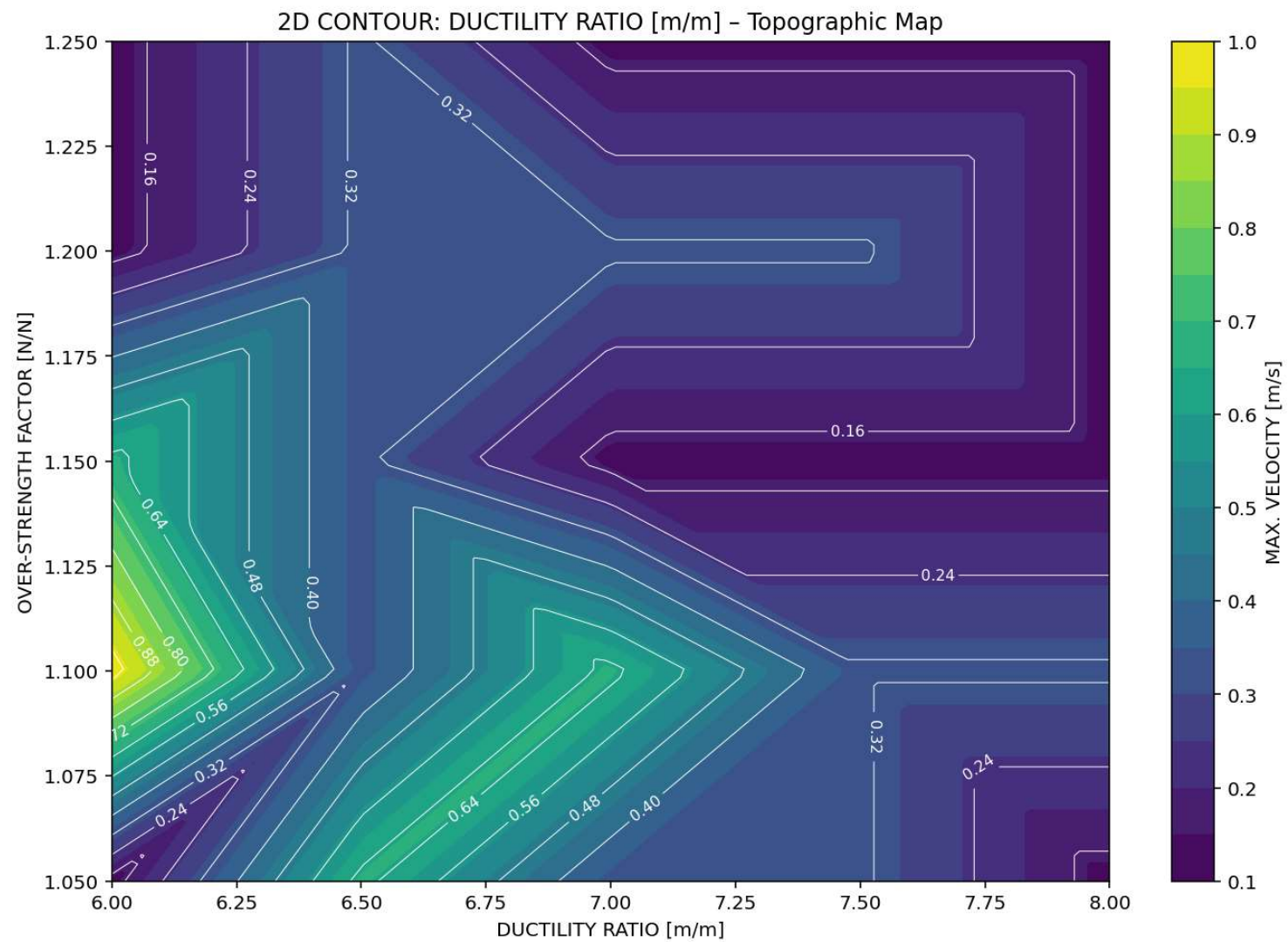
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. DISP. [m]



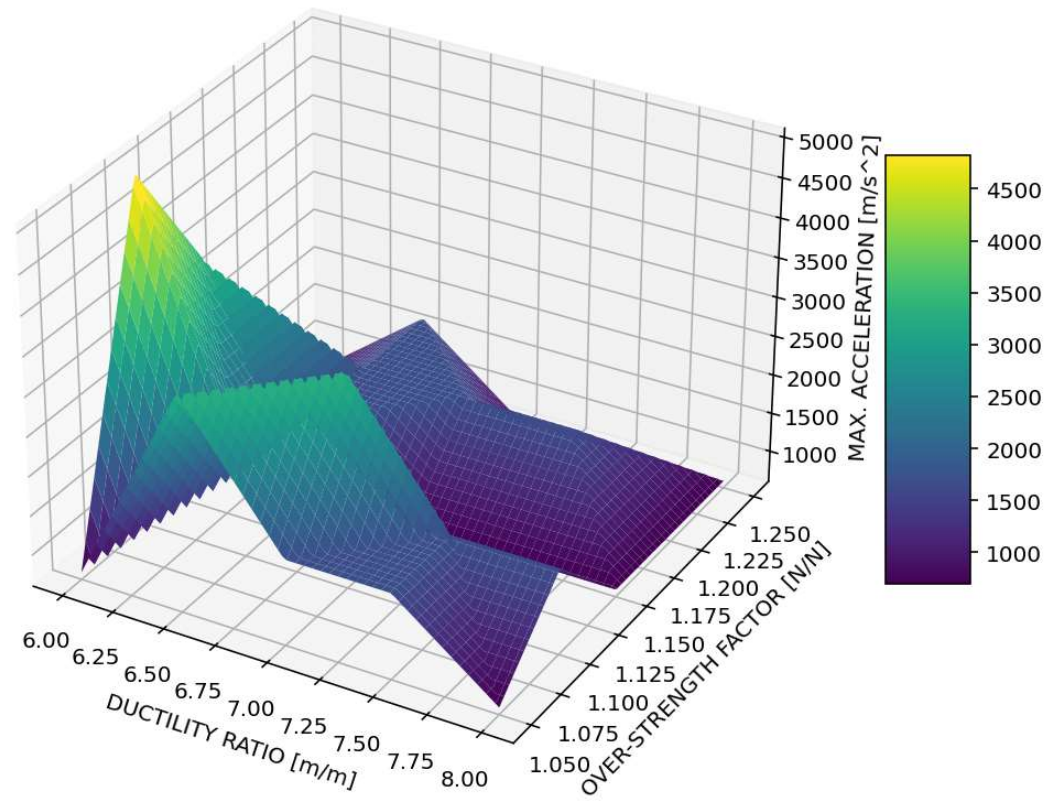


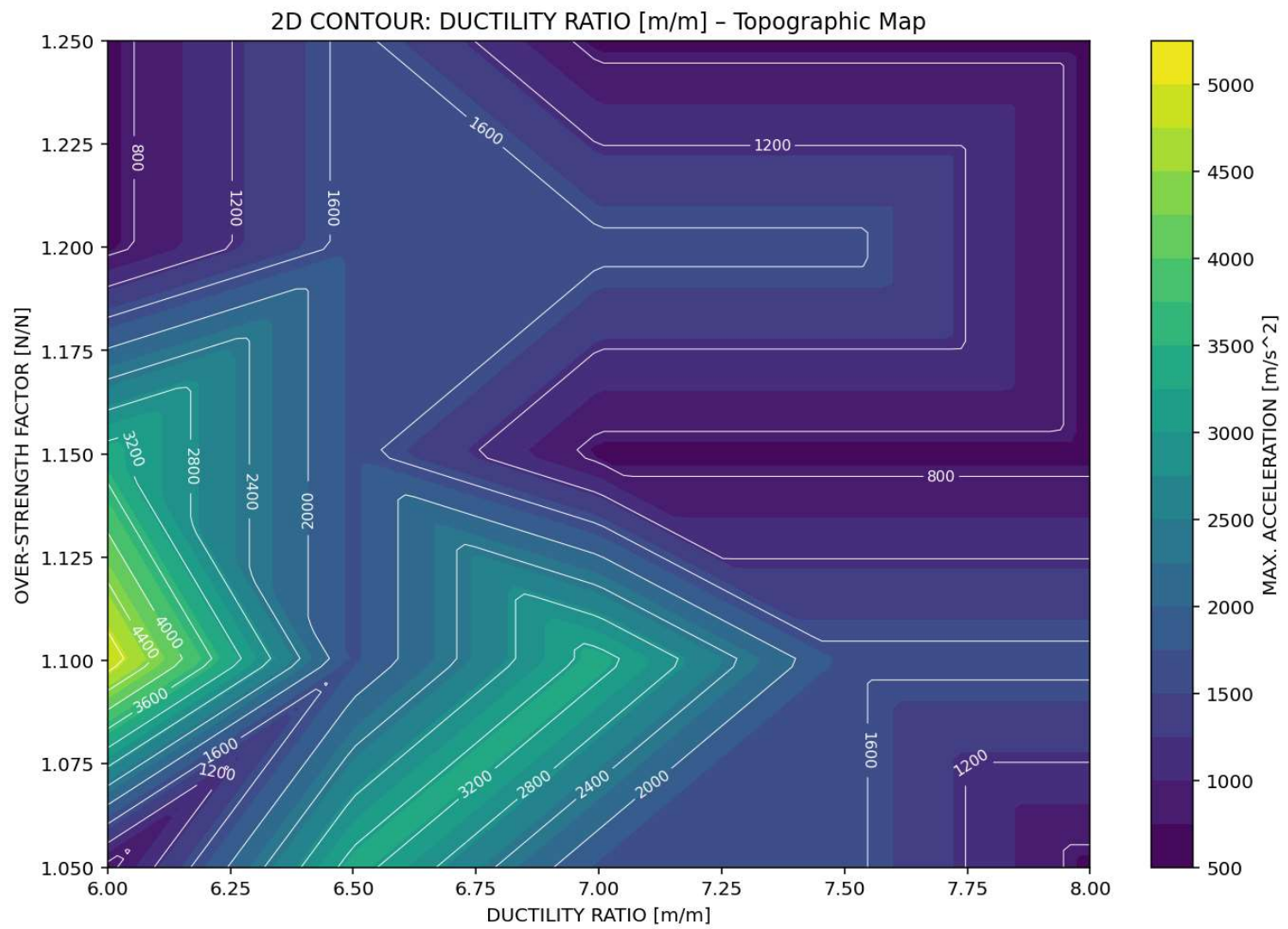
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. VELOCITY [m/s]



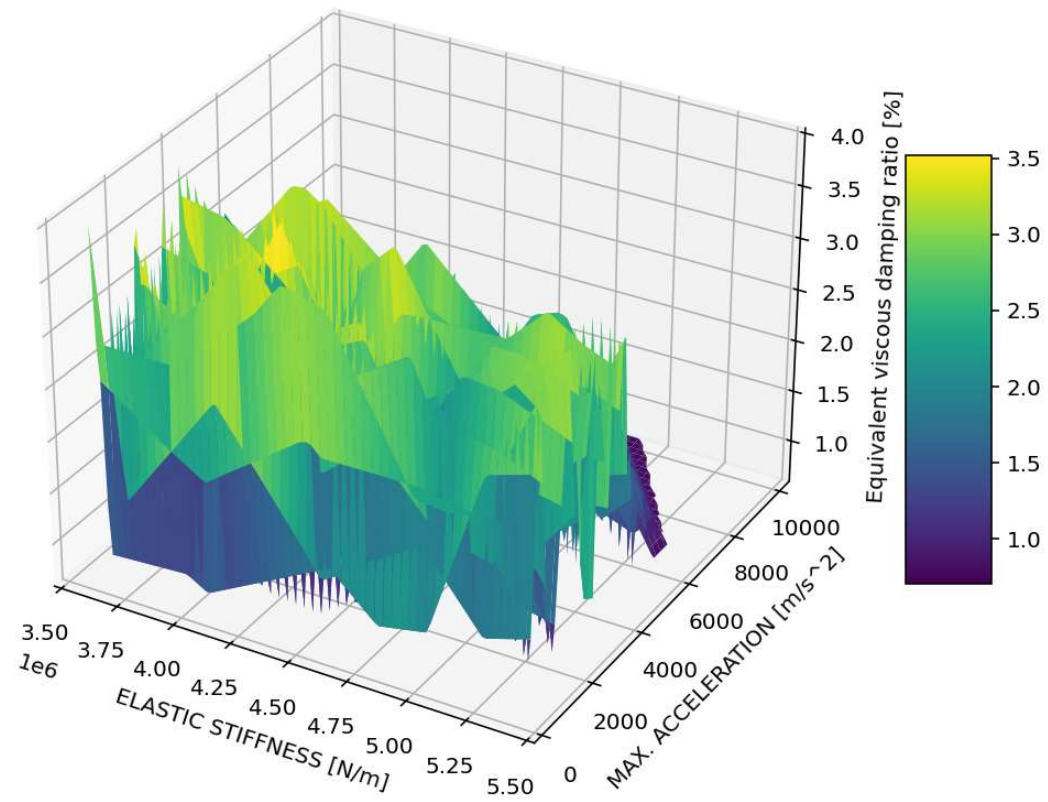


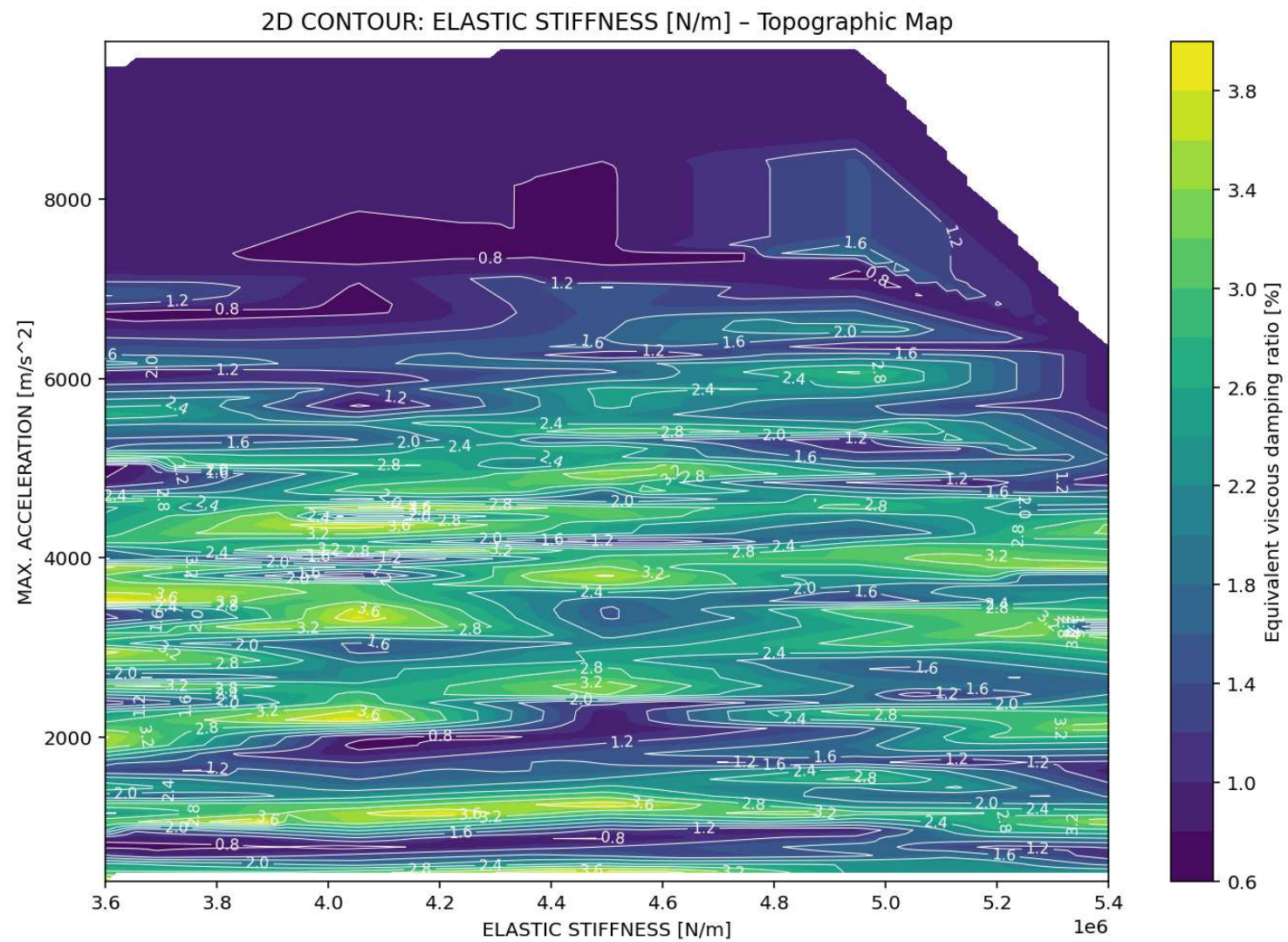
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. ACCELERATION [m/s²]



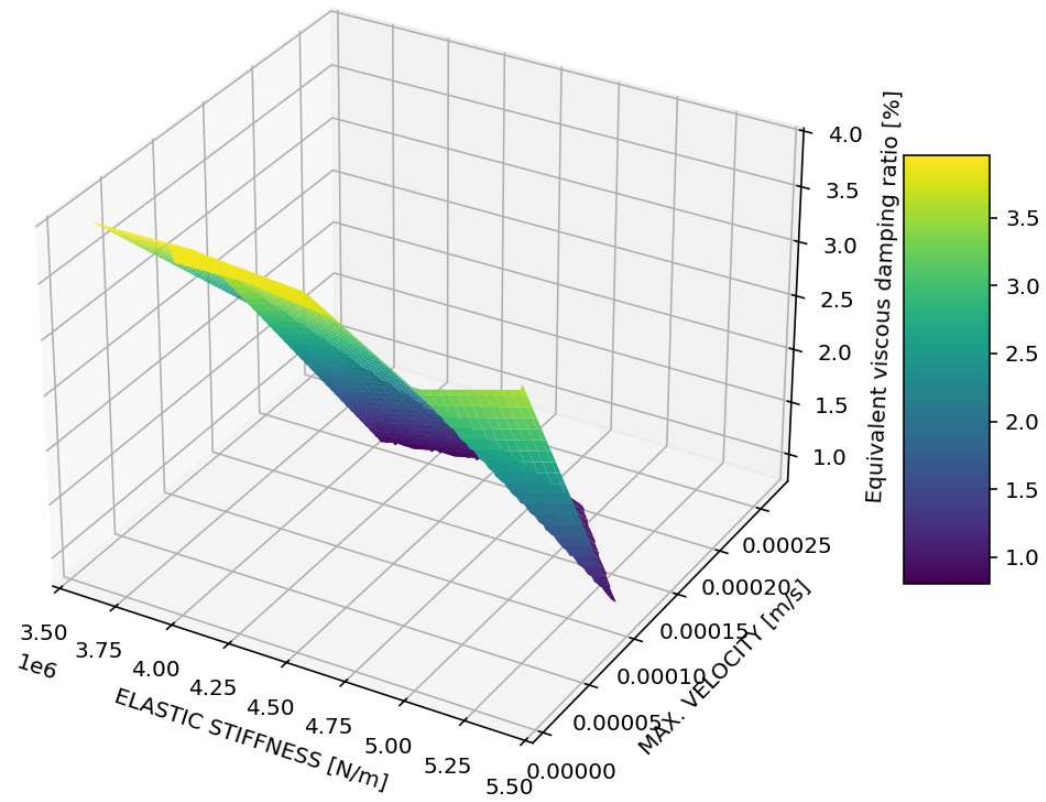


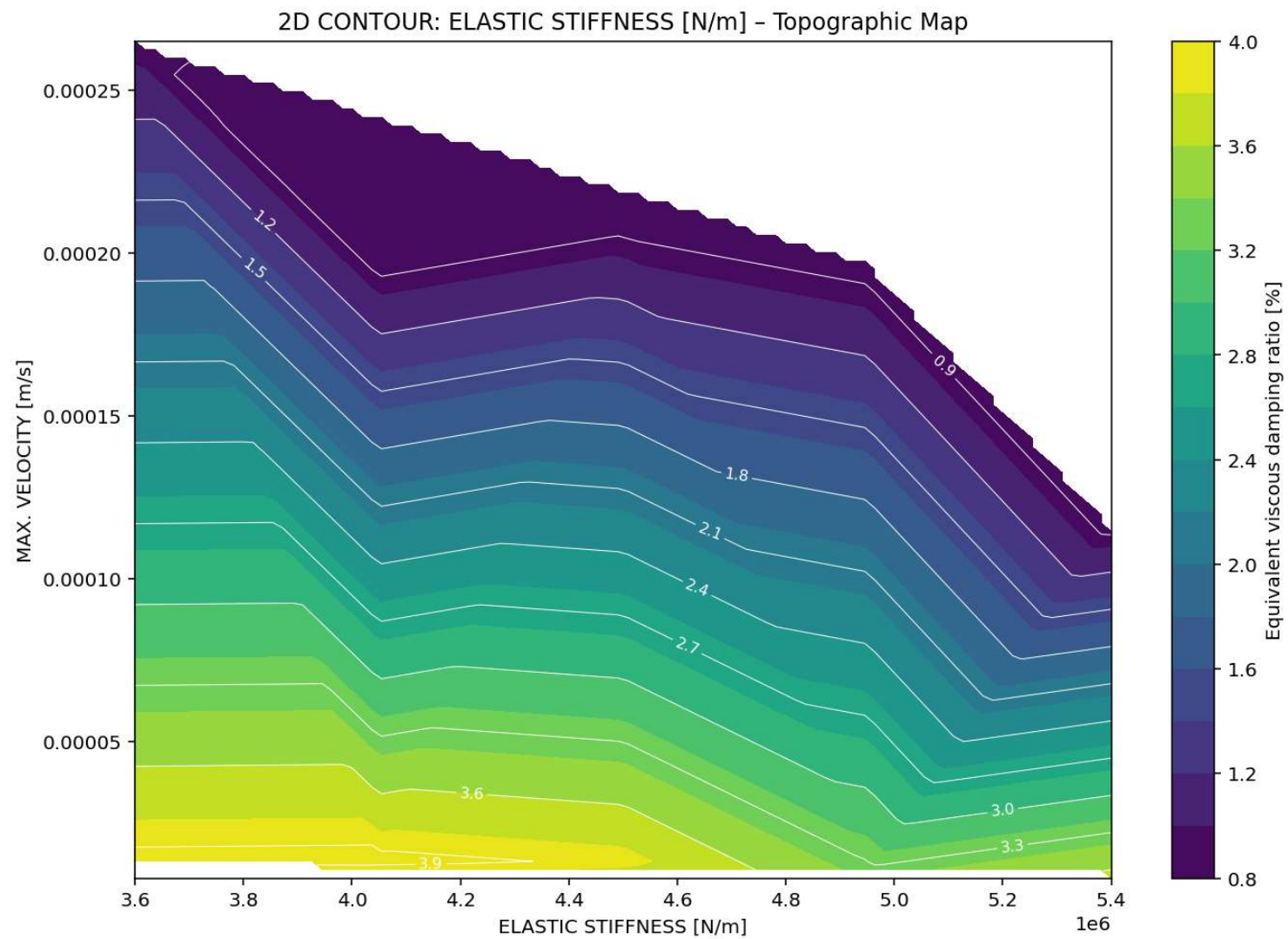
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. ACCELERATION [m/s²] - Equivalent viscous damping ratio [%]



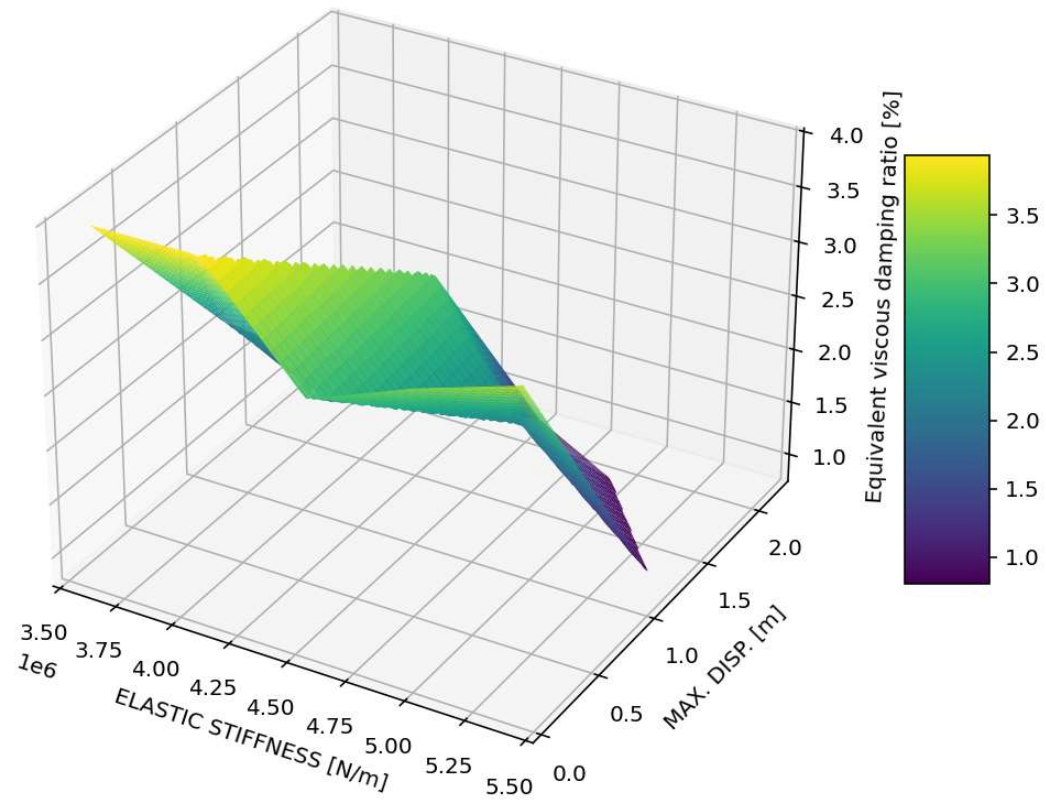


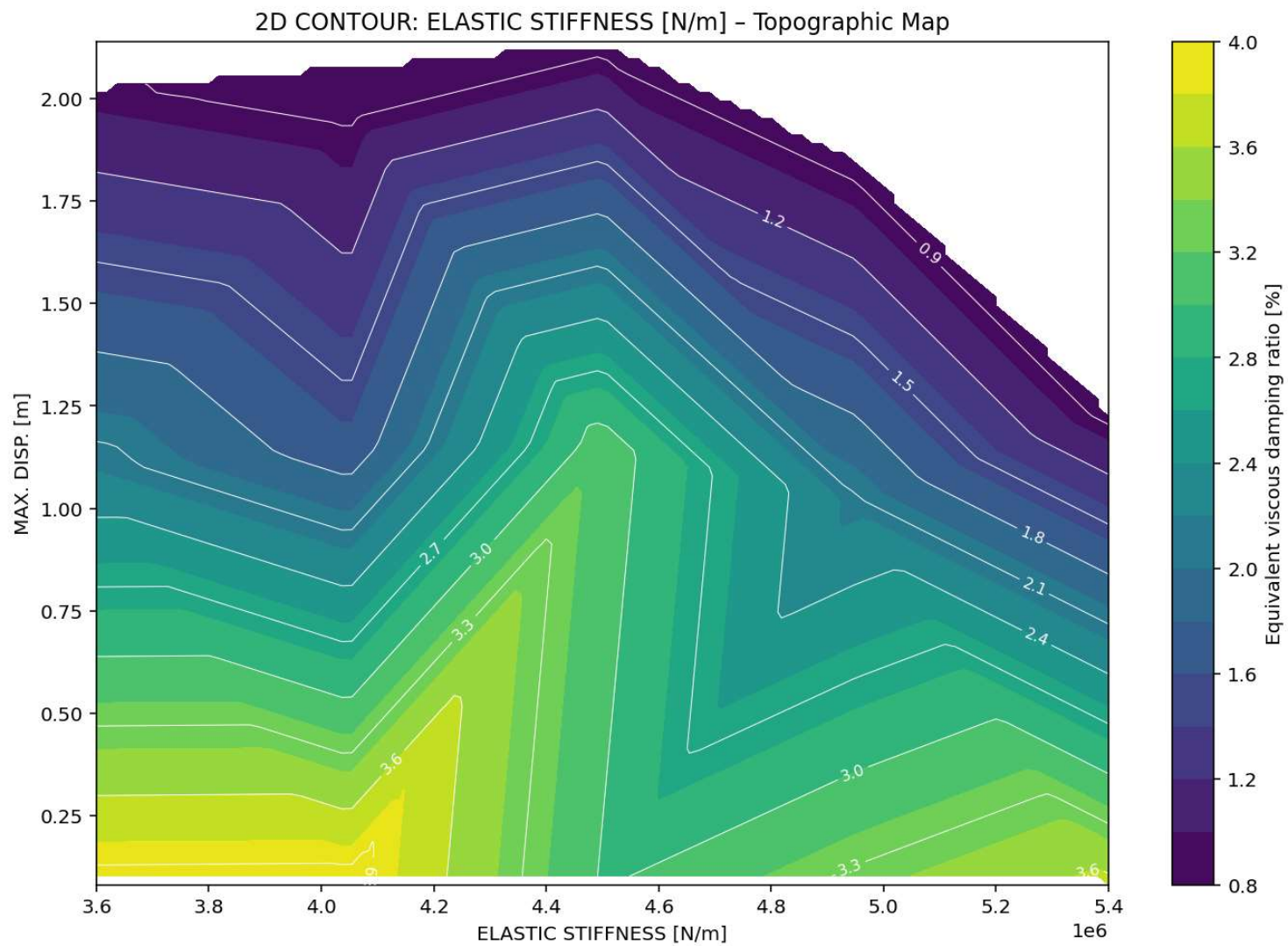
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. VELOCITY [m/s] - Equivalent viscous damping ratio [%]



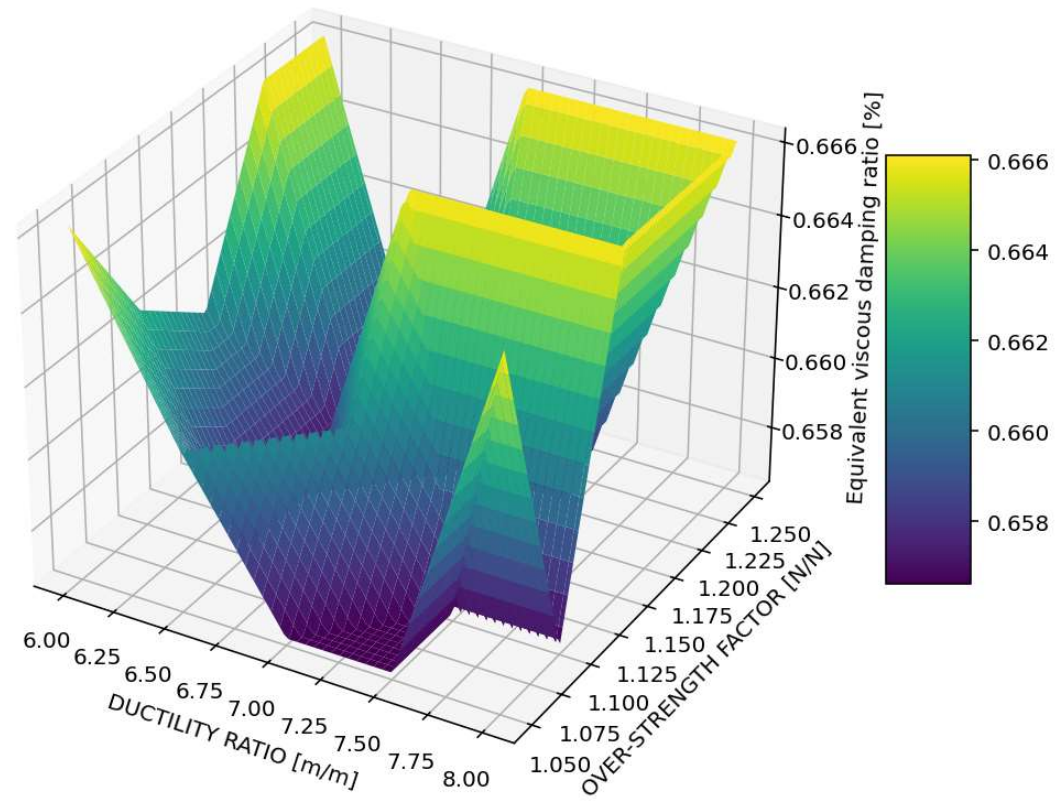


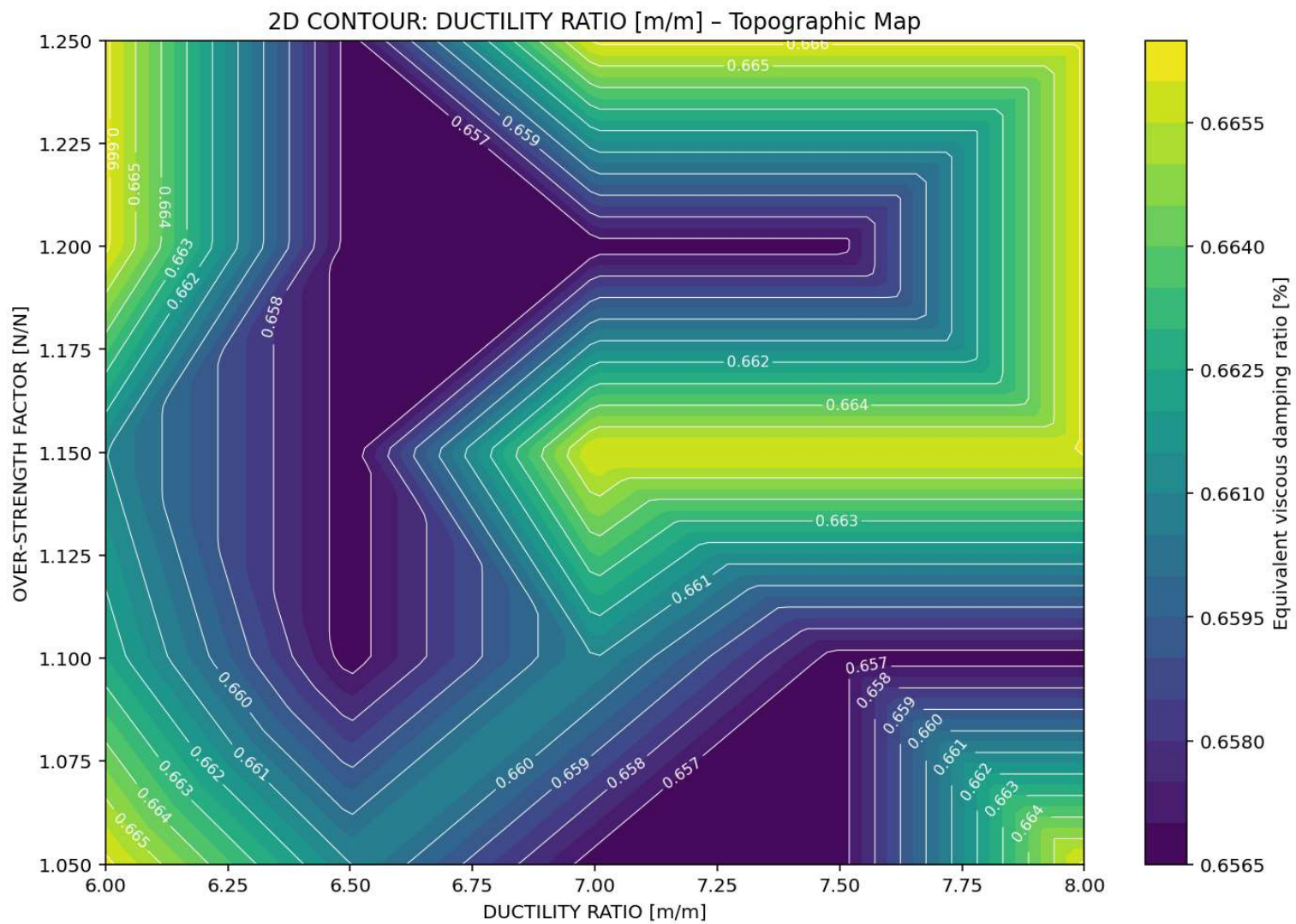
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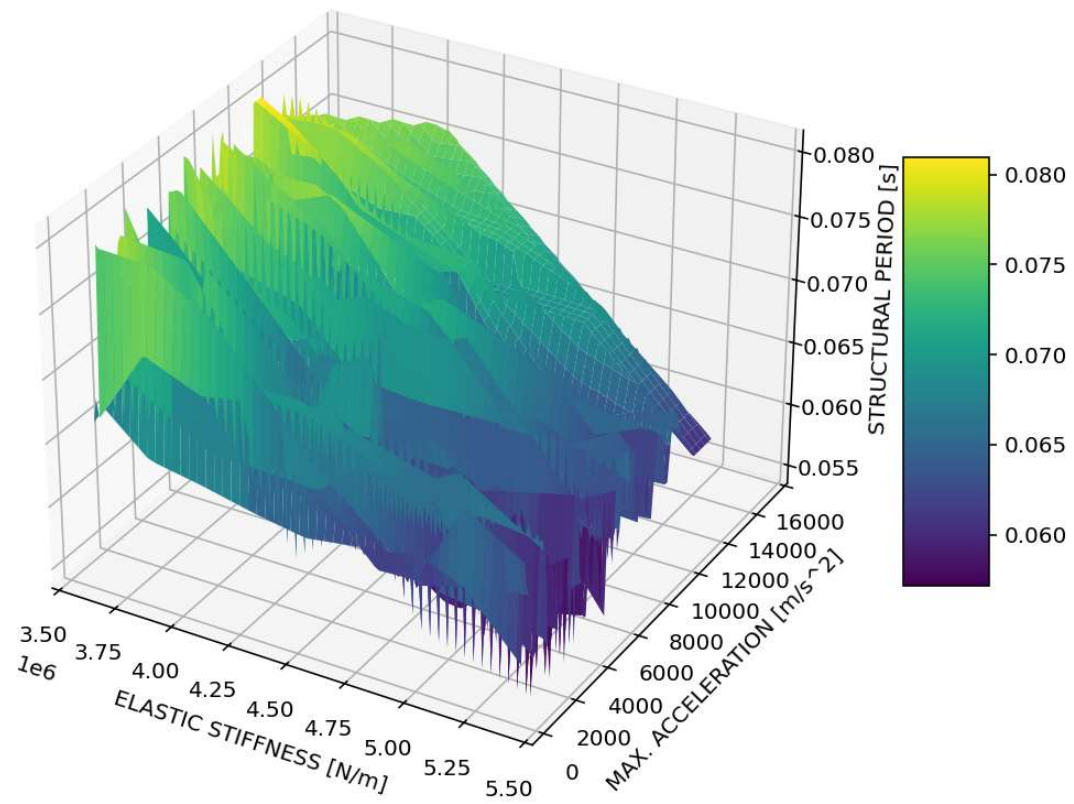


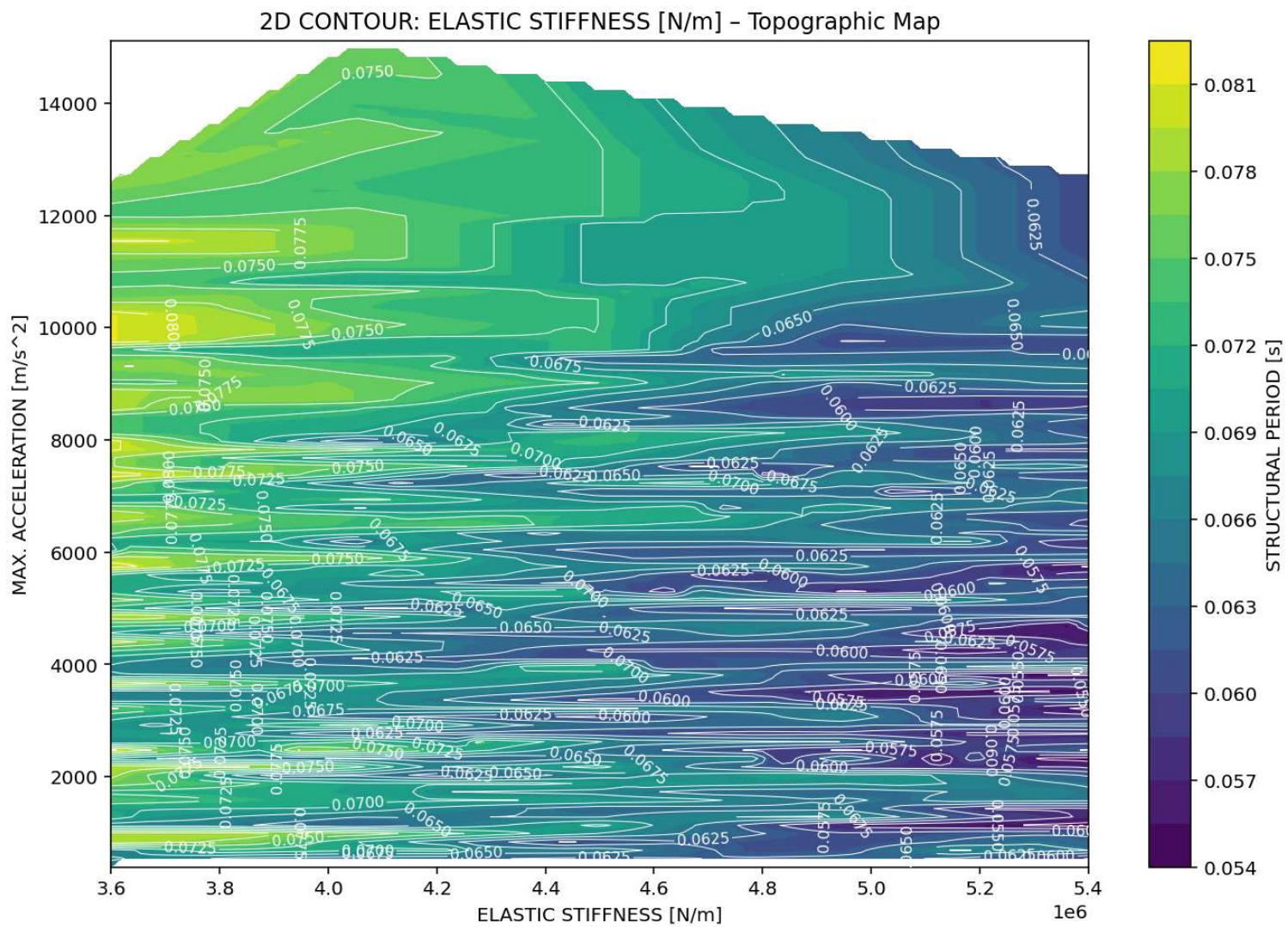
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - Equivalent viscous damping ratio [%]



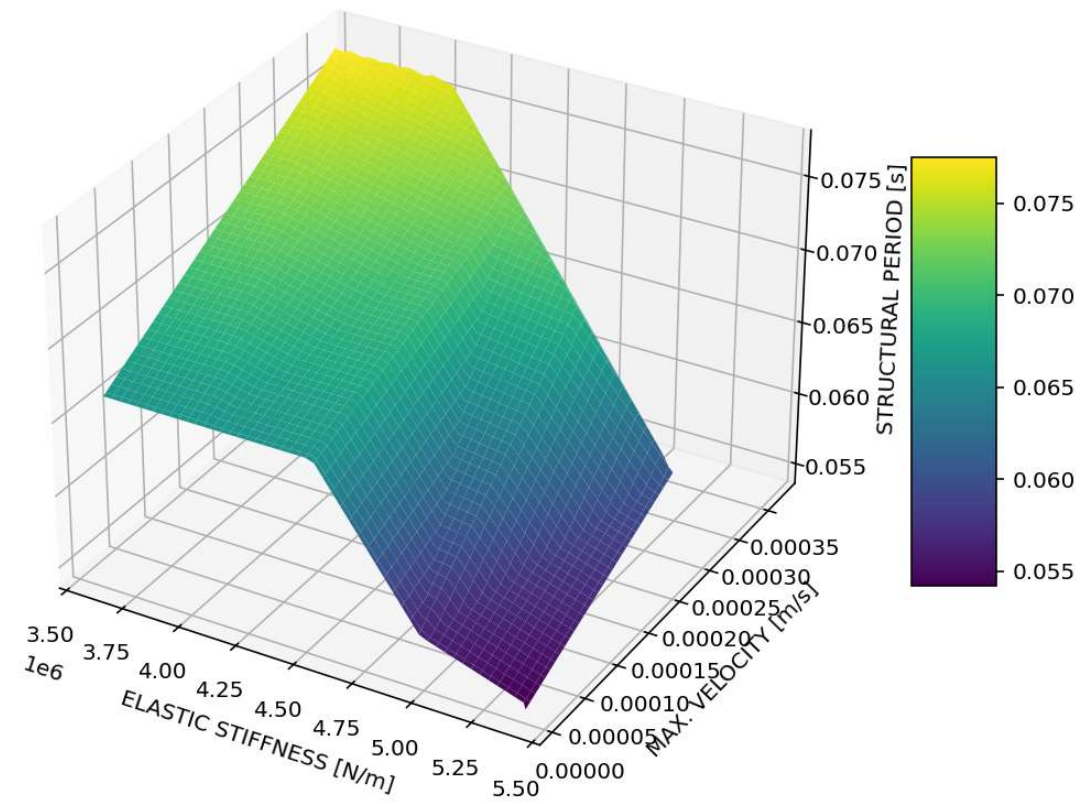


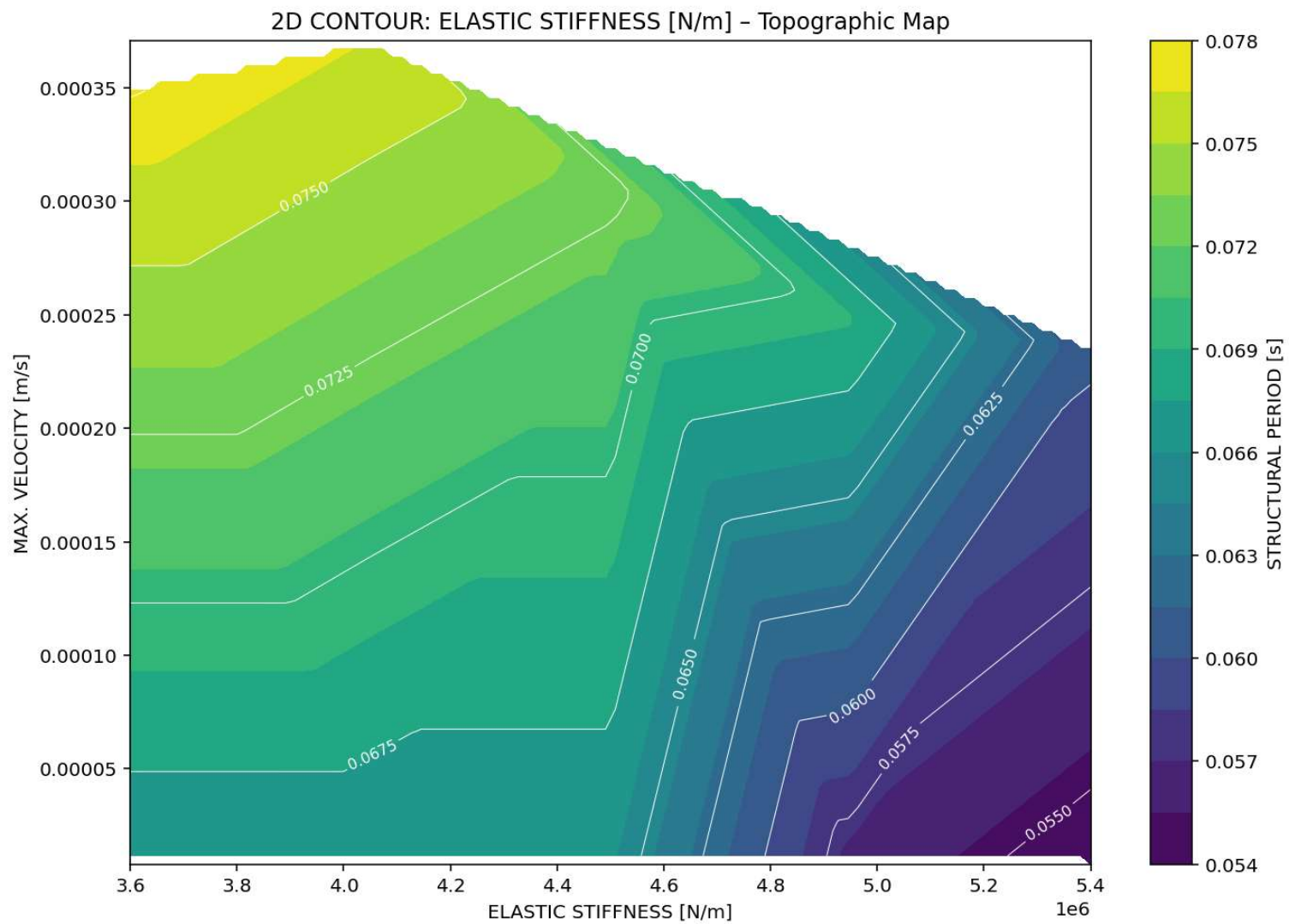
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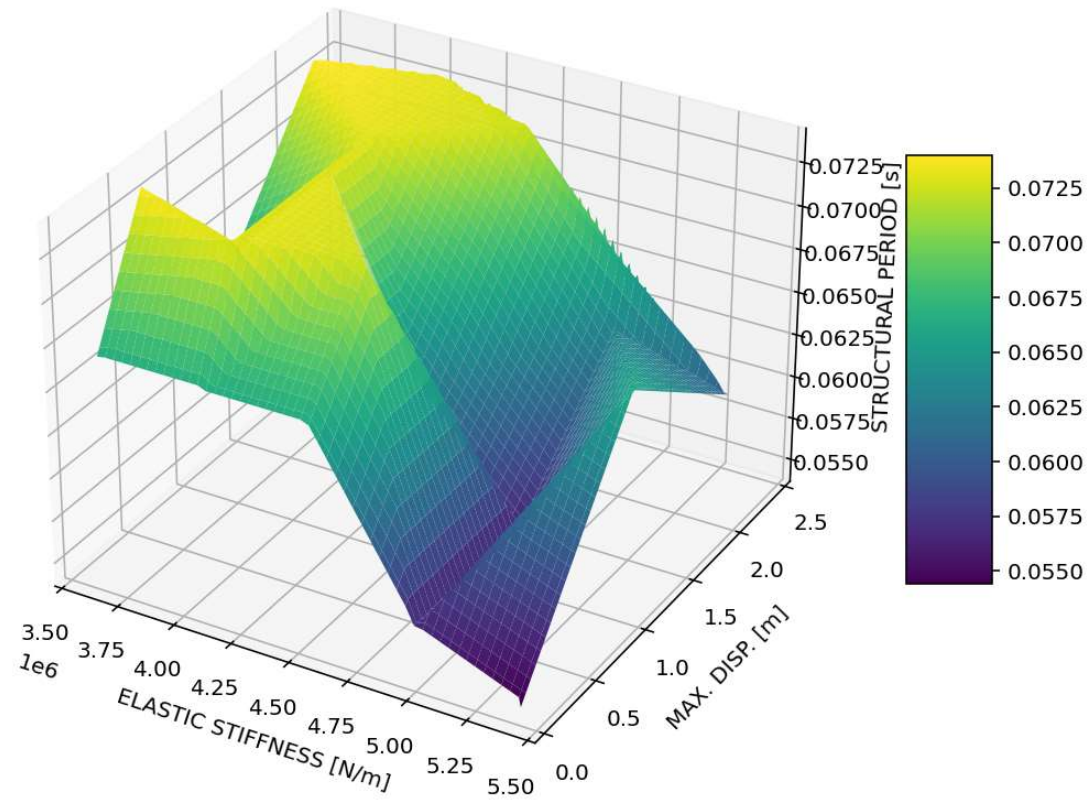


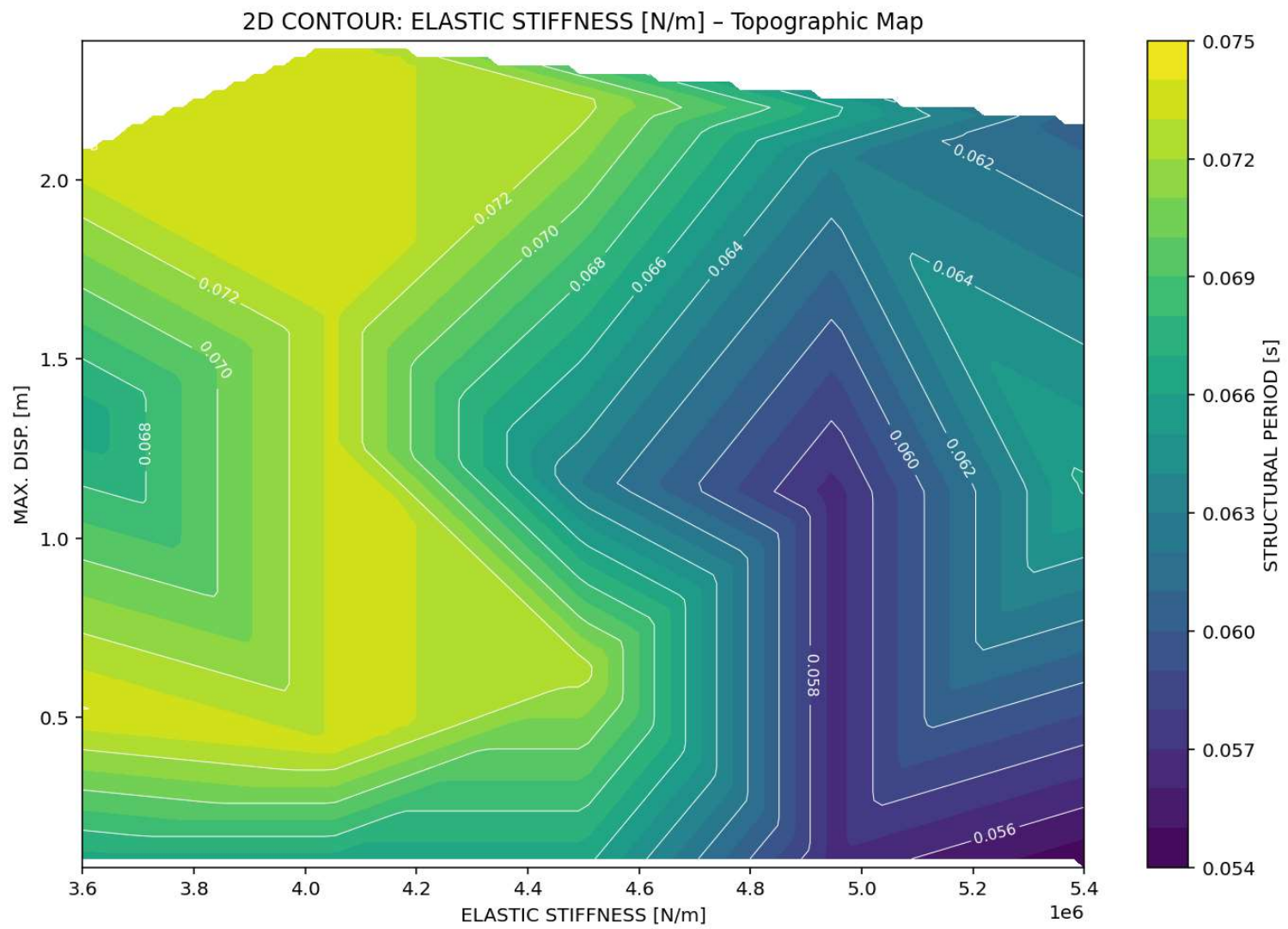
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. VELOCITY [m/s] - STRUCTURAL PERIOD [s]



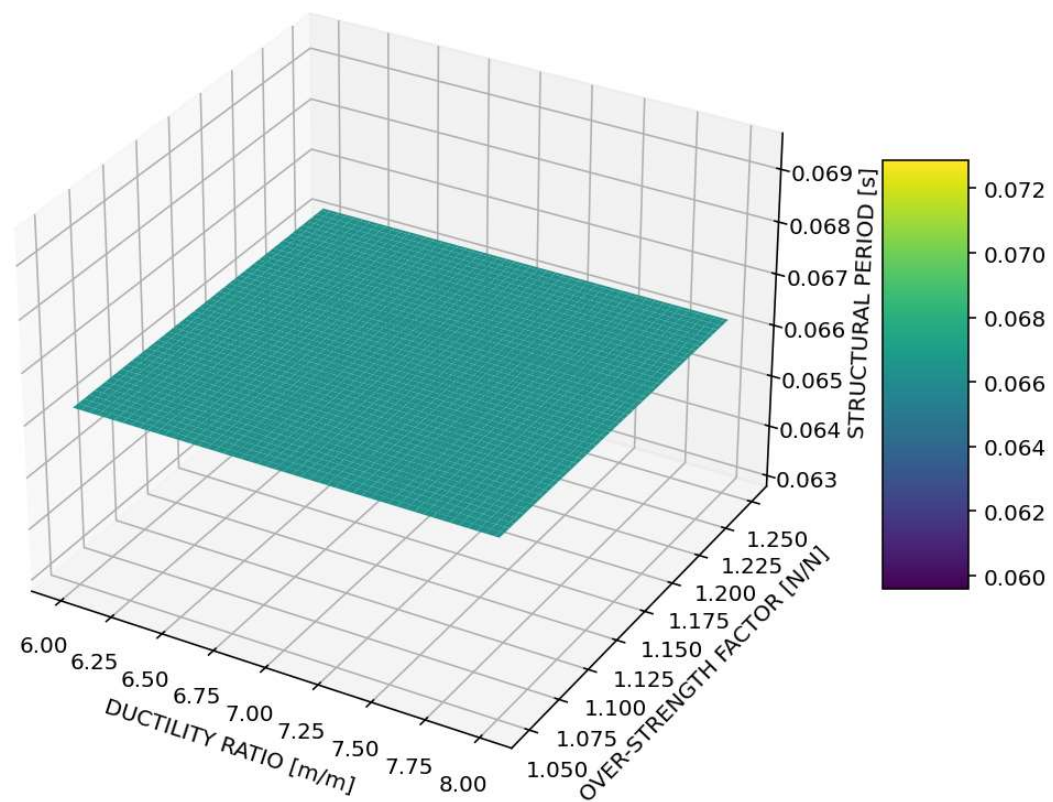


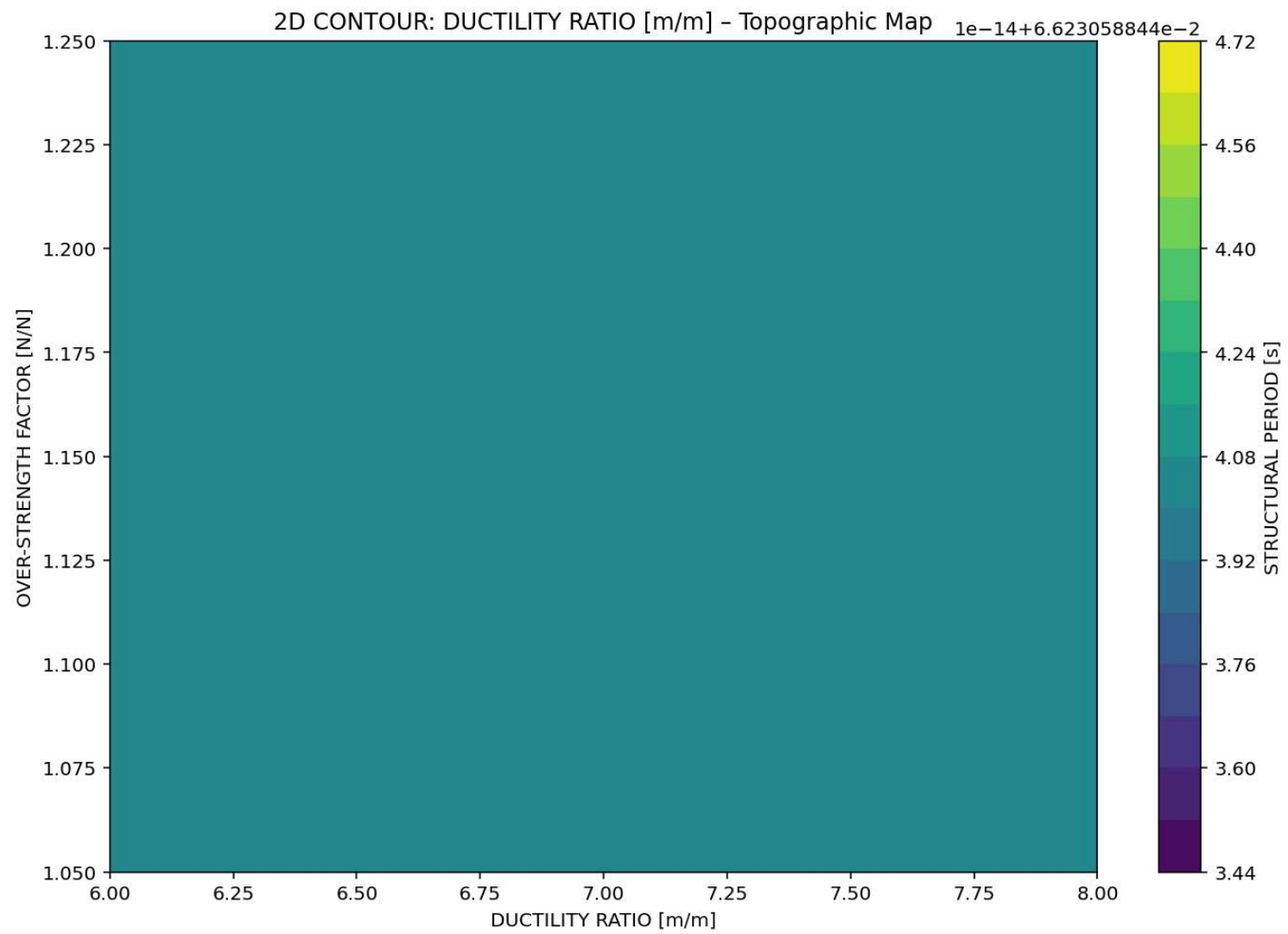
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. DISP. [m] - STRUCTURAL PERIOD [s]



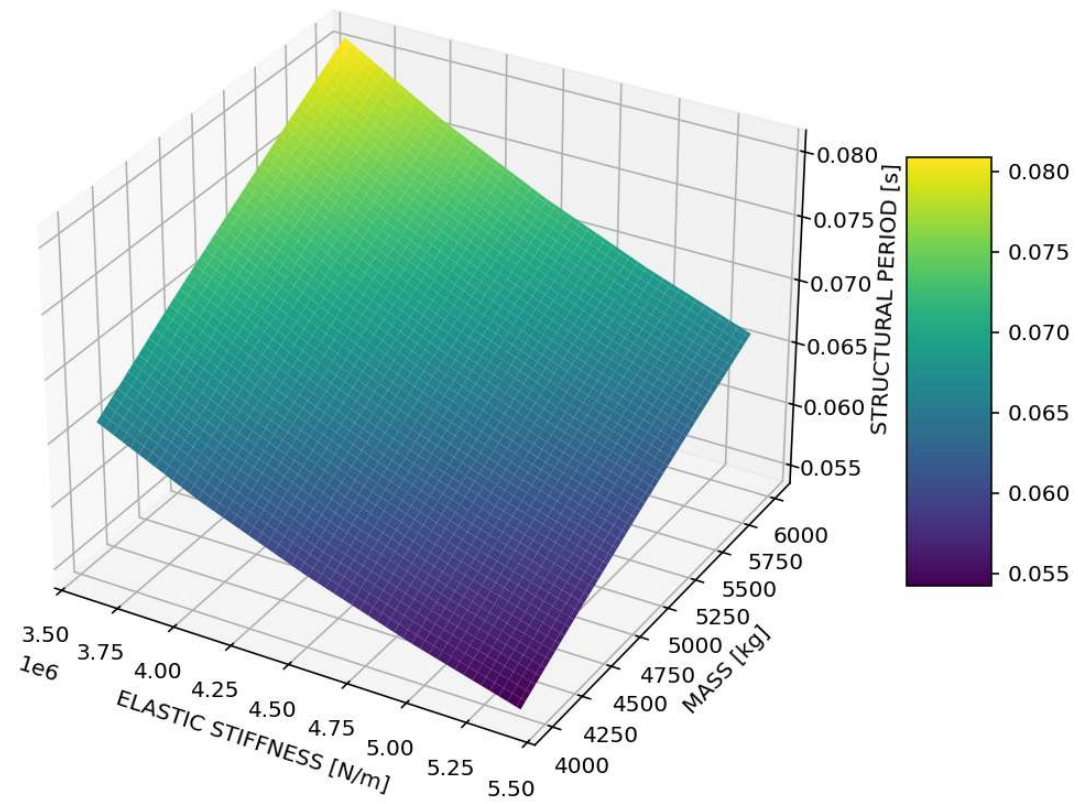


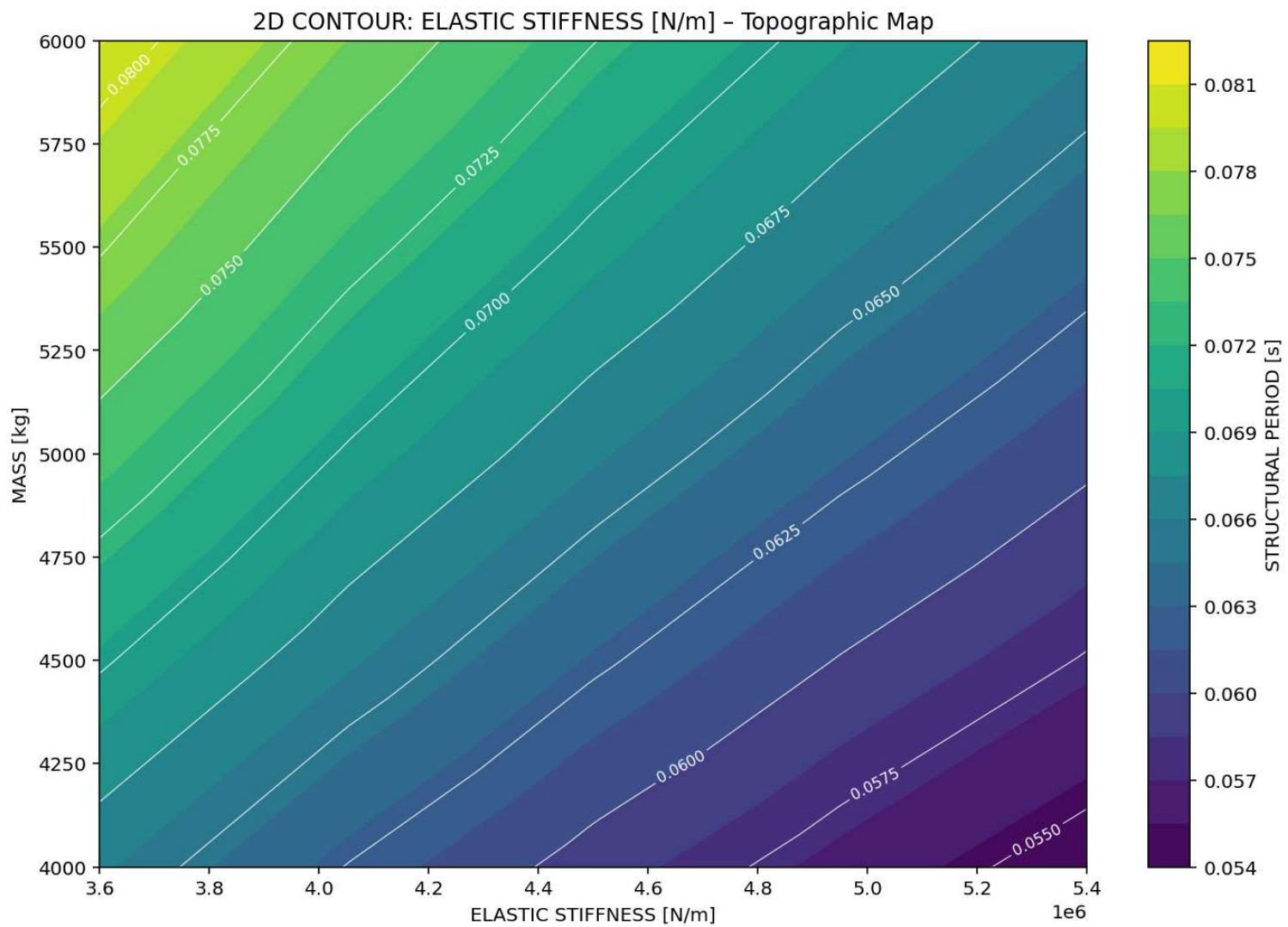
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - STRUCTURAL PERIOD [s]



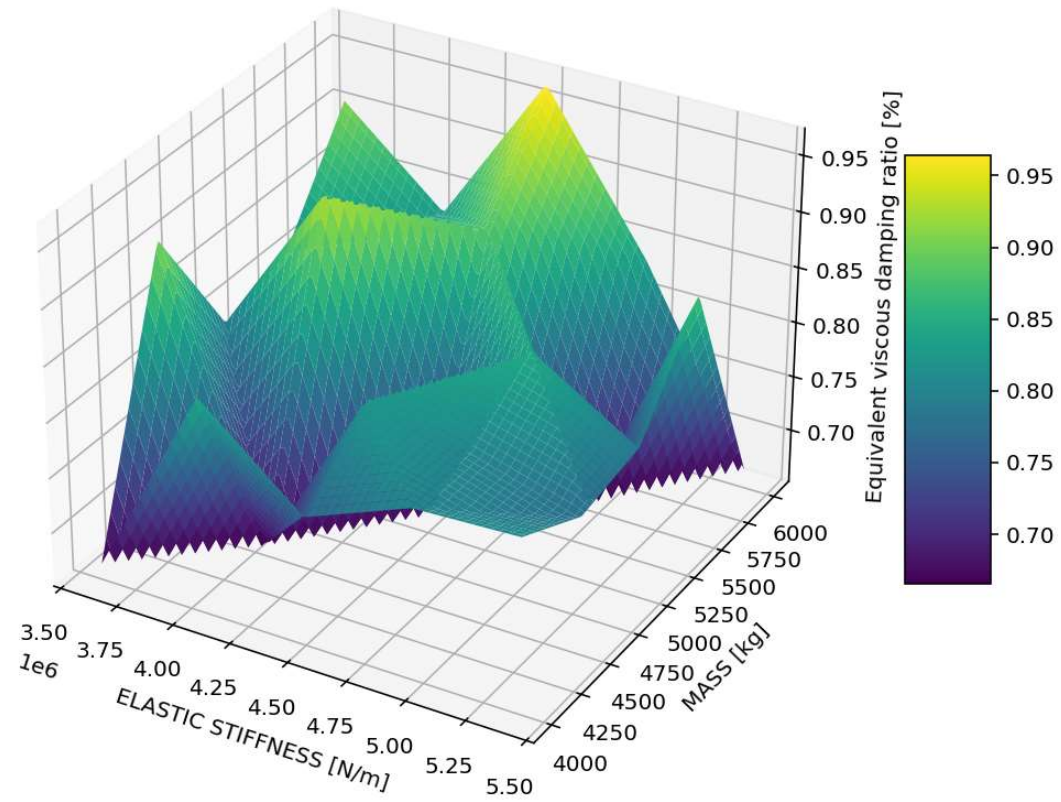


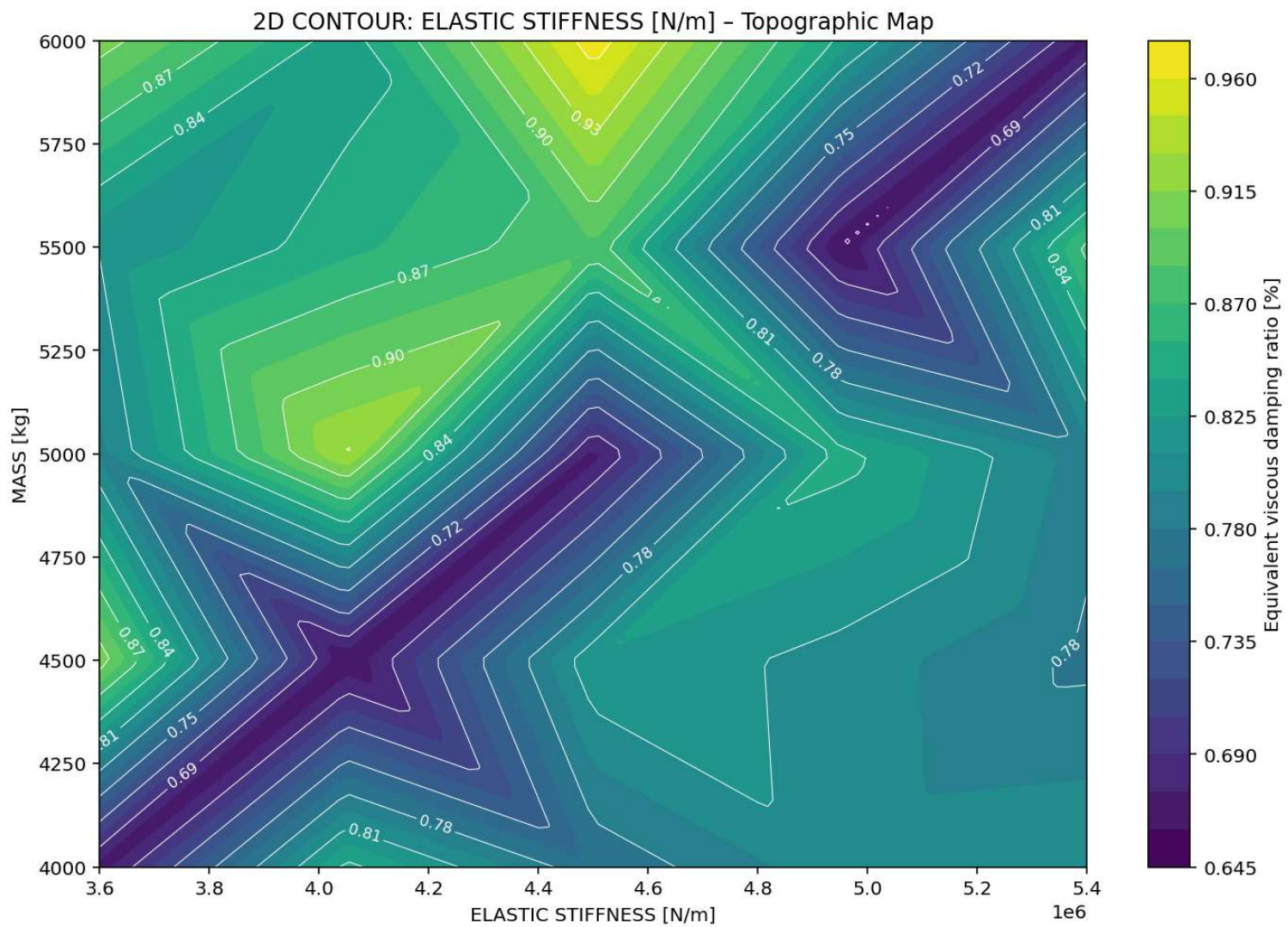
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MASS [kg] - STRUCTURAL PERIOD [s]



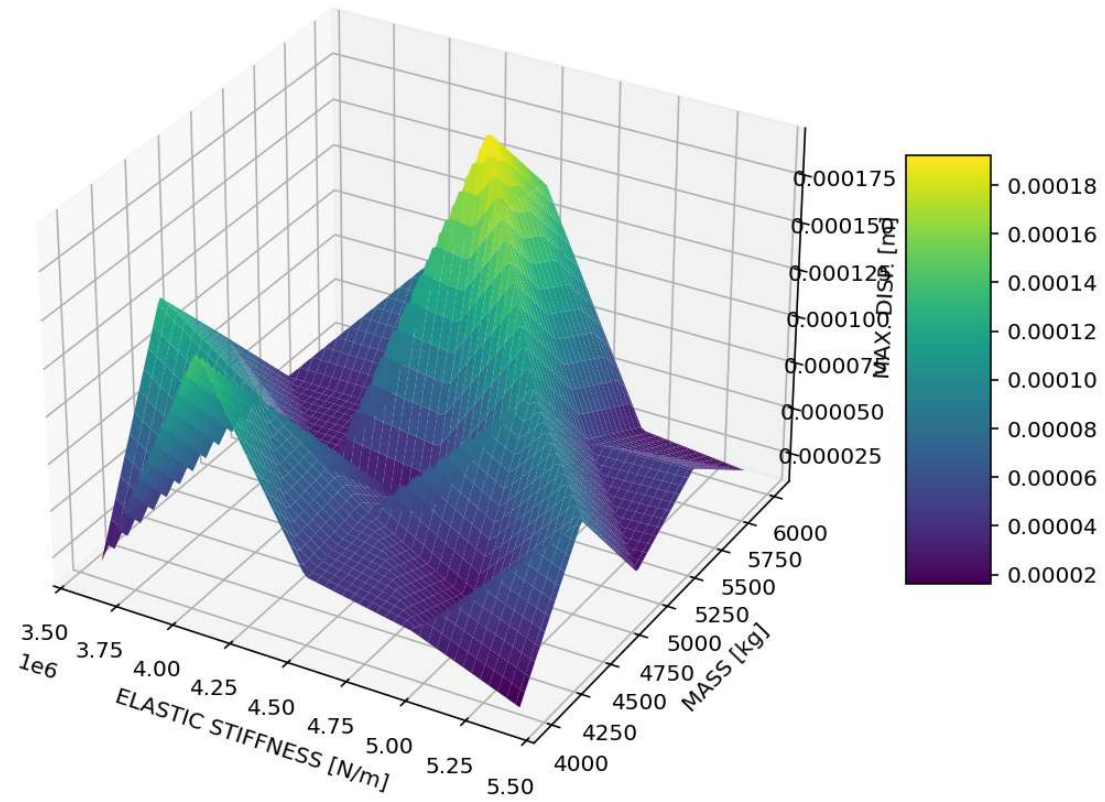


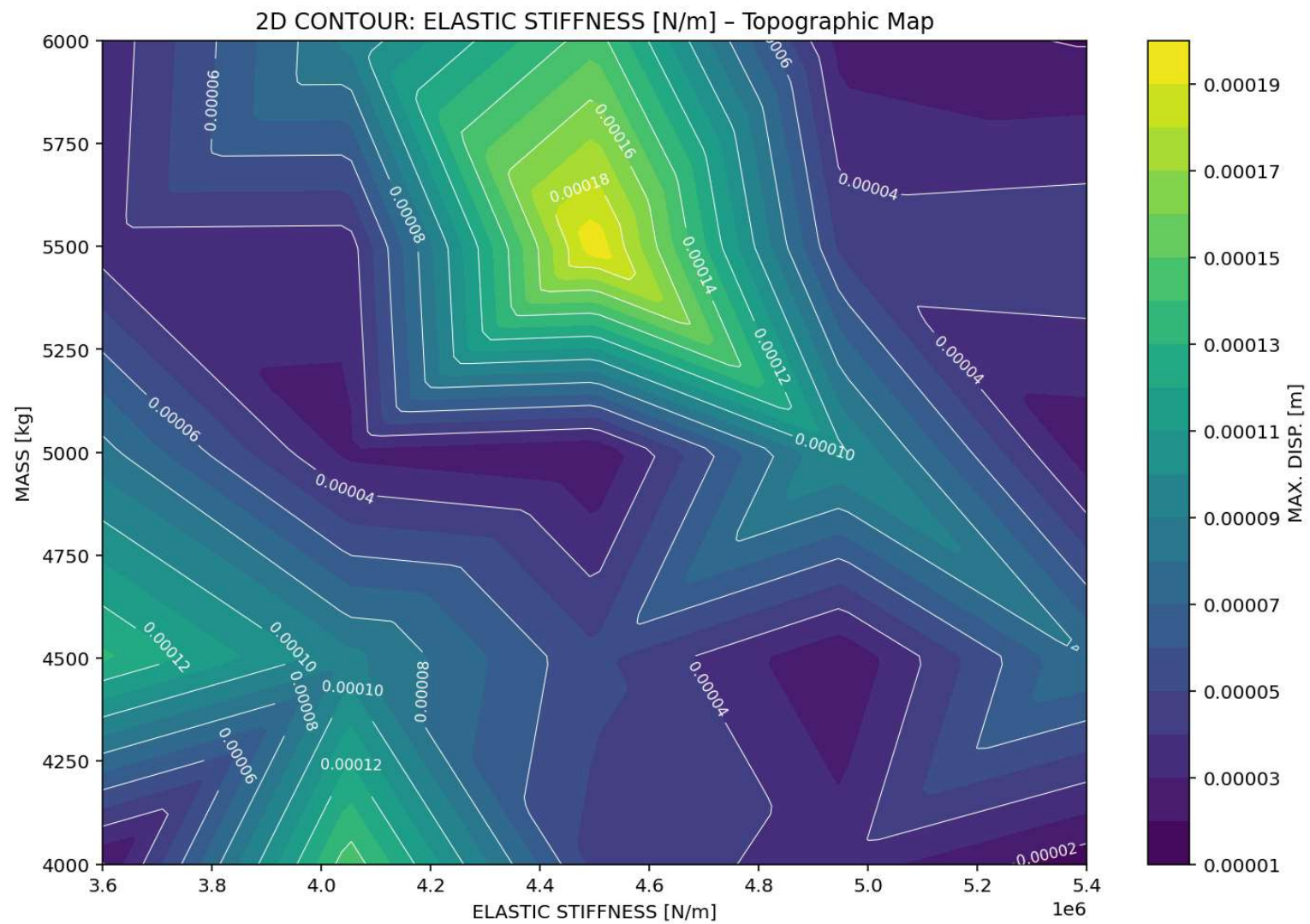
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MASS [kg] - Equivalent viscous damping ratio [%]



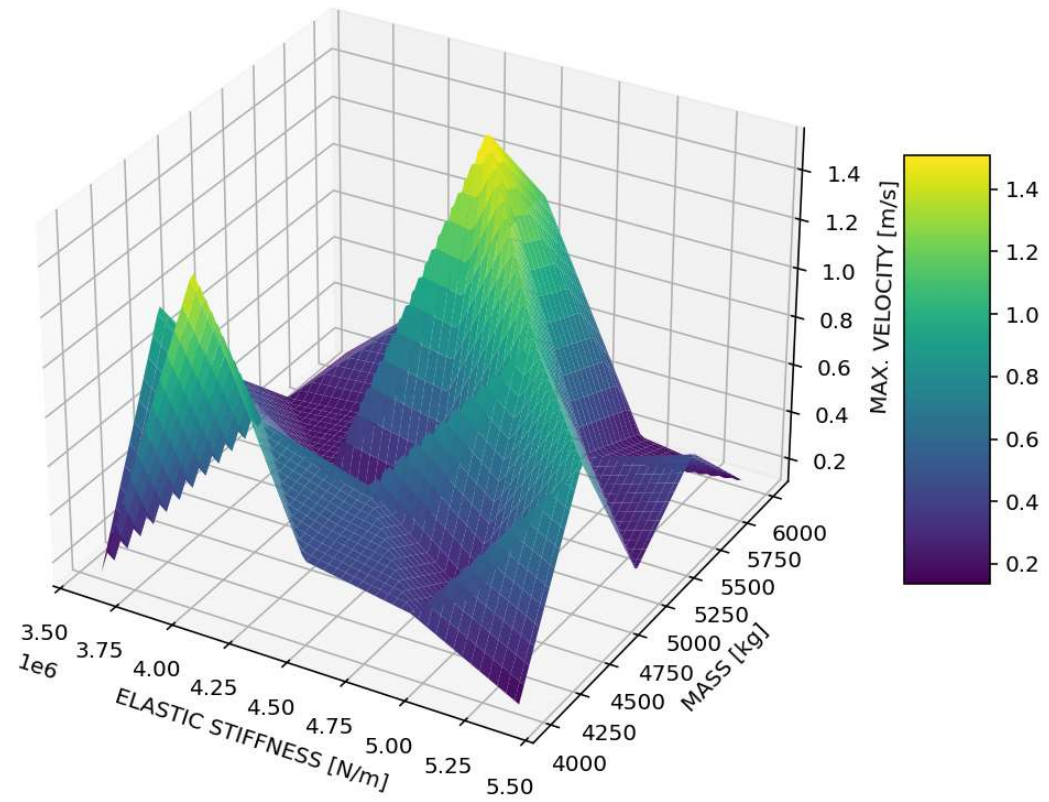


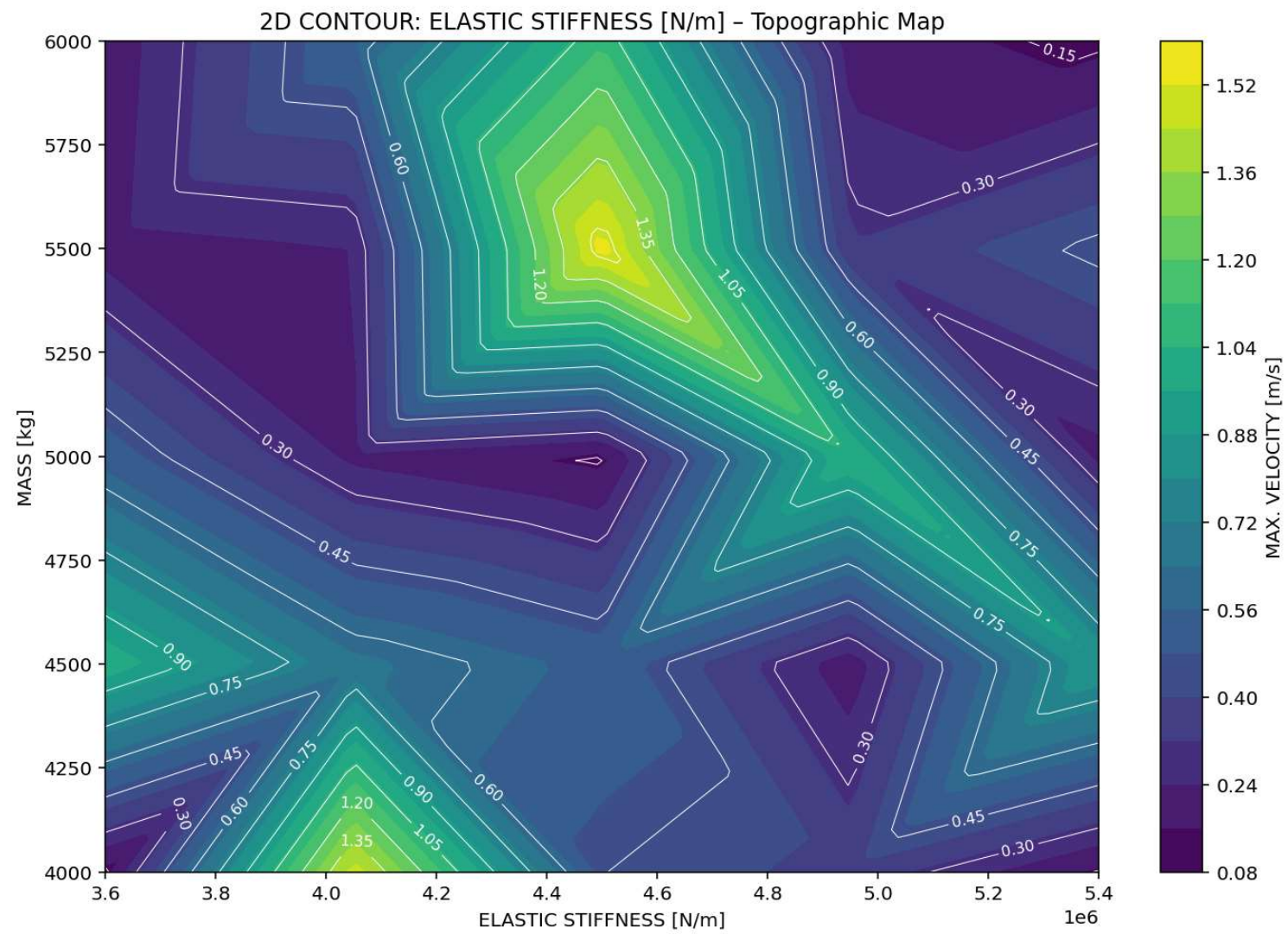
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MASS [kg] - MAX. DISP. [m]



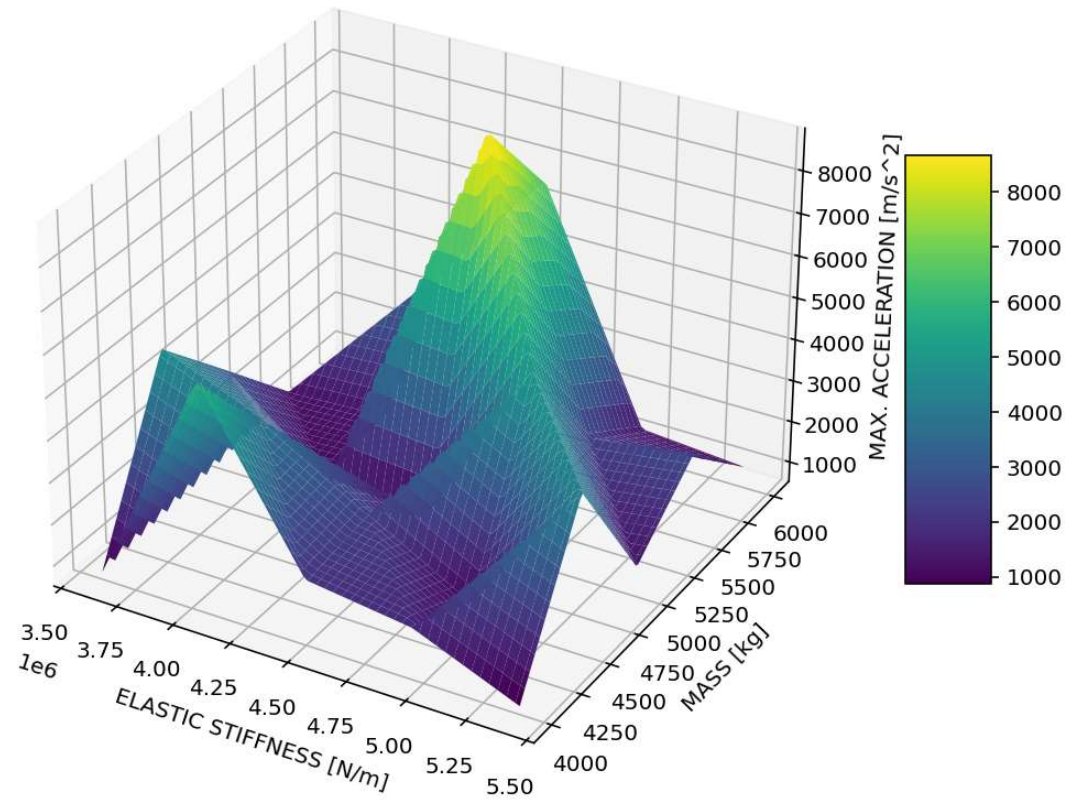


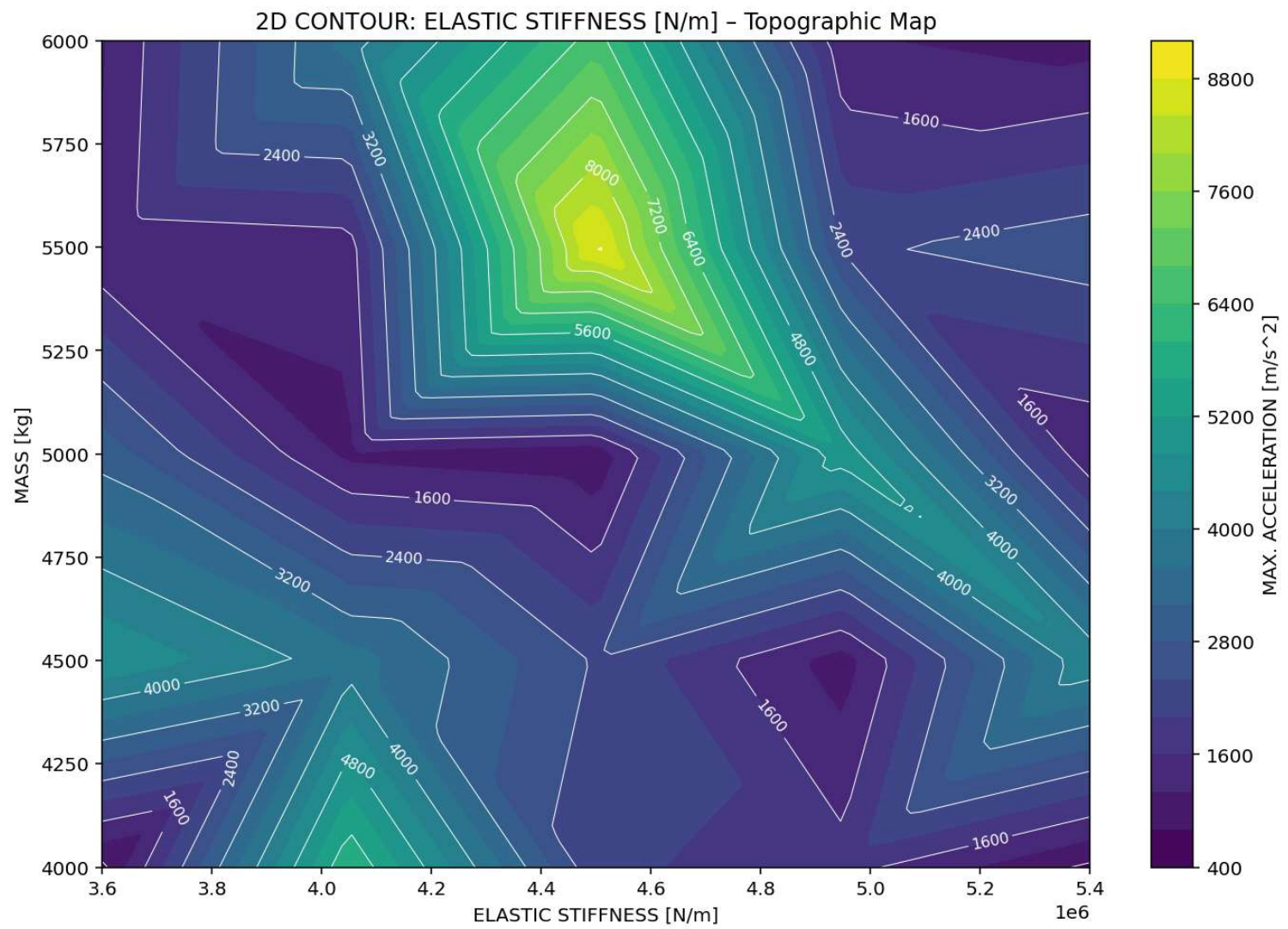
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MASS [kg] - MAX. VELOCITY [m/s]



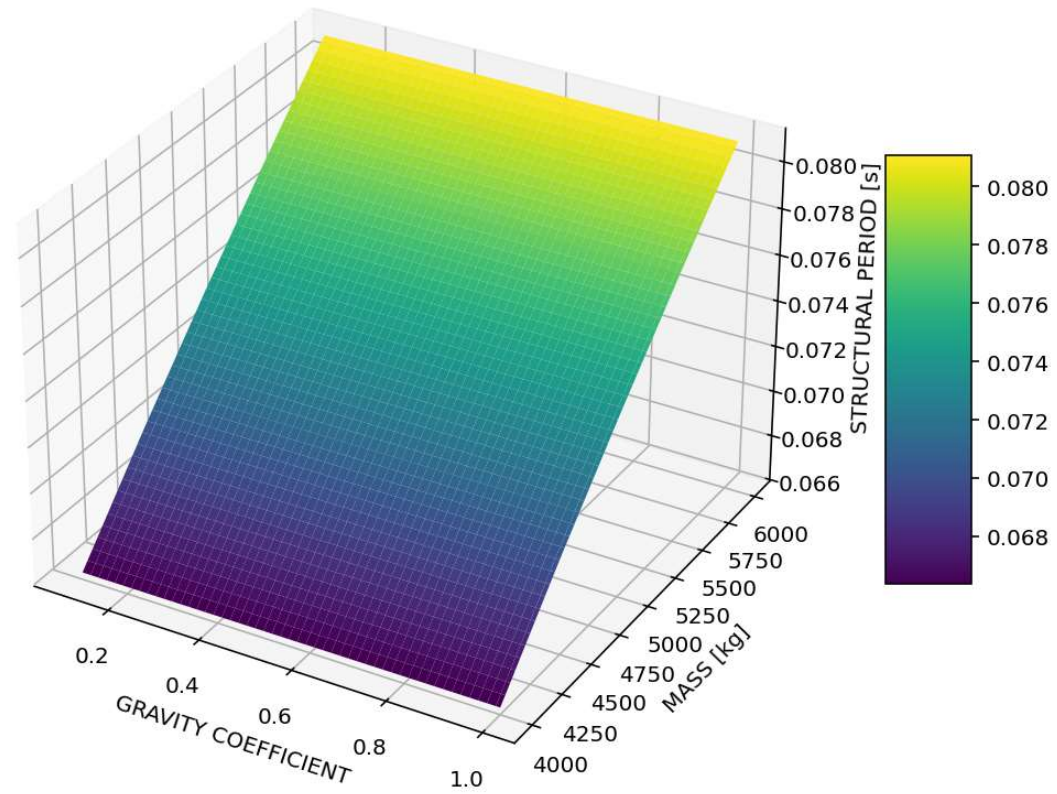


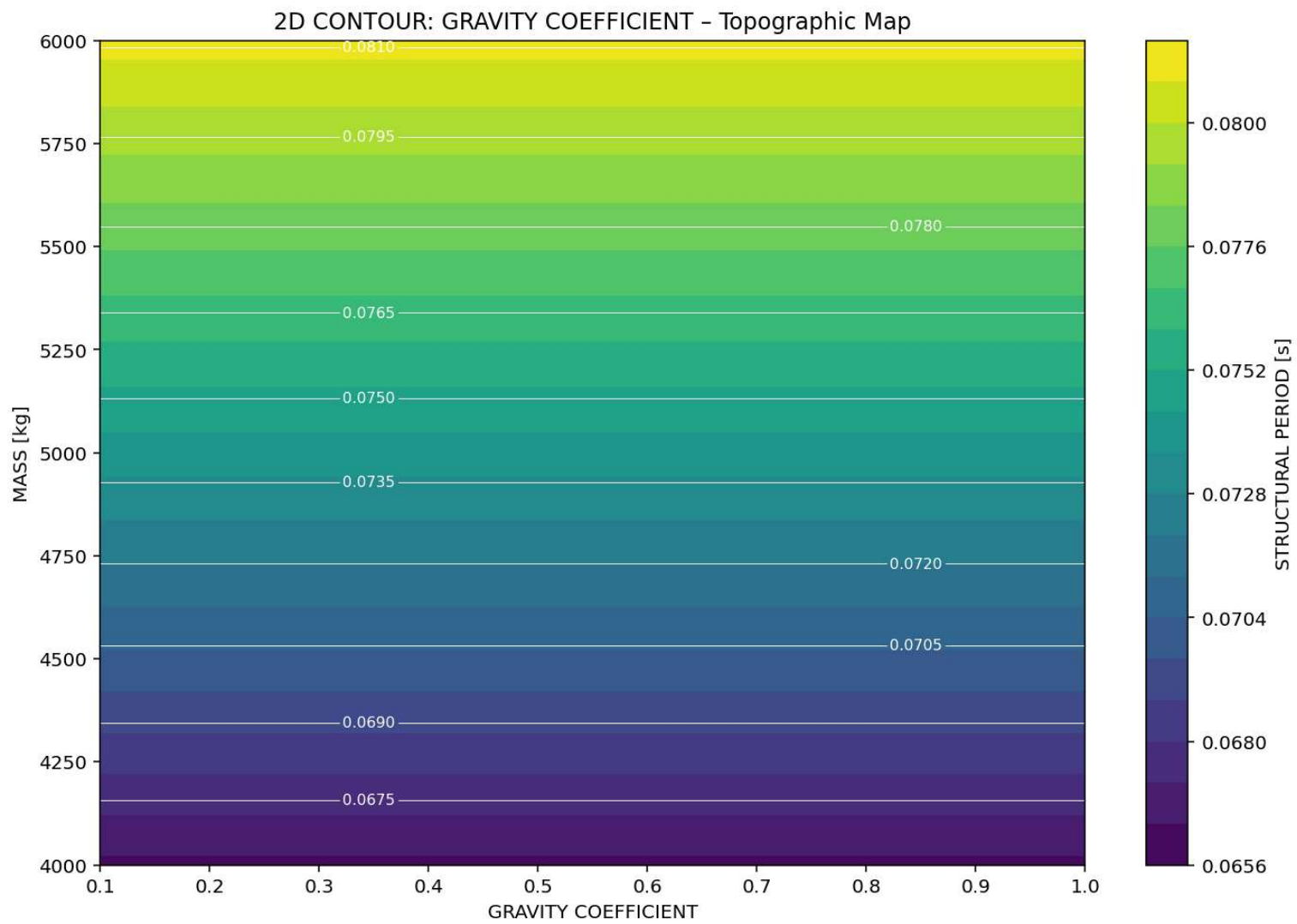
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MASS [kg] - MAX. ACCELERATION [m/s²]



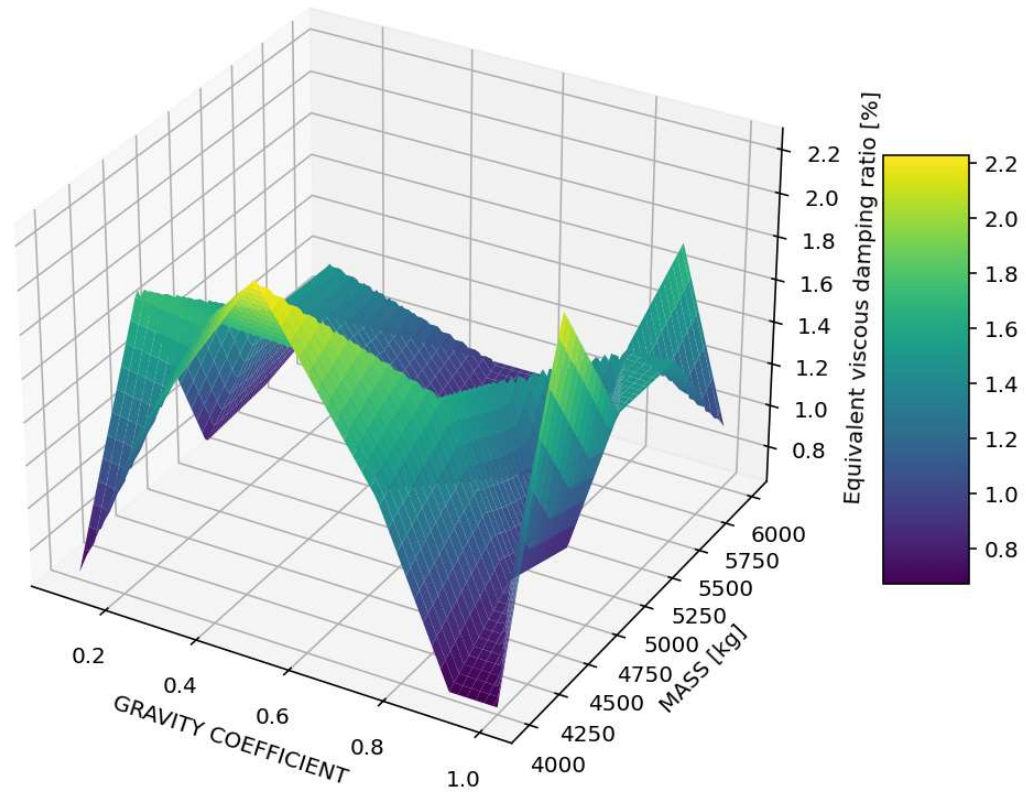


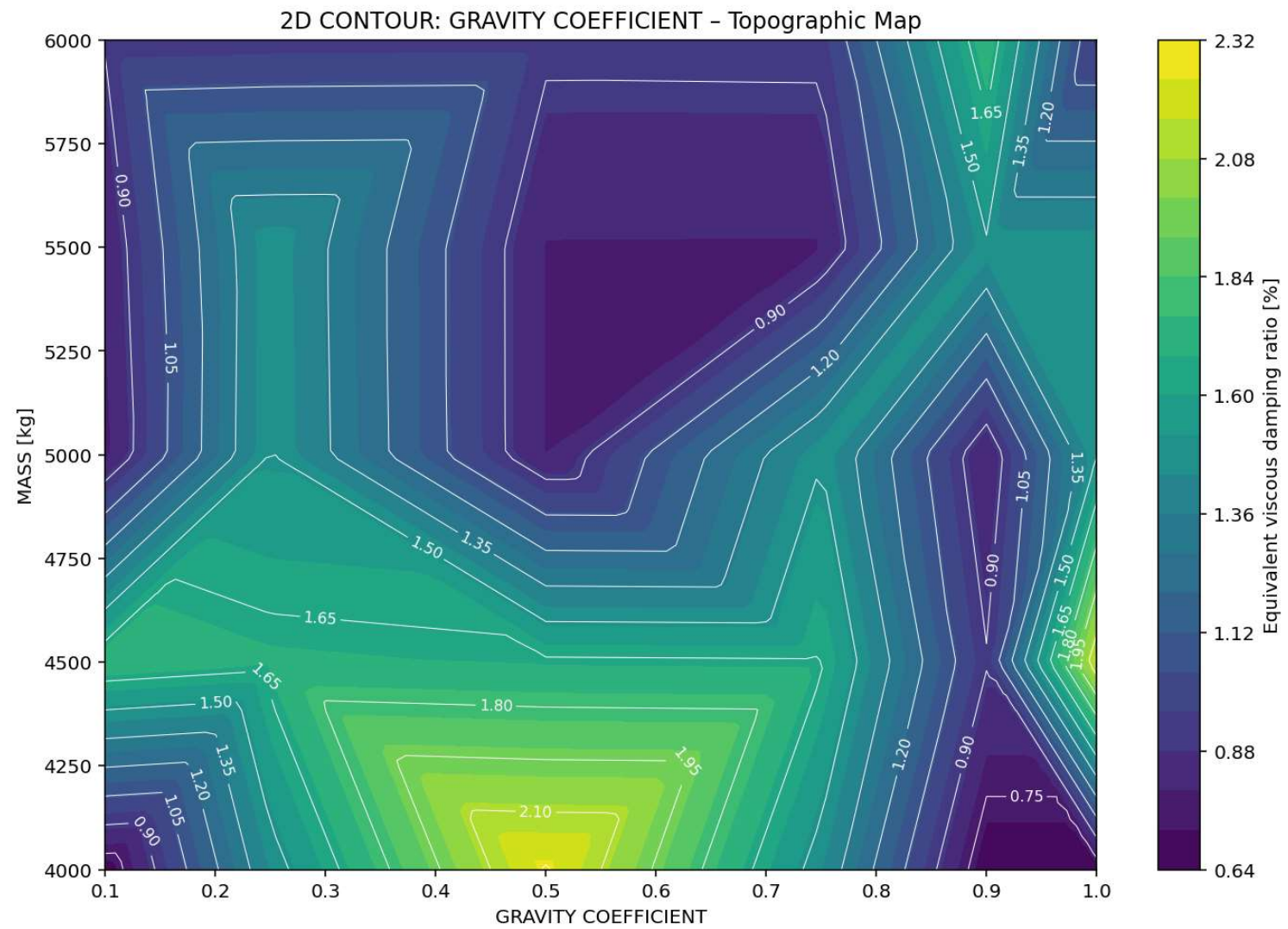
3D CONTOUR: GRAVITY COEFFICIENT - MASS [kg] - STRUCTURAL PERIOD [s]



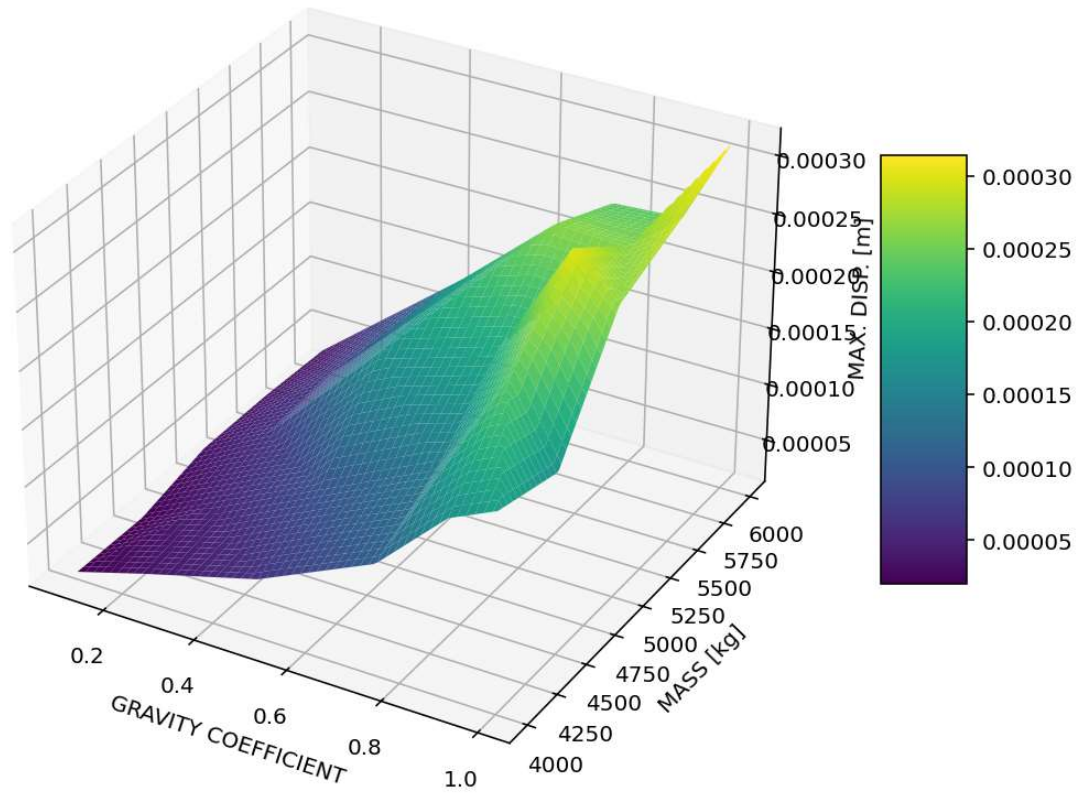


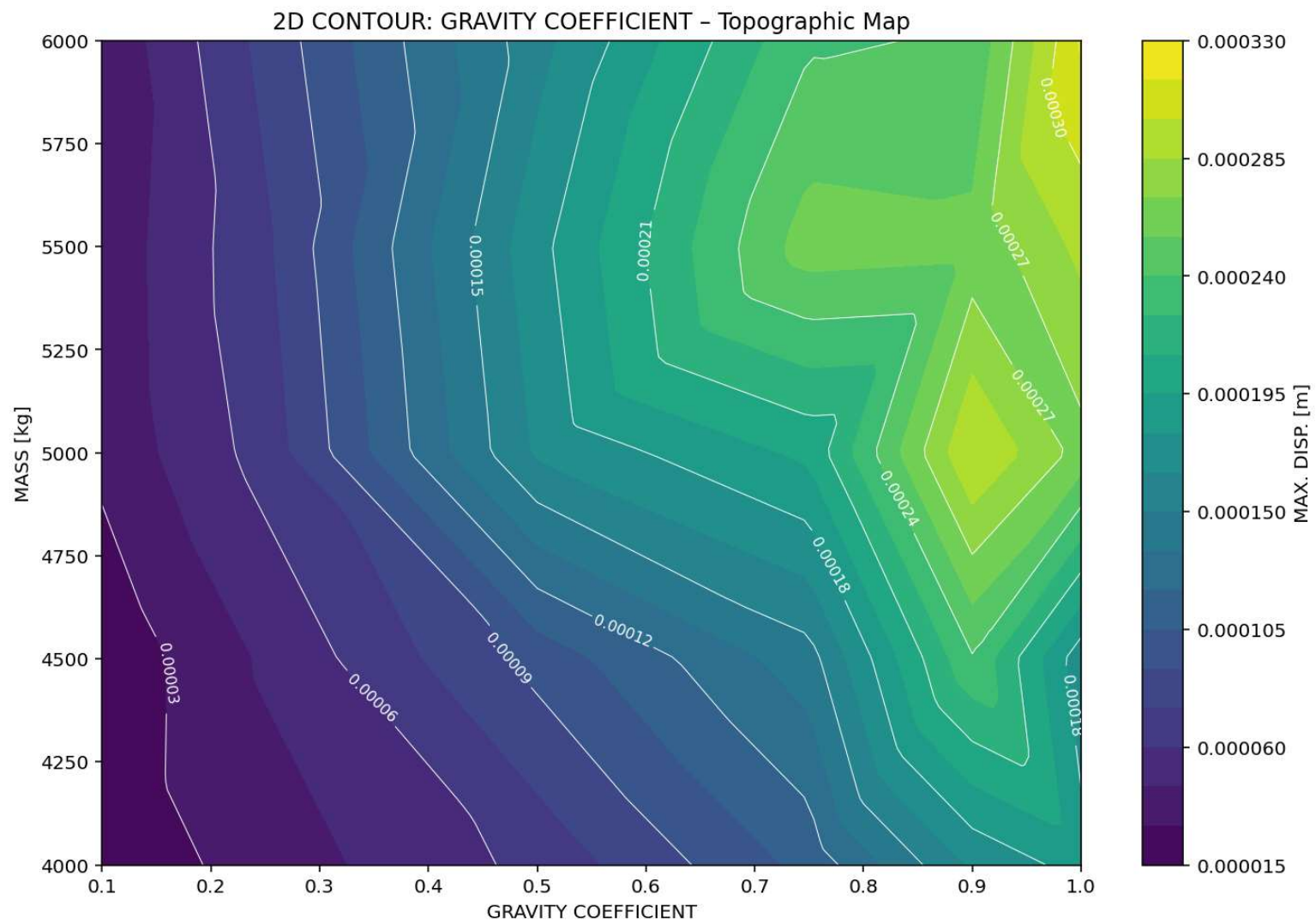
3D CONTOUR: GRAVITY COEFFICIENT - MASS [kg] - Equivalent viscous damping ratio [%]



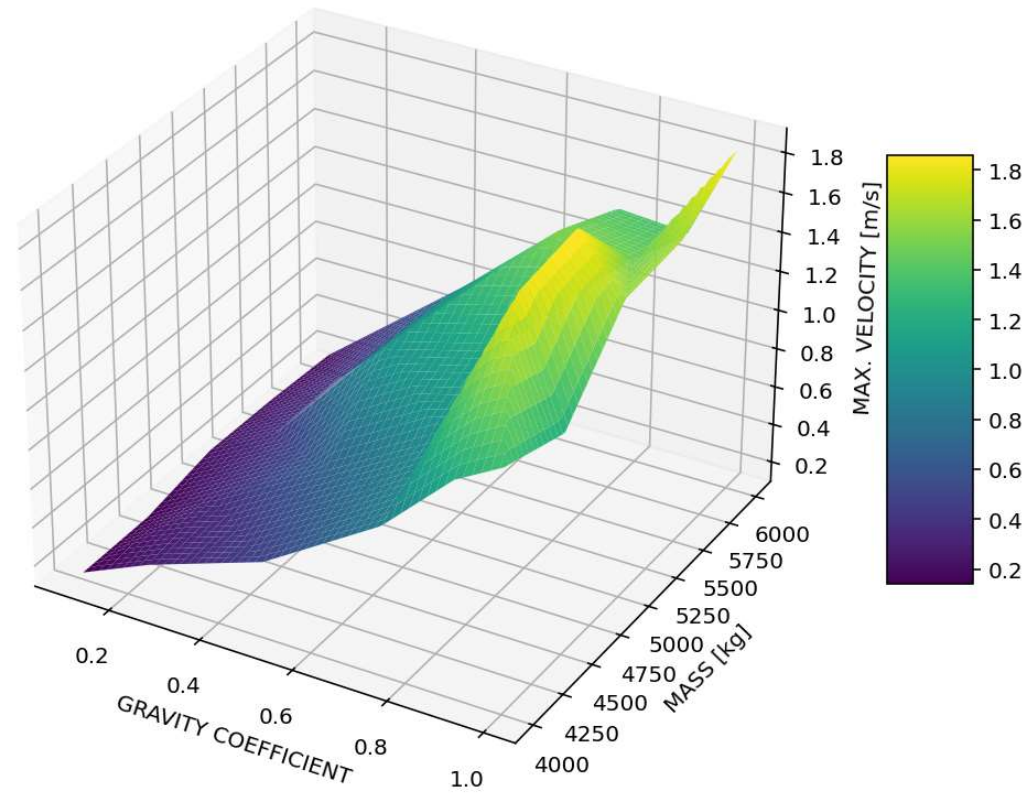


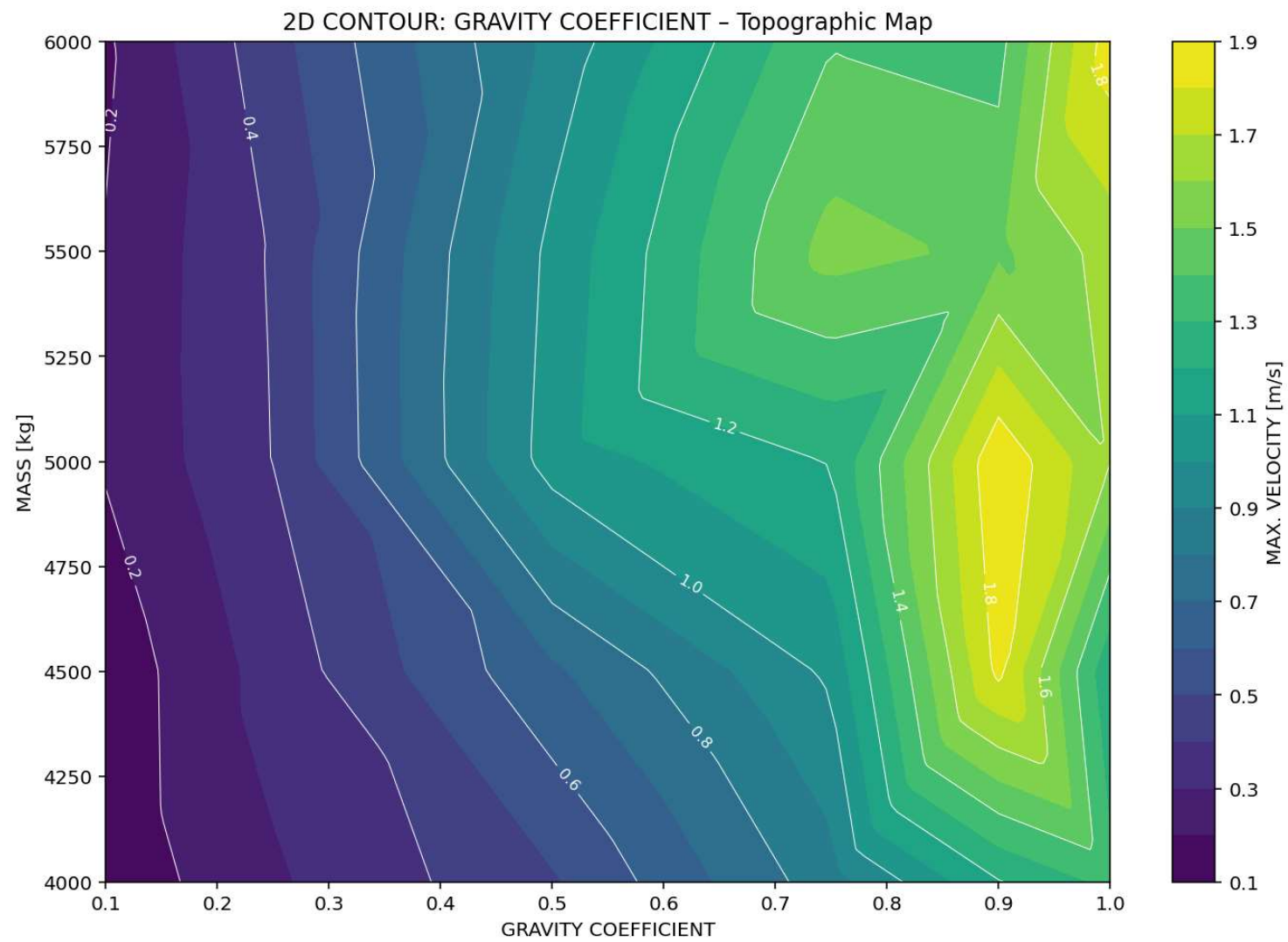
3D CONTOUR: GRAVITY COEFFICIENT - MASS [kg] - MAX. DISP. [m]



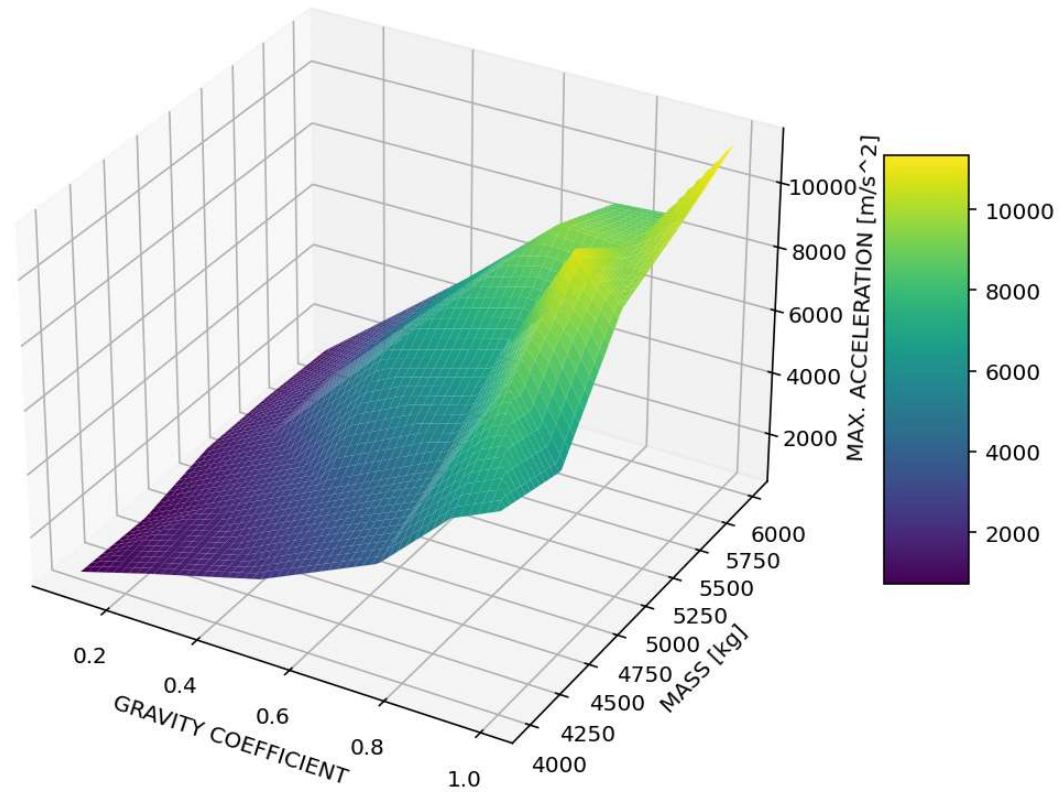


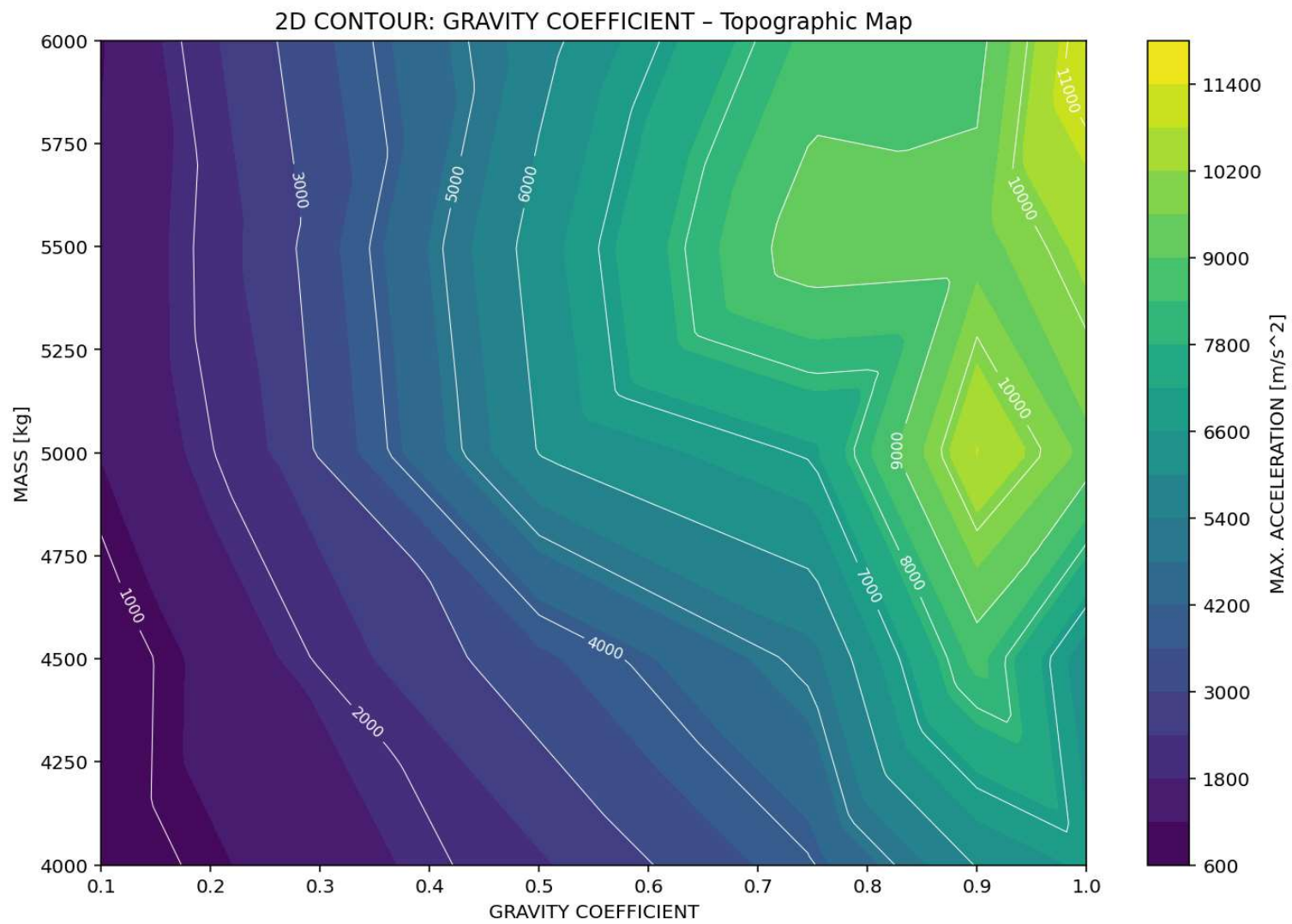
3D CONTOUR: GRAVITY COEFFICIENT - MASS [kg] - MAX. VELOCITY [m/s]

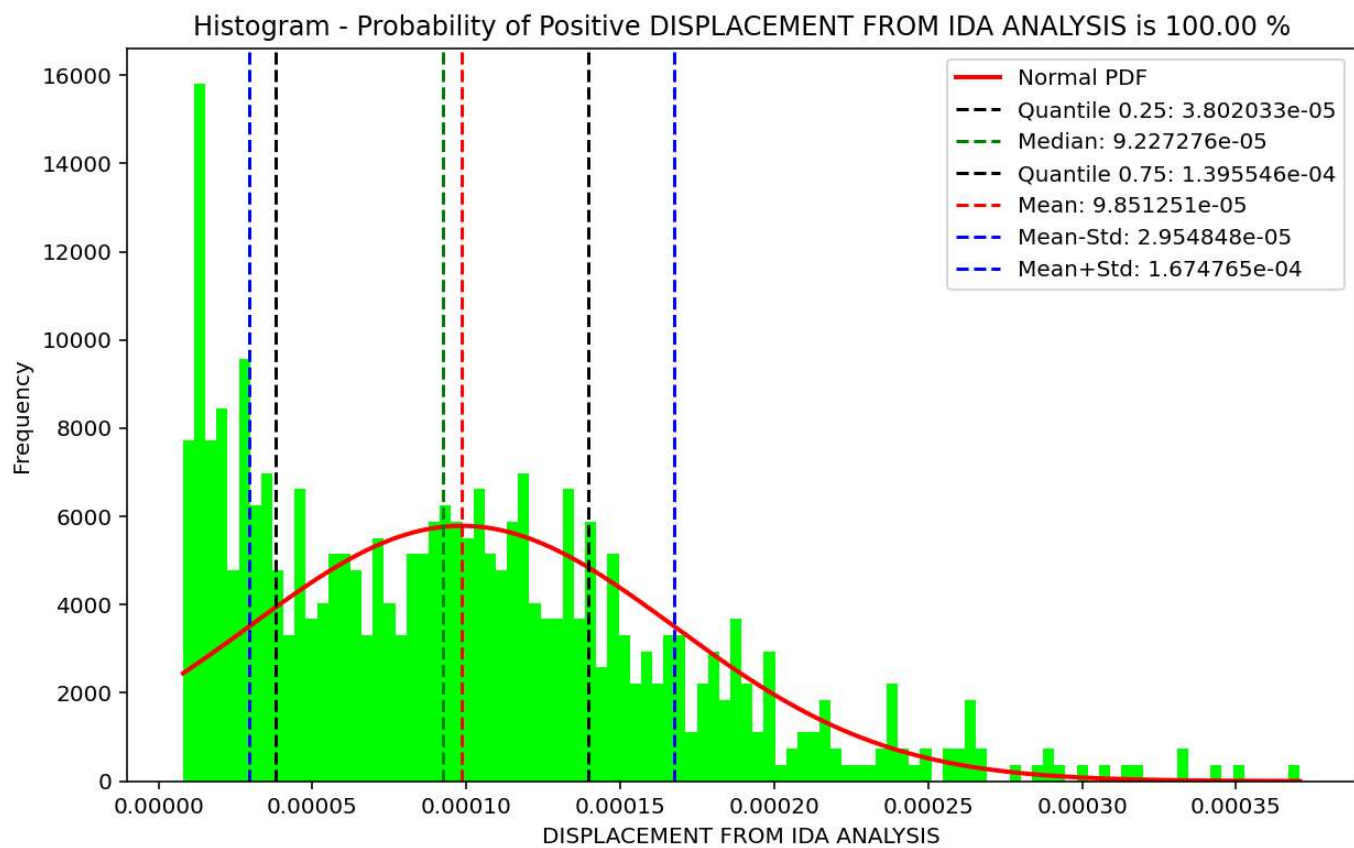


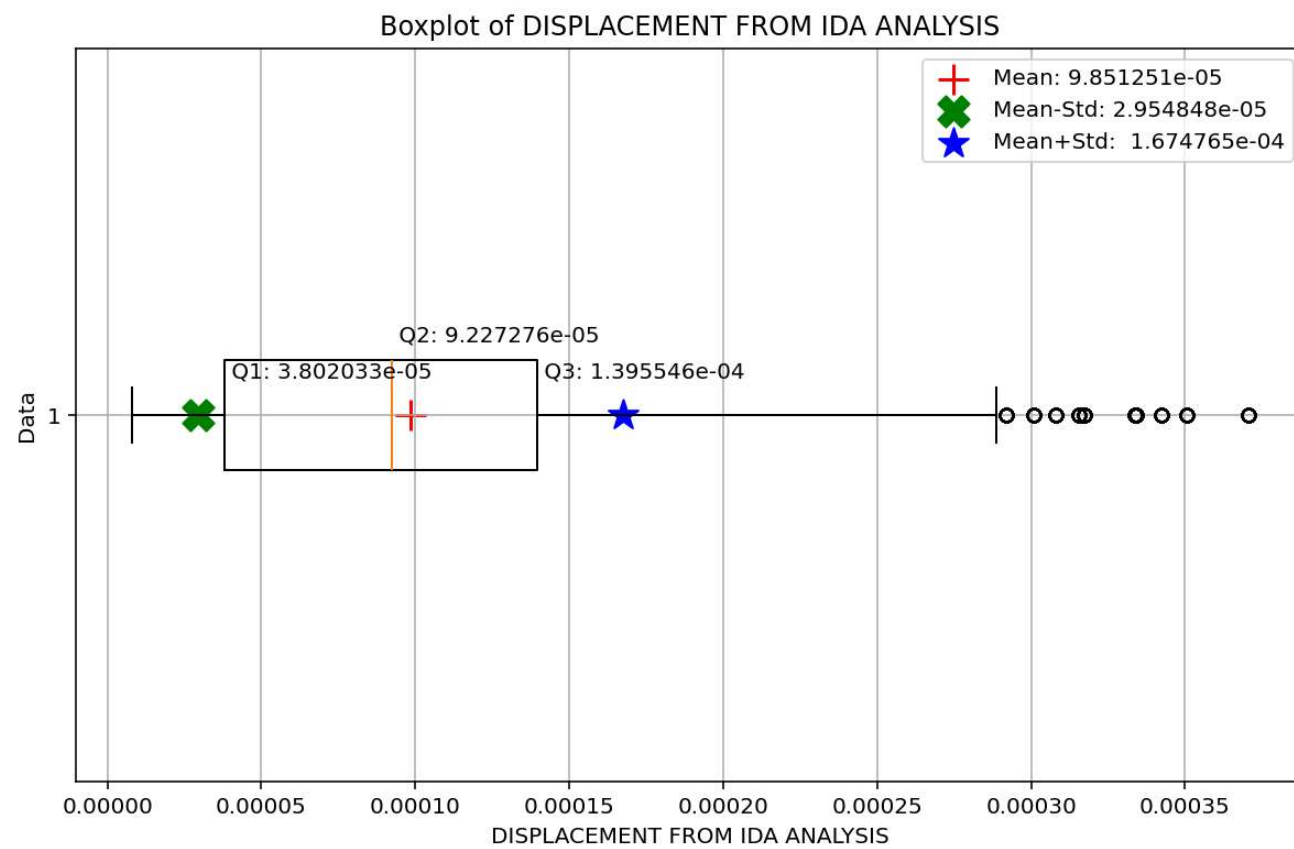


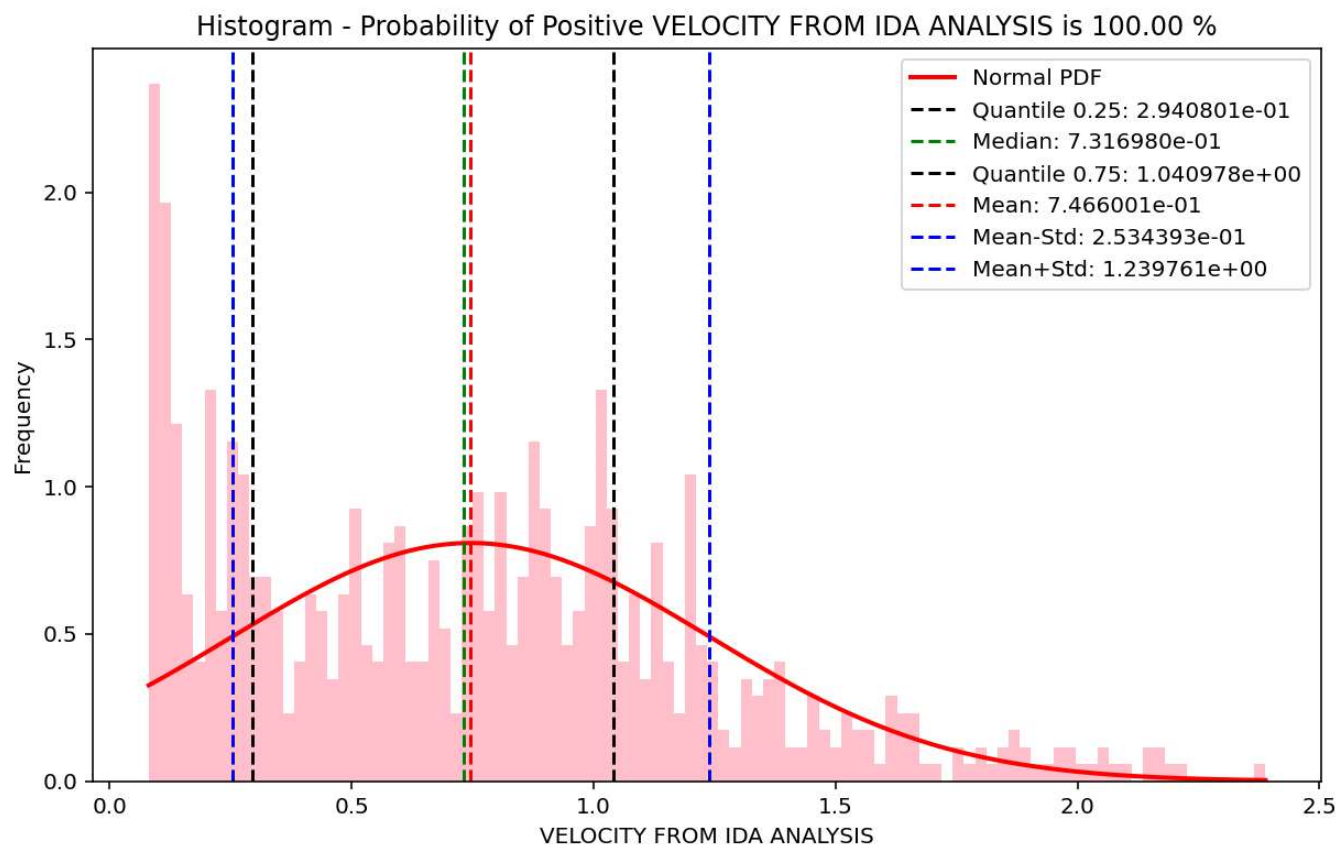
3D CONTOUR: GRAVITY COEFFICIENT - MASS [kg] - MAX. ACCELERATION [m/s²]



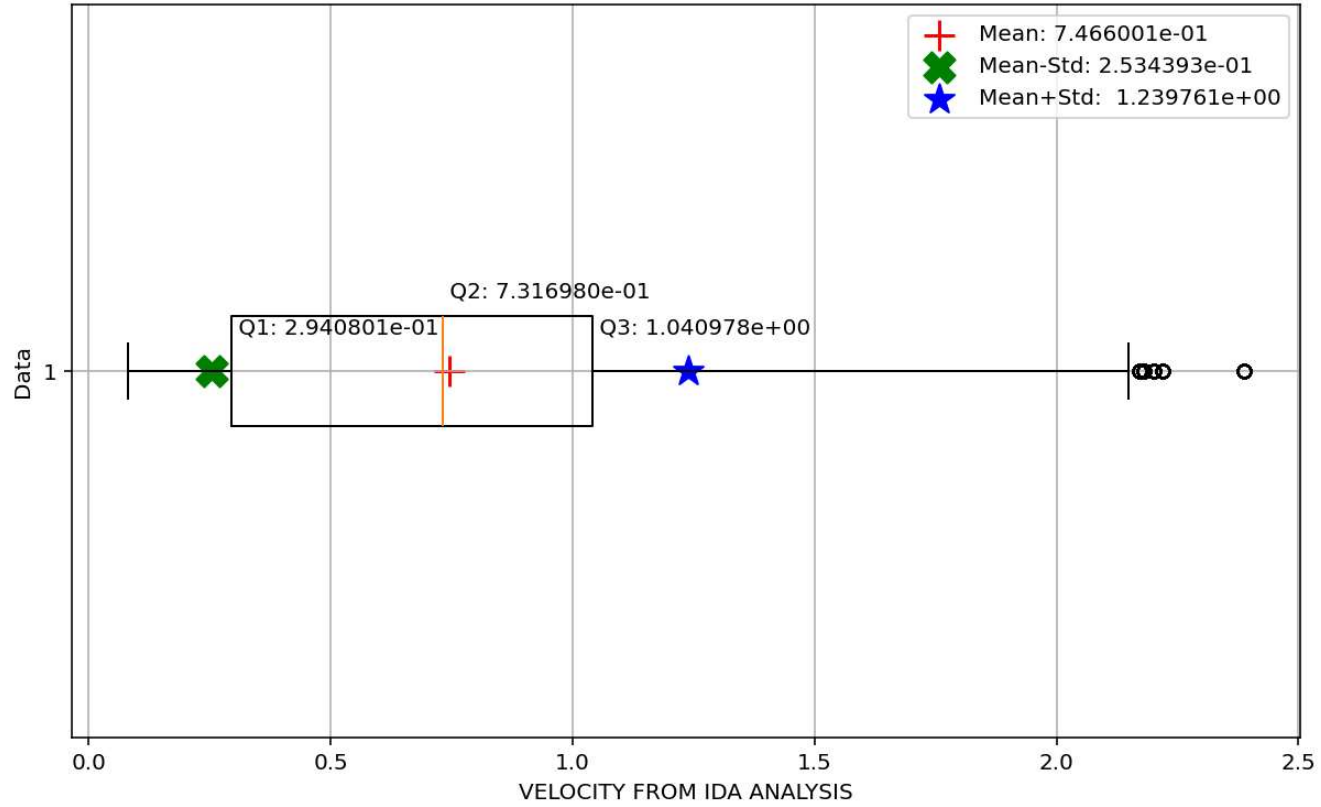


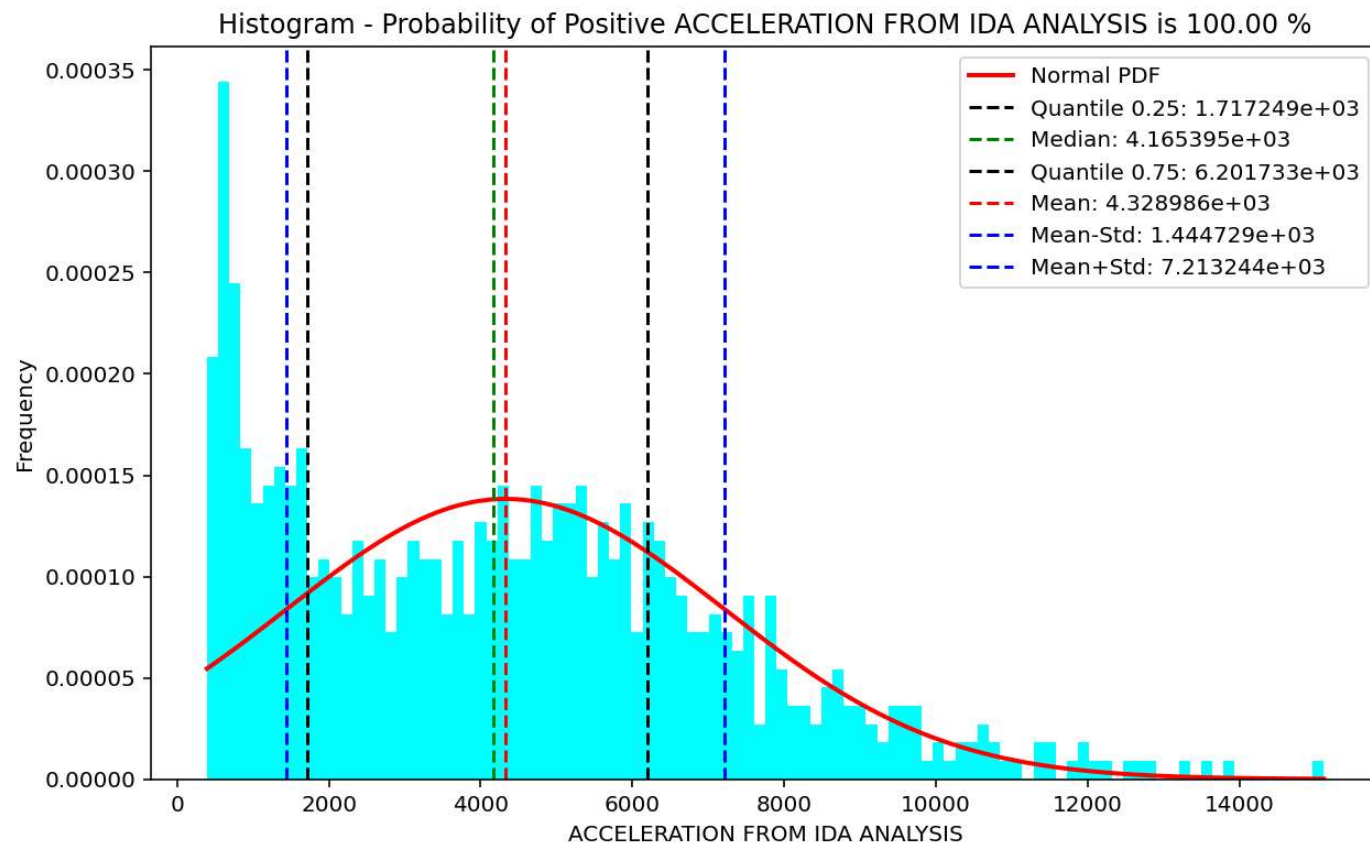


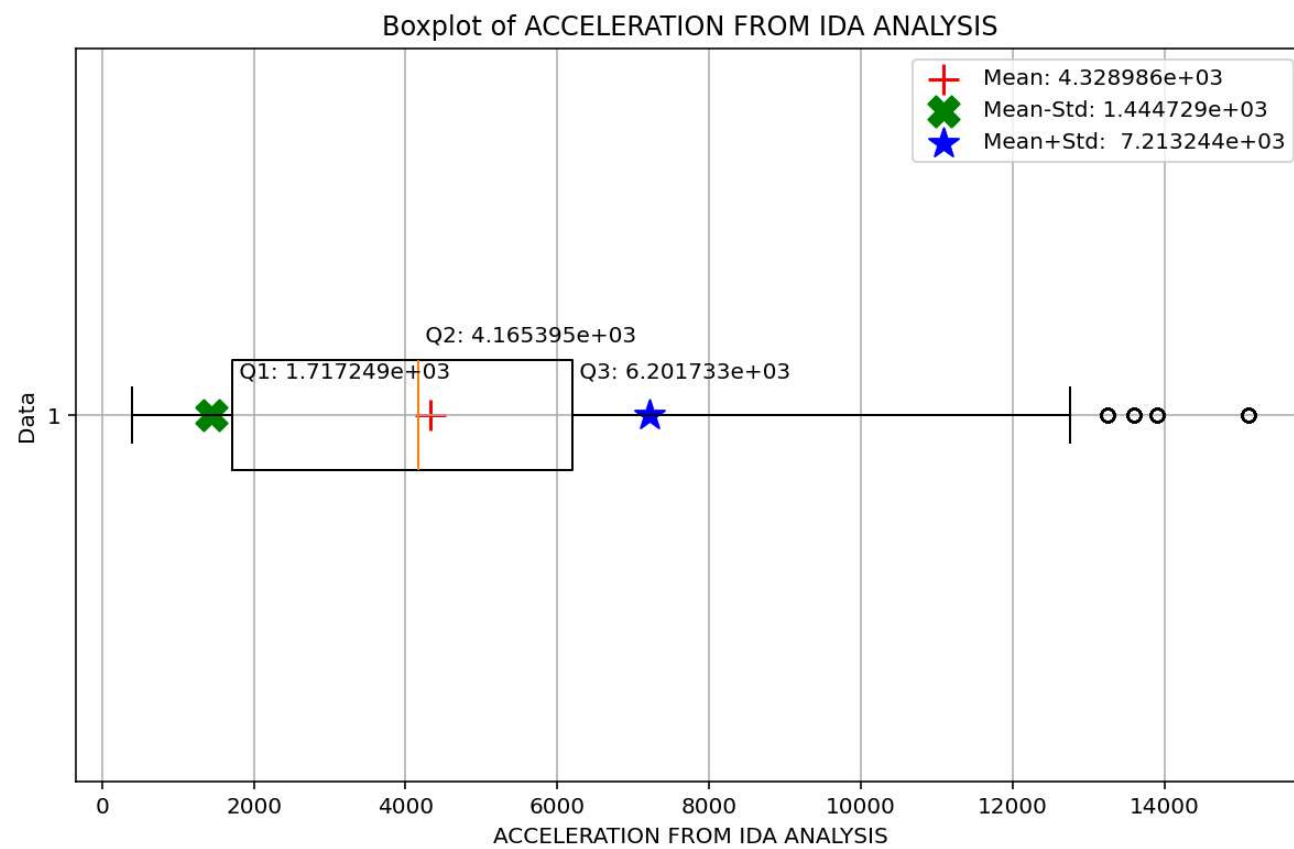


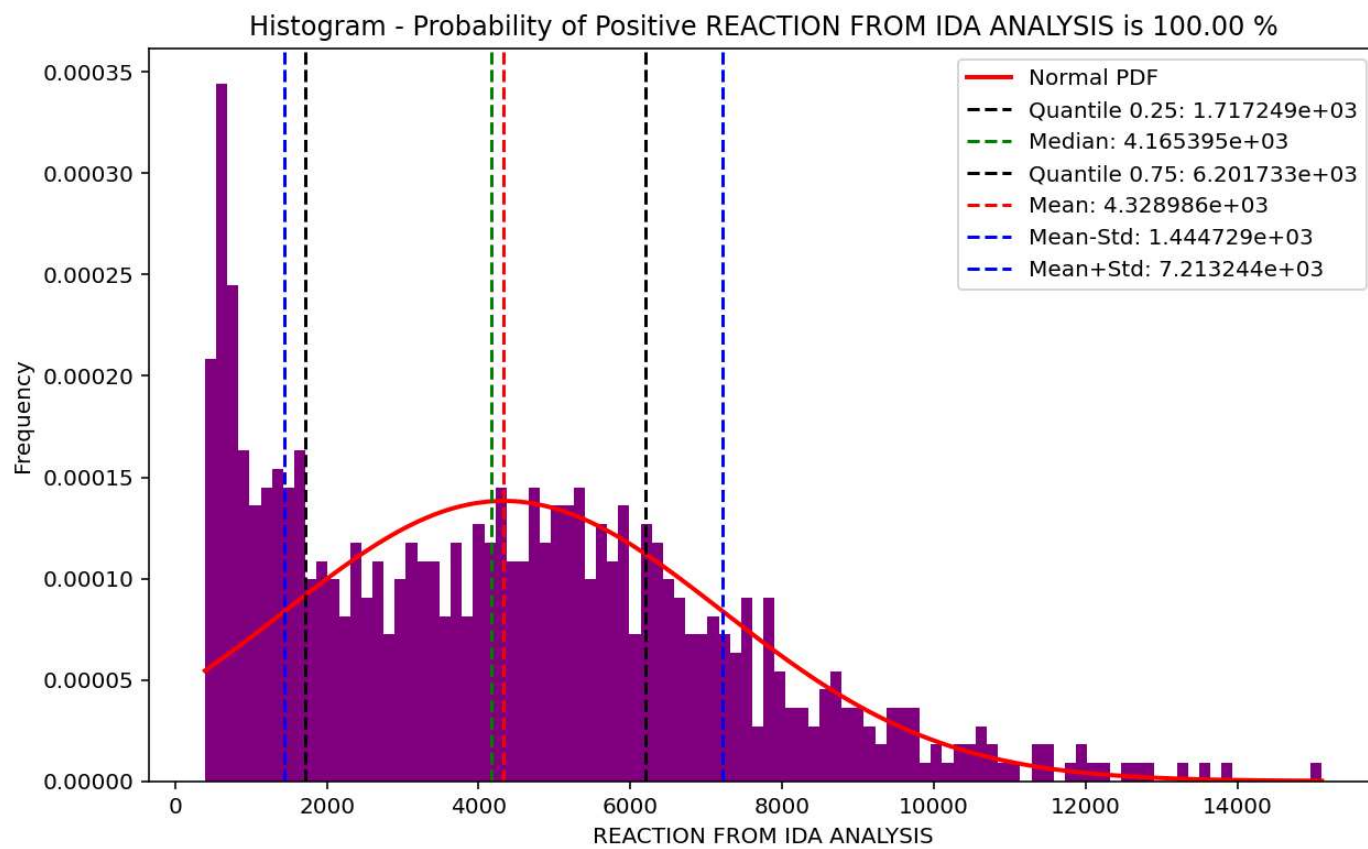


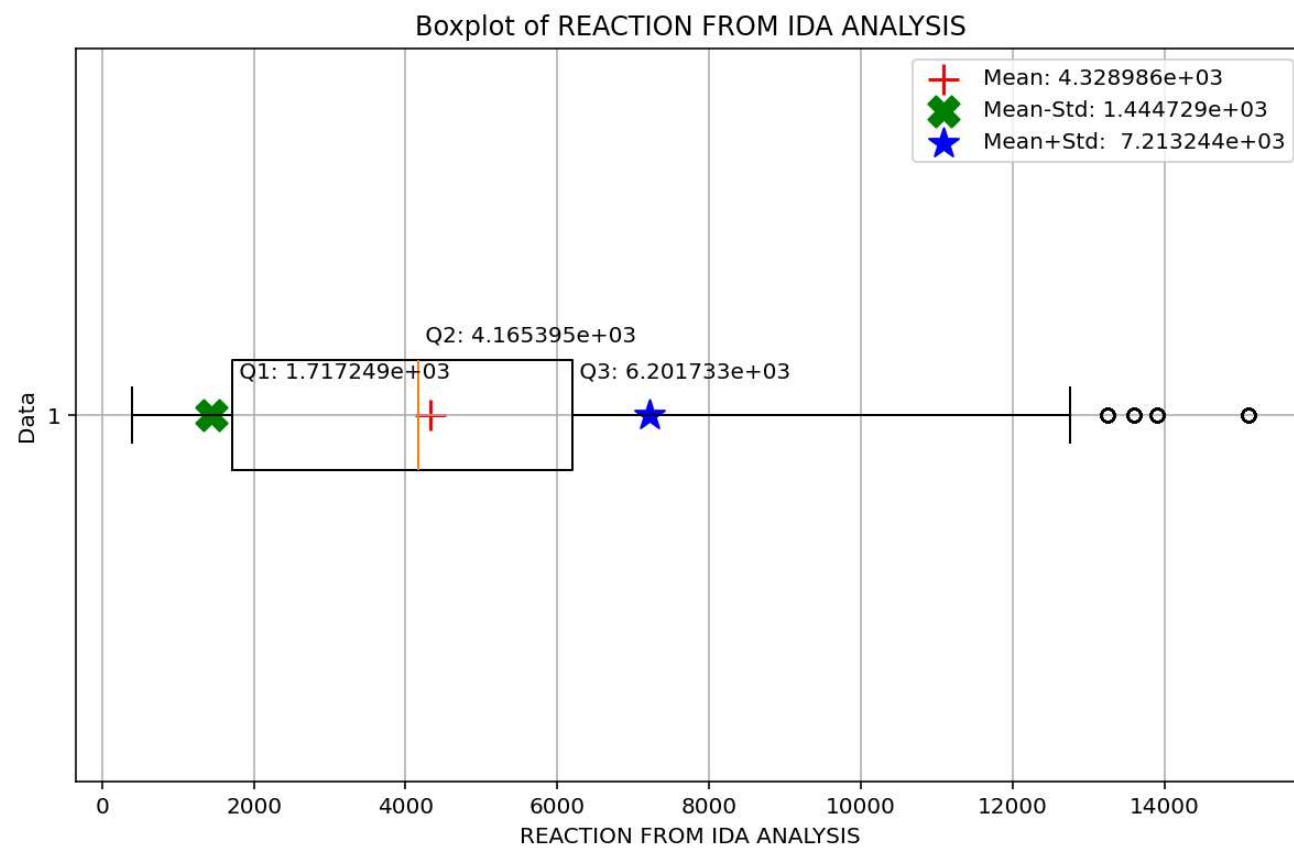
Boxplot of VELOCITY FROM IDA ANALYSIS

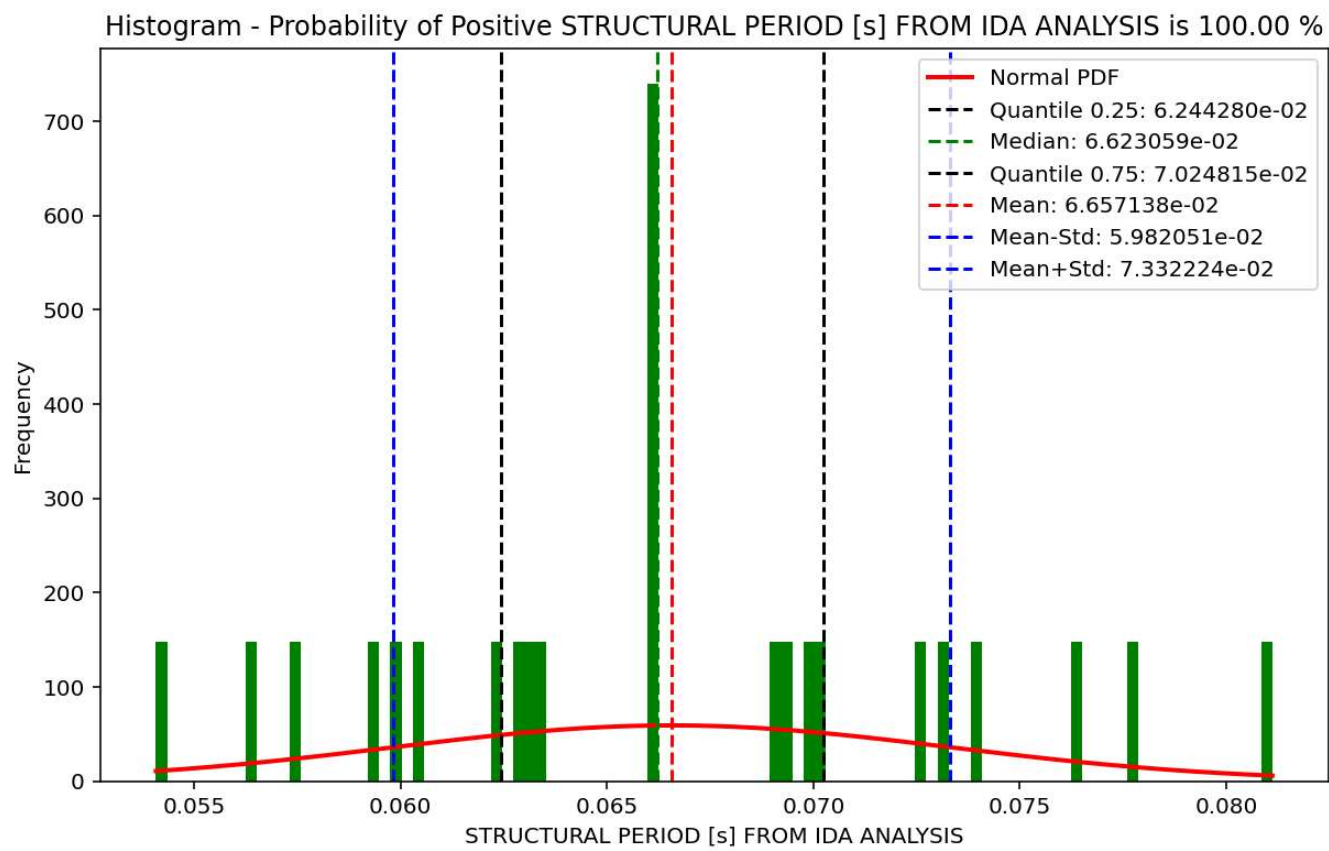


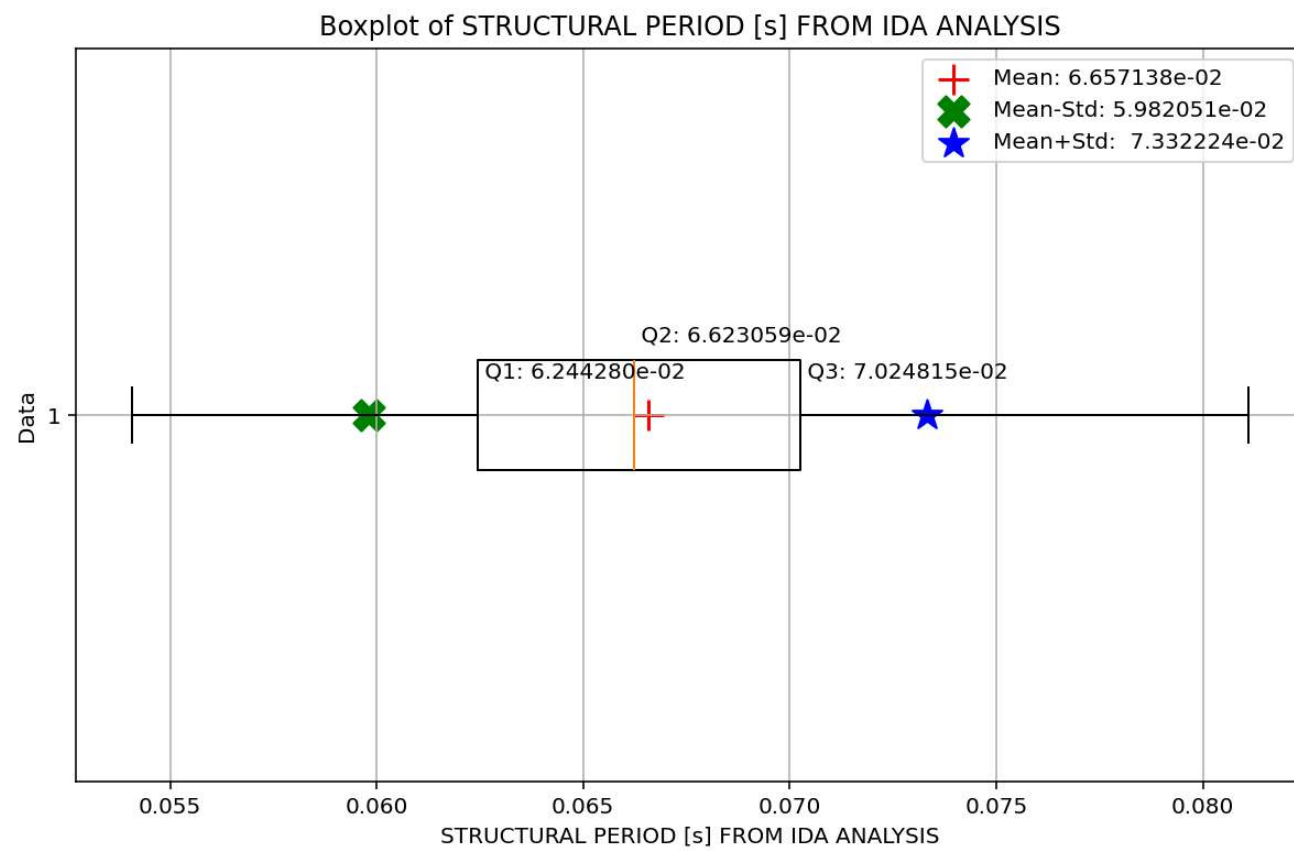




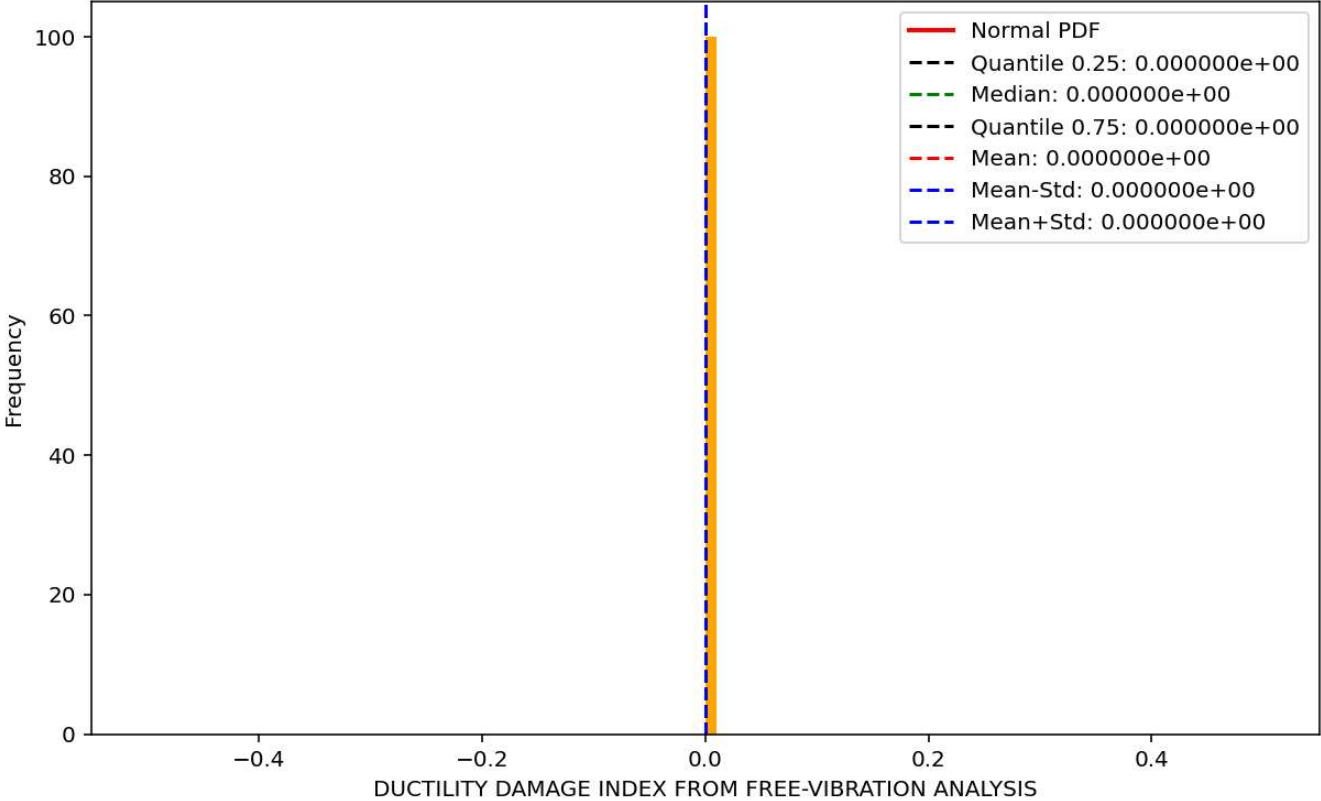


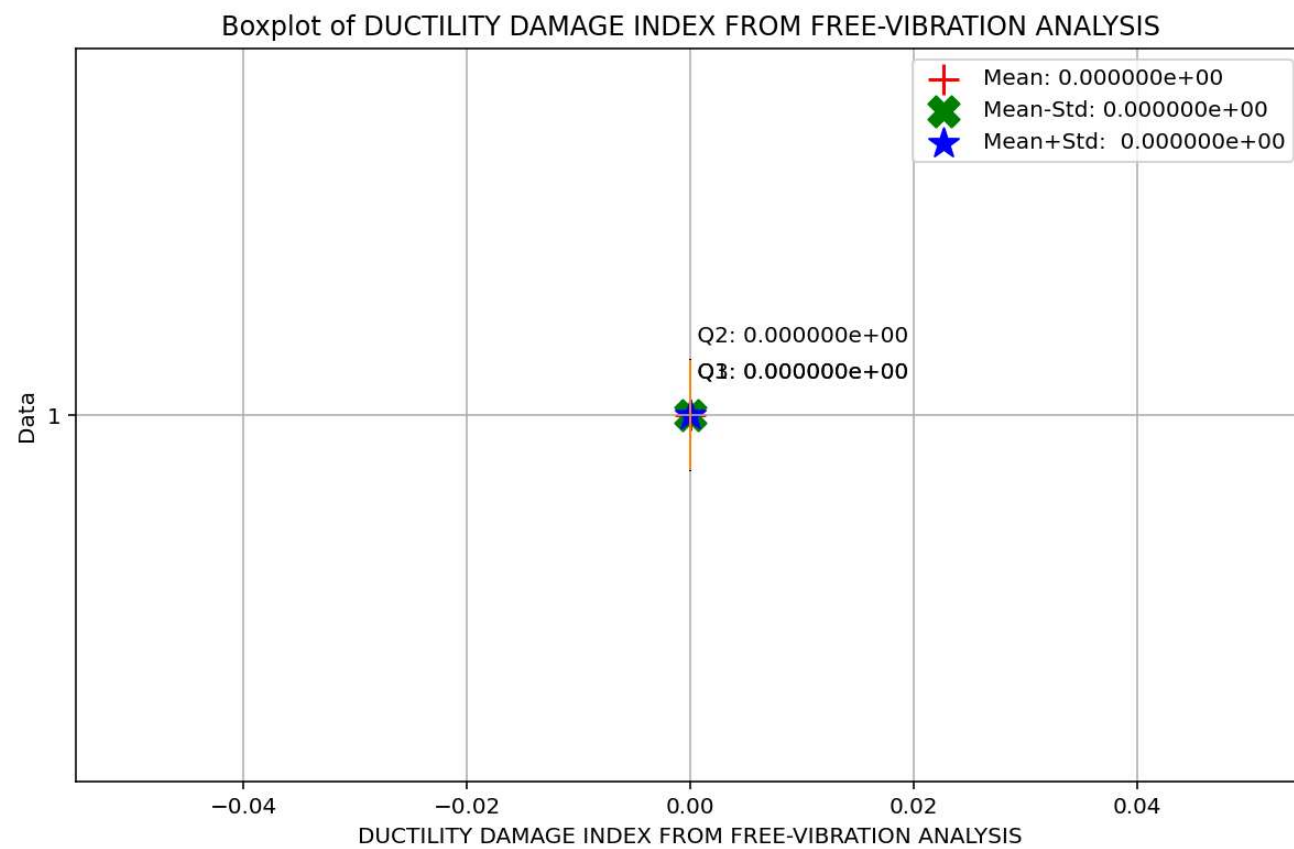




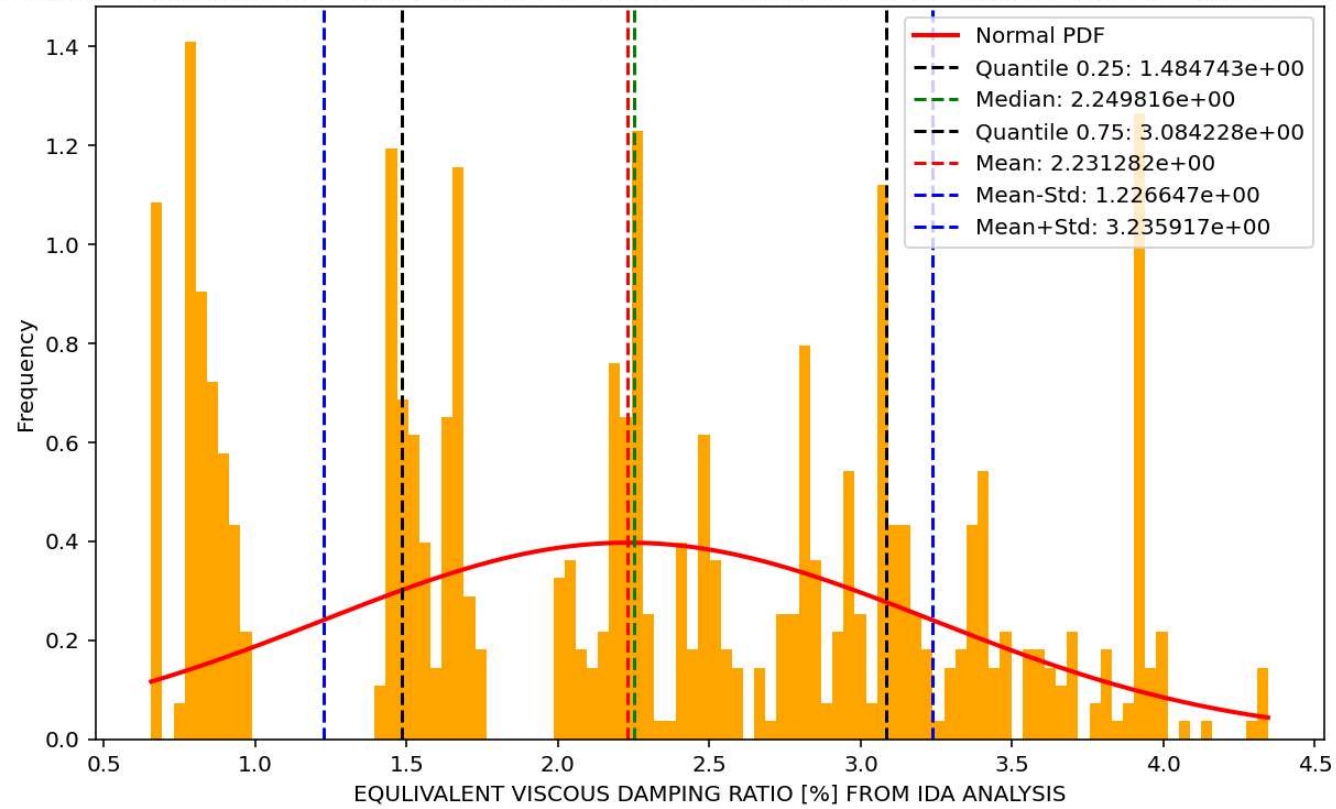


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %

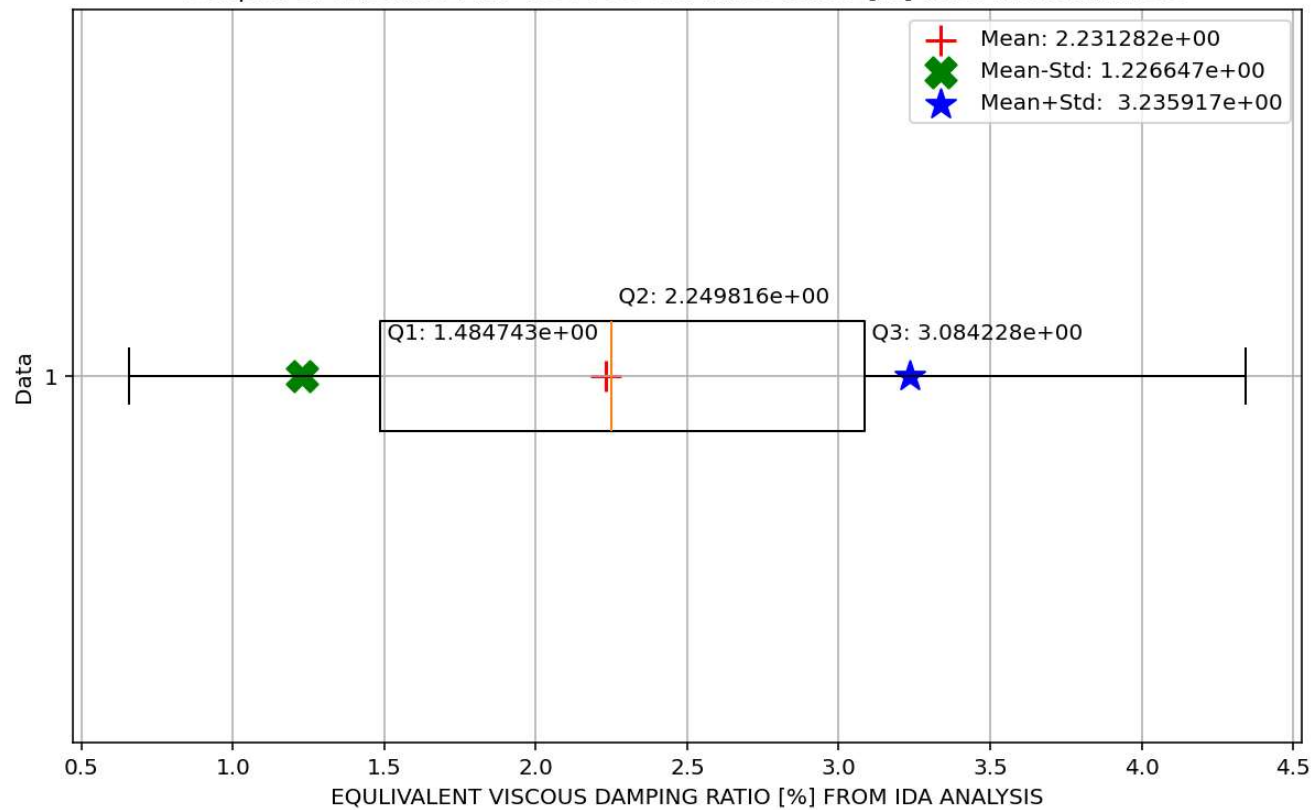




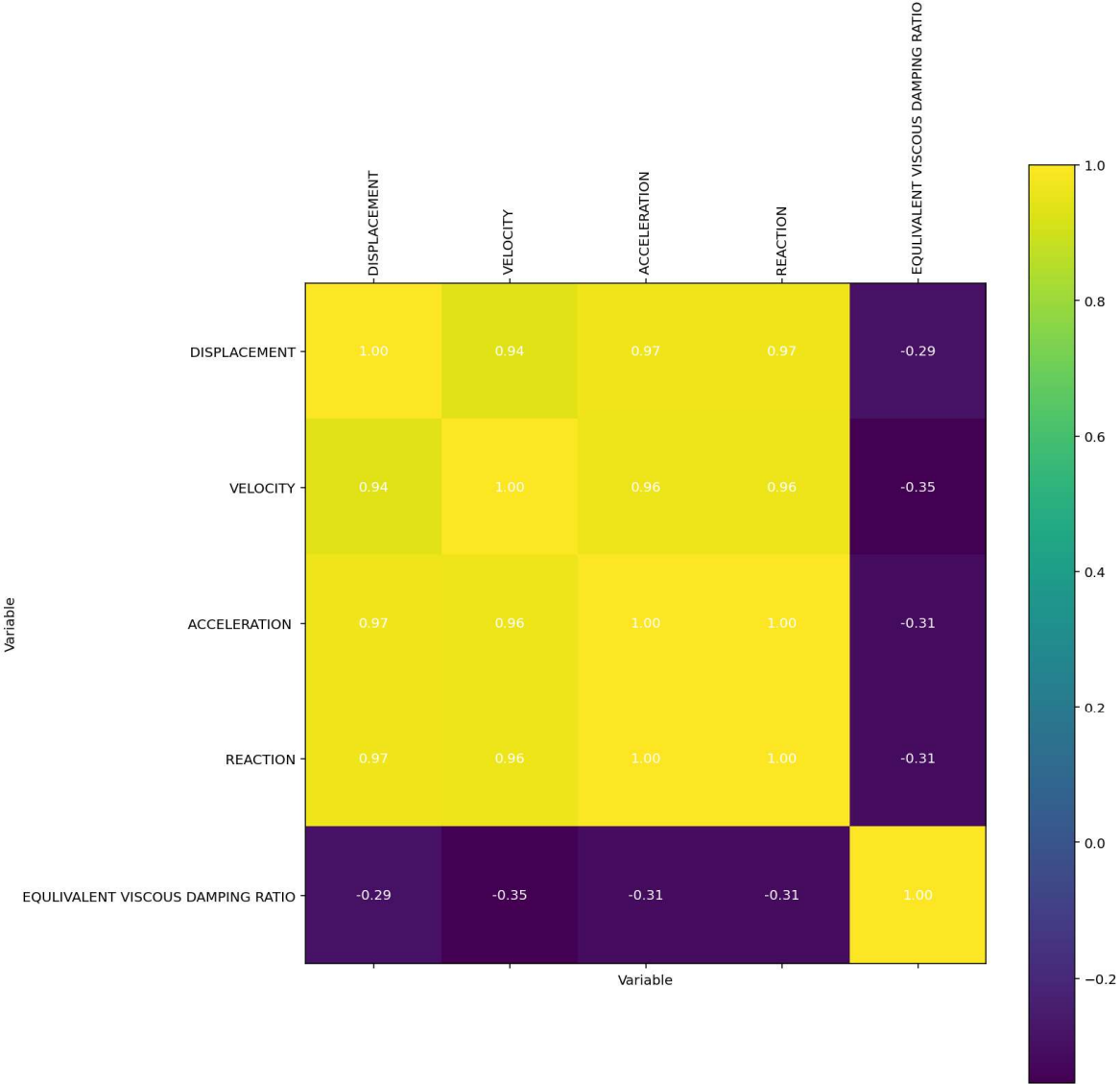
Histogram - Probability of Positive EQUIVALENT VISCOUS DAMPING RATIO [%] FROM IDA ANALYSIS is 100.00 %

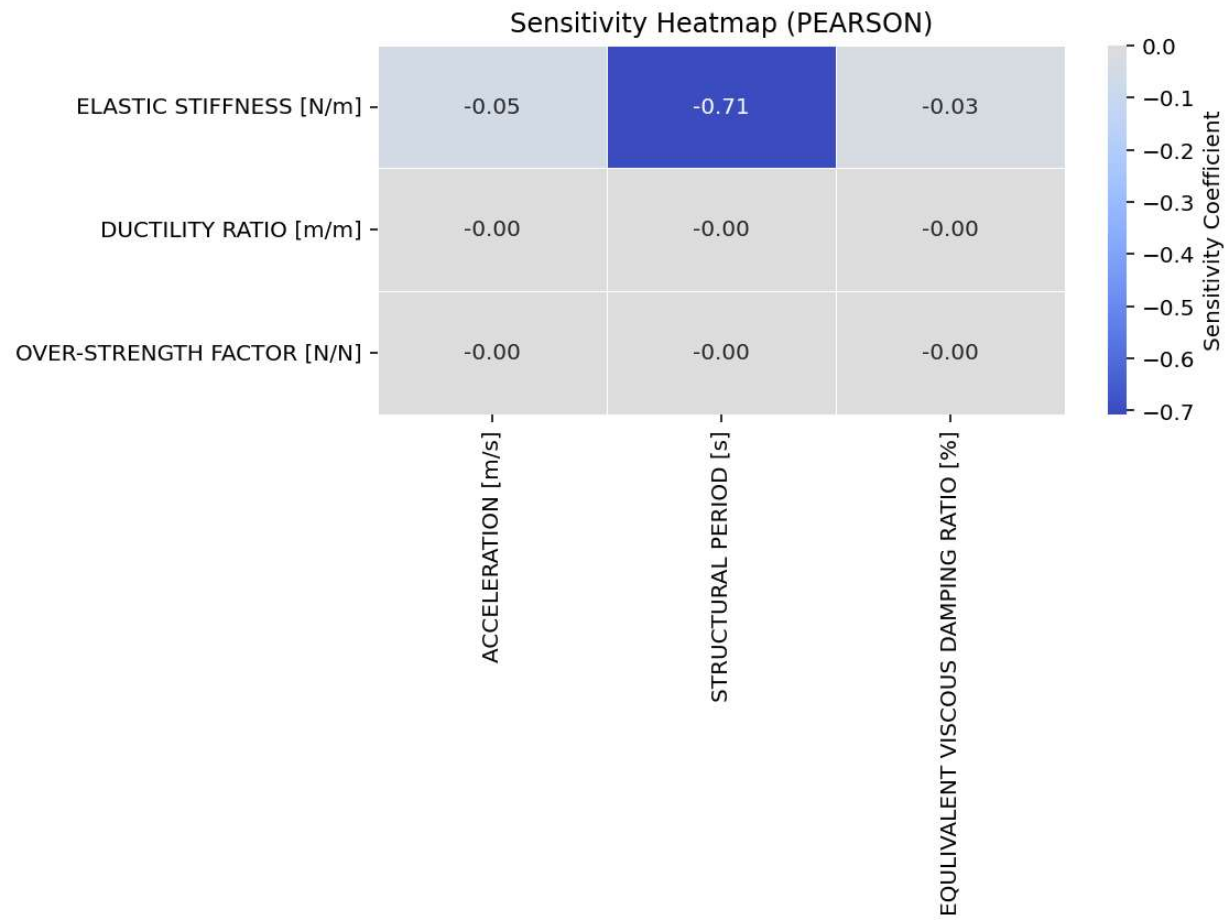


Boxplot of EQUIVALENT VISCOUS DAMPING RATIO [%] FROM IDA ANALYSIS

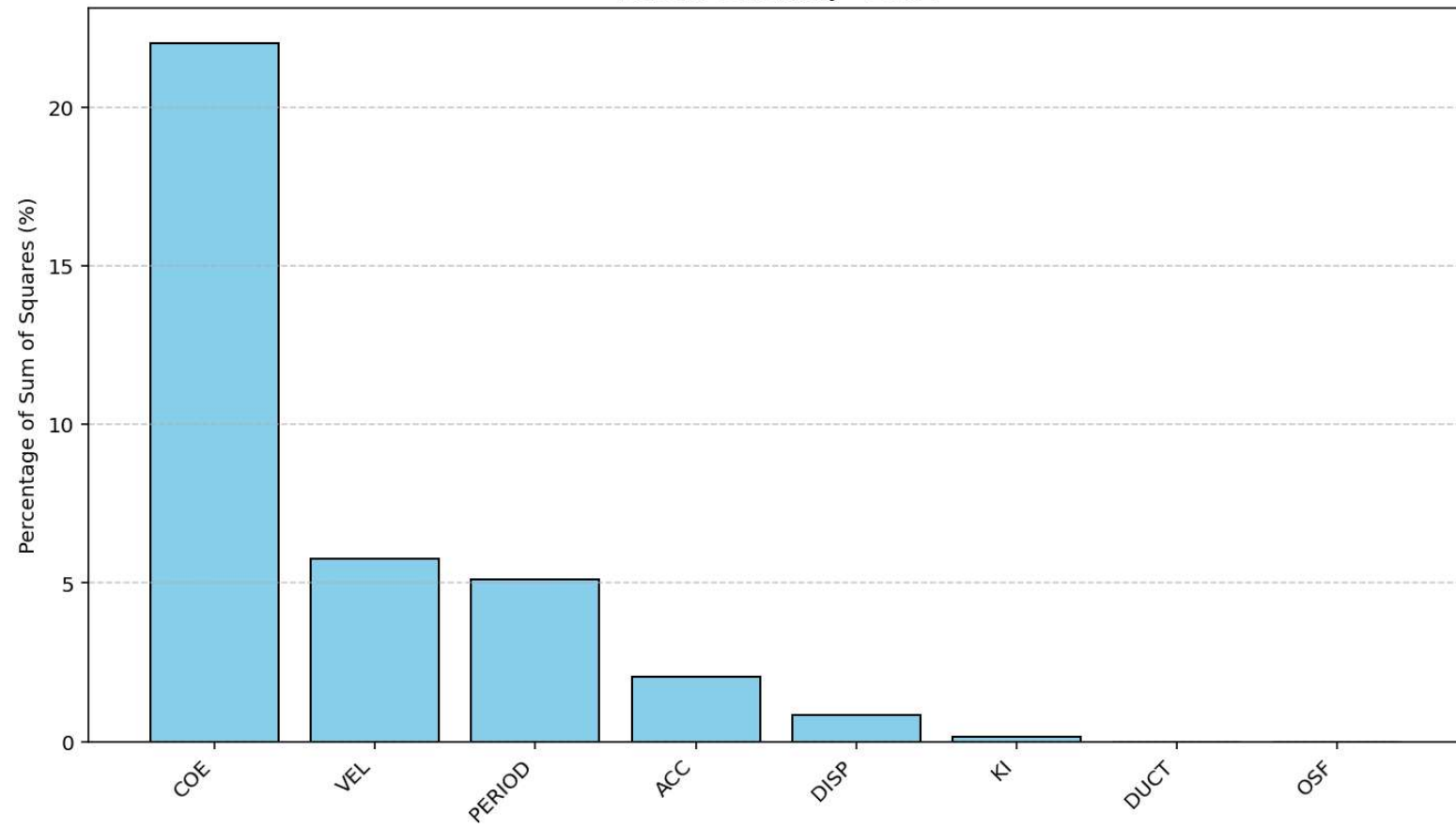


Correlation Heatmap





ANOVA Sensitivity - ZETA



ANOVA Sensitivity - ZETA

